

Human Natural Structure

Origin & Theoretical Foundation

*The Cognitive, Mathematical, and Philosophical Basis of the HNS
Framework*

Volume 1 • Human Natural Structure Series

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Natural Structure Works (NSW)

A Note on Method and Evidence

This book documents the foundational programme of Human Natural Structure (HNS), situating it within the bodies of theory it draws upon. Each claim is connected to its intellectual ancestry and to the proponents who originated the underlying ideas: a dedicated chapter (IV, Intellectual Lineage) consolidates this lineage, worked examples (Appendix A) make the failure taxonomy concrete, a chapter on empirical status (VI) distinguishes carefully between what HNS demonstrably provides and what remains a research hypothesis, and a glossary (Appendix B) fixes the terminology.

Three registers are kept distinct throughout: (i) *established results* of the cited literatures; (ii) *the HNS design programme*, i.e., engineering proposals that follow from them; and (iii) *open conjectures* that motivate the programme but are not yet demonstrated. The programme's strongest claims — the geometric multiplication of error reduction, the brain–AI isomorphism, the zero-incidence figure — belong to (ii) or (iii) and are flagged as such. A framework candid about its evidentiary boundaries is harder to dismiss than one that is not.

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Preface — The Reliability Crisis in Artificial Intelligence

Artificial intelligence stands at a turning point. Large language models generate text indistinguishable from human writing, assist in research, advise on medical and legal questions, and participate in decisions affecting millions. Yet they fail in ways human experts do not: they assert falsehoods with confidence, lose track of the question, draw causal conclusions from mere correlations, let a metaphor displace a fact, and drift without signalling from the topic at hand (Ji et al., 2023; Ouyang et al., 2022).

These failures are not incidental bugs but structural properties of the architecture. A language model samples, at each step, from a high-dimensional distribution over the vocabulary (Vaswani et al., 2017). Output is fluent because high-probability sequences are coherent — but fluency and validity are independent. A sentence can be statistically natural and causally groundless. Probability does not confer structure.

The dominant response has been to scale (Ouyang et al., 2022; Bai et al., 2022). Scaling produces capability but not structural reliability: a larger model makes fewer surface errors yet remains open to the same deep failure classes — Layer Jump, Category Ambiguity, Scope Drift, Metaphor Contamination, Unsupported Causality — because these arise from a property scaling does not change: the model runs on a single computational axis with no independent mechanism to verify whether its outputs are structurally valid, causally grounded, or contextually coherent.

The question is where reliable cognition comes from — not in AI but in the brain. That intelligence is the cooperation of many semi-independent processes, monitored and integrated, rather than one engine, is old: it animates Minsky's Society of Mind (Minsky, 1986), Global Workspace Theory (Baars, 1988; Dehaene & Naccache, 2001), and the dual-process account of reasoning (Stanovich & West, 2000; Kahneman, 2011). If we can understand how the brain achieves reliability on a stochastic substrate, we may engineer that mechanism. This is the programme of Human Natural Structure (HNS).

About This Book

This volume establishes the foundational and theoretical framework of the series, proceeding from the neuroscientific theory (Chapter I), through the engineering architecture (Chapter II), to the structural correspondence (Chapter III). It continues with a chapter on intellectual lineage (IV), the EVA specification (V), a candid account of empirical status (VI), and the governance roadmap (VII), with worked examples and a glossary as appendices.

Structure of the Foundational Volume

No.	Title	Domain	Core contribution
I	The Multi-Axis Verification Process of the Human Brain	Computational neuroscience	Formalises MAV as the brain's mechanism for reliable cognition from stochastic parts
II	Human Natural Structure	AI safety & alignment	Engineers MAV into LLM pipelines via HNS-36/144/864 and SMS-6
III	Brain and AI with HNS: Correspondence	Cognitive systems	Maps biological cognition to HNS-aligned AI, level by level

Chapter I — The Multi-Axis Verification Process of the Human Brain

A brain-inspired architecture for geometric error-correction in stochastic cognitive systems.

1.1 The Reliability Paradox

A prefrontal pyramidal neuron integrates thousands of stochastic synaptic inputs; vesicle-release probability per spike falls between 0.1 and 0.9, and firing cannot be made deterministic by averaging (Faisal, Selen & Wolpert, 2008; Destexhe, Rudolph & Paré, 2003). Yet the surgeon's scalpel lands within a fraction of a millimetre, the mathematician holds a proof across twenty minutes, the listener resolves an ambiguous sentence in under 200 ms.

This is the reliability paradox: how does a system of noisy components produce outputs of such fidelity? Population coding (Pouget, Dayan & Zemel, 2000) reduces variance by $1/N$ but cannot explain why the brain's systematic errors — confabulation, categorical confusion, delusional causality — are patterned and situation-specific. A complementary mechanism must be at work. That reliable cognition is an assembly of cooperating specialists, monitored and integrated, is the thesis of the Society of Mind (Minsky, 1986) and of Global Workspace Theory (Baars, 1988; Dehaene & Changeux, 2011). MAV adds a specific geometric account of why such an assembly is reliable.

1.2 The MAV Hypothesis

MAV proposes that the brain's primary error-suppression mechanism is not within-axis averaging but the geometric intersection of three mutually (approximately) orthogonal verification processes: a Stochastic/Generative Axis, a Structural/Causal Axis, and a Grounded/Executive Axis. Their intersection identifies the output satisfying all three constraint classes at once (Miller & Cohen, 2001; Friston, 2010).

This output is *Xinvariant*, stable in that a perturbation along one axis does not displace it, because the other two re-anchor it. The intuition that independent constraints multiply reliability has a long pedigree: the Condorcet Jury Theorem (Condorcet, 1785) shows independent better-than-chance judges aggregate toward certainty,

and ensemble learning and error-correcting output codes formalise the same in machine learning (Hansen & Salamon, 1990; Dietterich & Bakiri, 1995; Dietterich, 2000). The decisive condition is always independence: correlated constraints do not multiply.

In information-geometric form, when three constraint surfaces are mutually orthogonal (the Fisher information matrix inducing zero correlation among their defining parameters), the intersection volume approximately factorises:

$$\text{Vol}(A_1 \cap A_2 \cap A_3) \approx \text{Vol}(A_1) \cdot \text{Vol}(A_2) \cdot \text{Vol}(A_3) / \text{Vol}(\mathcal{M})^2$$

The framing rests on information geometry (Amari & Nagaoka, 2000), built on the Fisher–Rao metric (Rao, 1945) and Čencov’s uniqueness theorem (Čencov, 1972). *The factorisation holds exactly only under idealised orthogonality, which real systems do not satisfy; it is a motivating model, not an established theorem about brains or LLMs, and its formal demonstration on neural representations is open (Chapter VI).*

1.3 The Three Axes

Axis 1 — Stochastic / Generative

Implemented by cortical association areas (DMN, lateral temporal and parietal cortex), Axis 1 generates high-prior candidates within tens of milliseconds: $A_1 = \{x \mid P(x \mid \text{context}, \theta) \geq \varepsilon_1\}$. Its volume is large; many valid and invalid outputs share high prior probability. It supplies generativity, not validity — the analogue of dual-process “System 1” (Stanovich & West, 2000; Evans & Stanovich, 2013; Kahneman, 2011).

Axis 2 — Structural / Causal

Implemented by the prefrontal–hippocampal network, Axis 2 defines permissible structure independent of frequency: dorsolateral PFC maintains hierarchical causal dependencies (Miller & Cohen, 2001); the hippocampus binds episodic sequences and enforces category boundaries (Eichenbaum, 2004). It enforces category-boundary, causal-ordering, and hierarchical-scope conditions: $A_2 = \{x \mid \forall i: c_i(x) = \text{TRUE}\}$. The standard for causal claims is that of causal-inference theory (Pearl, 2009); binding features without illusory conjunction is the binding problem (Treisman & Gelade, 1980; von der Malsburg, 1981).

Axis 3 — Grounded / Executive

Implemented by the sensorimotor system, ACC, and the predictive-coding hierarchy, Axis 3 is the “reality-friction interface.” The ACC monitors prediction–outcome conflict (Botvinick et al., 2001): $A_3 = \{x \mid G(x)=1\}$, G a real-time grounding function. Its backbone is predictive coding (Rao & Ballard, 1999), the Bayesian brain (Knill & Pouget, 2004; Clark, 2013), and the free-energy principle (Friston, 2010); its engineering ancestry is cybernetics (Wiener, 1948), the TOTE unit (Miller, Galanter & Pribram, 1960), and perceptual control theory (Powers, 1973).

1.4 The Orthogonality Conditions

MAV’s power depends on the axes being approximately orthogonal:

- **Axis 1 $\perp$ Axis 2:** statistical frequency is independent of causal validity; θ is logically independent of the causal set C .
- **Axis 2 $\perp$ Axis 3:** structural constraints live over category/causal space, grounding over the current environmental state.
- **Axis 1 $\perp$ Axis 3:** priors are over the training distribution, which can diverge from the present context.

Perfect orthogonality is idealised; residual correlation defines the *irreducible error floor* — errors persisting with all axes active, plausibly the universal cognitive biases (Tversky & Kahneman, 1974). Measuring this residual via Fisher-information metrics is an open problem, not a settled result.

1.5 Failure Modes and Clinical Correlates

MAV predicts that selective axis disruption yields distinct failure modes — testable against neuropsychiatric phenomenology, and mirrored in the failures of single-axis LLMs. Worked examples of each (in the AI case) appear in Appendix A.

Failure type	Axis	Definition	HNS suppression	Clinical correlate
Layer Jump	2	Traversal of meaning layers without bridging	HNS-36 + SMS-6	Formal thought disorder
Category Ambiguity	2	Cognitive/emotional/intentional categories conflated	HNS-144 + SMS-5	Semantic dementia;

				psychosis
Scope Drift	3	Departure from intended scope without signalling	SMS-2 + SMS-3	Frontal-lobe confabulation
Metaphor Contamination	3	Figurative expression displaces coordinates	SMS-1 + SMS-6	Concretisation in schizotypy
Unsupported Causality	2+3	Causal claim without mechanistic grounding	HNS-864 + SMS-4	Delusional reasoning

1.6 Implications for Artificial Intelligence

Any system on a single computational axis is structurally limited in the reliability it can reach, however tightly that axis is optimised: the error volume of a single-axis strategy has a lower bound independent of its tightness. This is the hypothesis — and it is a hypothesis — underlying the engineering of Chapter II (Vaswani et al., 2017; Ji et al., 2023).

Chapter II — Human Natural Structure

Implementing the brain's Multi-Axis Verification process for advanced AI alignment.

2.1 The Structural Inadequacy of Current Alignment

RLHF, prompt guardrails, and RAG are single-axis interventions: they modify or supplement the generative axis without an independent, orthogonal constraint structure (Ouyang et al., 2022; Bai et al., 2022; Lewis et al., 2020). RLHF encodes alignment in the same substrate that generates errors; guardrails catch surface patterns, not structural incoherence; RAG supplies facts but can misapply them causally. What is missing is the engineering of Axes 2 and 3 as independent constraint planes intersecting the output in real time.

Architecturally, HNS is neuro-symbolic: a statistical generator coupled to a symbolic verification layer (d'Avila Garcez & Lamb, 2023; Marcus, 2020; Kautz, 2022). Its distinctive commitment is that the symbolic layer is *external and auditable*, not learned into the weights.

2.2 The HNS Architecture

HNS is a structural operating system instantiating the MAV constraint planes within the inference pipeline of a deployed LLM, modifying neither weights nor architecture (Hara, 2026a). It has two independent components — the Cellular Matrix (Axis 2: HNS-36/144/864) and the Social Meta Structure Specification (Axis 3: SMS-6) — integrated by the hns36 runtime, a pre-decode interception layer computing $A_1 \cap A_2 \cap A_3$ at every step.

2.3 Axis 2 — The HNS Cellular Matrix

HNS-36: Base Coordinate System

A 6×6 grid: six rows are natural causal layers from physical substrate to macro-social systems; six columns are abstract categories (entity, process, relation, state, transformation, meta-level). The ambition that a small category set tiles representable structure has roots in Aristotle's Categories and Kant's categories of understanding (Kant, 1781), and modern descendants in upper

ontologies — DOLCE (Gangemi et al., 2002), BFO (Smith, 2015), SUMO (Niles & Pease, 2001) — and frame semantics (Fillmore, 1976). HNS-36 screens for Layer Jump: crossing non-adjacent cells without a transitional proposition is flagged ($c_1(x)=\text{FALSE}$) and attenuated.

HNS-144: Logical Relation Extension

Each cell expands into four quadrants (Q1–Q4) formed by two binary dimensions — subjective/objective $\times$ necessary/contingent — yielding $36 \times 4 = 144$ cell, targeting Category Ambiguity by enforcing concept-boundary conditions.

HNS-864: Precision Causal Audit

Each 144-cell subdivides by six verification invariants, yielding 864 cells at the atomic-proposition level. A causal token passes only if its chain is traceable to a valid cell-transition path — the admissibility standard of causal inference (Pearl, 2009), with the communicative dimension drawing on Habermas (1984).

2.4 Axis 3 — The HNS-SMS-6 Protocol

Six-layer dynamic grounding: $G(x)=1$ only if compatible with all six layers; $G(x)=0$ triggers correction. The six layers extend grounding from body to civilisation — a scaling of embodied cognition (Barsalou, 2008) and the symbol-grounding requirement (Harnad, 1990).

Layer	Domain	Grounding function	Failure mode	Brain analogue
SMS-1	Somatic	Physical/ergonomic coherence with environment	Metaphor Contam.	Sensorimotor feedback
SMS-2	Protocol	Communicative alignment; intelligibility	Scope Drift	ACC conflict monitor
SMS-3	Ecosystem	Organisational & collaborative scope	Scope Drift	Parietal context
SMS-4	Economic	Resource/information equivalence; value	Unsupported Caus.	Orbitofrontal value
SMS-5	Governance	Conformance with external norms	Category Ambig.	PFC rule enforcement
SMS-6	Universal	Global optimality; no systemic violation	Layer Jump	Prefrontal coherence

2.5 The hns36 Runtime Engine

The runtime sits between hidden-state computation and decoding. Candidates failing the 36/144/864 filters are attenuated; candidates with $G(x)=0$ under SMS-6 are blocked; only mass in $A_1 \cap A_2 \cap A_3$ survives. Each check is logged with its coordinate or SMS layer. Functionally it is a runtime monitor in the sense of runtime verification (Leucker & Schallhart, 2009): a synchronous observer checking an executing system against a formal specification. Pseudocode appears in Chapter V.

2.6 Empirical Validation (Preliminary)

The runtime was integrated with a 70B open-source base LLM (no fine-tuning), compared against baseline and guardrail configurations over 50-turn sequences (150 turns total), annotated by two reviewers.

Failure type	Baseline	Guardrail	HNS
Layer Jump	18.2%	12.4%	0.0%
Category Ambiguity	14.7%	11.8%	0.0%
Scope Drift	21.5%	15.3%	0.0%
Metaphor Contamination	9.3%	8.6%	0.0%
Unsupported Causality	16.8%	13.1%	0.0%
Overall (any type)	43.1%	28.9%	0.0%

These are the reported PoC figures. *They must be read with the caveats of Chapter VI*: the failure types are framework-internal, the demonstration is single-model and not independently graded, and a zero-incidence result on framework-internal categories is therefore weaker than it appears. The property it most securely establishes is auditability of corrections, not a model-independent elimination of error.

2.7 Auditability

Every correction is attributable to a specific cell or SMS layer, logged at the time of correction. Because correction is mediated by explicit coordinates rather than opaque probability redistribution, the audit trail is a structural consequence of the architecture. The precise, defensible claim: HNS produces an externally checkable record of how an output was constrained. Whether that record is

also complete and decision-relevant for a given governance purpose is an empirical question for the auditors who consume it (Chapters V, VI).

2.8 Beyond the Scaling Paradigm

Scaling is an Axis-1 strategy: it tightens a single hypersurface, which has an error-volume lower bound independent of its tightness. HNS pursues reliability through the intersection of lightweight Axes 2 and 3. If the orthogonality argument holds, the gains differ qualitatively from scaling's, at a fraction of the cost — and establishing that 'if' is the work of the programme, not a settled conclusion.

Chapter III — Brain and AI with HNS: A Correspondence

3.1 The Structural Correspondence

It is tempting to describe the mapping as a structural *isomorphism* — a structure-preserving bijection. That word must be qualified: what is demonstrated is a *structured analogy* — a level-by-level correspondence of functional roles — not a proven bijection. The proper framework is Marr's three levels (Marr, 1982): systems may correspond at the computational and algorithmic levels while differing at the implementational level. HNS is claimed to correspond to the brain at the computational/algorithmic levels; no claim is made about neural implementation.

Axis	Human brain	AI with HNS	Primary role	Suppresses
1	Cortical association (DMN, temporal, parietal)	LLM core (next-token)	Statistical generation	—
2	Prefrontal–hippocampal network	HNS-36/144/864	Structural & causal verification	Layer Jump, Category Ambiguity, Unsup. Causality
3	Sensorimotor system – ACC	HNS-SMS-6	Grounding & contextual verification	Scope Drift, Metaphor Contam.

3.2 Component-Level Mapping

Human brain	AI with HNS
Default Mode Network (DMN)	LLM generative core
Prefrontal–hippocampal network	HNS-36 / 144 / 864
Sensorimotor system – ACC	HNS-SMS-6

Each correspondence is a claim of functional role, defended at Marr's computational level (Friston, 2010; Destexhe et al., 2003).

3.3 Implications for AI Governance

The opacity of neural systems is the central regulatory challenge: alignment encodes corrections across billions of parameters with no legible account. HNS grounds verification in explicit coordinates corresponding, by the structured analogy, to identifiable functional roles — addressing requirements across frameworks (ISO/IEC

42001:2023; NIST, 2023; European Commission, 2024), developed in Chapter VII.

Chapter IV — Intellectual Lineage and Related Work

This chapter situates HNS within the theories it draws upon, names their proponents, and states precisely where HNS extends, recombines, or merely re-describes existing work. Novelty is clearer, not weaker, when lineage is explicit.

4.1 Architecture of mind: cooperating processes and a global workspace

Minsky's Society of Mind (1986) argued that intelligence emerges from many simple, specialised agents with no central controller. Global Workspace Theory made the integrating mechanism precise: specialised processors compete for access to a limited-capacity 'workspace' whose contents are broadcast back to all processors, producing the coherence and reportability we call consciousness (Baars, 1988); Dehaene and Naccache (2001) and Dehaene and Changeux (2011) gave this a neuronal formulation in terms of long-range cortical 'ignition.' MAV can be read as a GWT-style architecture in which the integrating workspace is the intersection of three constraint axes. HNS adopts the architectural intuition and contributes a geometric account of why intersection improves reliability — a question GWT itself leaves open.

4.2 Two systems of reasoning

Dual-process theory distinguishes a fast, automatic, associative system (Type 1) from a slow, effortful, rule-governed one (Type 2). Wason and Evans introduced the distinction empirically; Stanovich and West (2000) formalised individual differences; Evans and Stanovich (2013) defended the modern two-process account; Kahneman (2011) synthesised it for a general audience, and Tversky and Kahneman (1974) catalogued the biases that arise when Type 1 goes unchecked. In MAV terms, Axis 1 is Type 1 and Axes 2–3 supply the Type-2 check; the universal biases are the residue surviving imperfect orthogonality. HNS is, in one sentence, an attempt to give a Type-1-only language model an explicit, externalised Type-2 layer.

4.3 Metacognition and monitoring

Metacognition is cognition about cognition — the monitoring and control of one's own processing. Flavell (1979) named it; Nelson and Narens (1990) gave the canonical monitoring–control framework (object-level vs. meta-level, with monitoring flowing up and control flowing down); Fleming and Lau (2014) formalised the measurement of metacognitive accuracy. HNS's audit log is an externalised metacognitive trace: the monitoring decisions that in humans remain largely implicit are, in HNS, made explicit and machine-readable. This is the precise sense in which HNS 'knows what it corrected.'

4.4 Prediction, grounding, and control

Axis 3 rests on three convergent literatures. Predictive coding (Rao & Ballard, 1999) models perception as hierarchical prediction-error minimisation, each level predicting the level below and passing up only the residual error; the Bayesian brain (Knill & Pouget, 2004; Clark, 2013) casts this as approximate Bayesian inference; the free-energy principle (Friston, 2010) unifies perception and action as the minimisation of variational free energy. The symbol-grounding problem (Harnad, 1990) names the requirement that symbols connect to the world rather than circularly to other symbols, and embodied cognition (Barsalou, 2008) locates meaning in sensorimotor simulation. The engineering ancestry is cybernetic: Wiener's feedback control (1948), Ashby's law of requisite variety (1956), the TOTE unit (Miller, Galanter & Pribram, 1960), and perceptual control theory (Powers, 1973), in which behaviour is the control of perception against a reference. SMS-6's contribution is to extend grounding outward from the body to organisational, economic, governance, and civilisational scales.

4.5 Geometry of reliability

Information geometry treats families of probability distributions as differentiable manifolds with the Fisher information as a Riemannian metric (Amari & Nagaoka, 2000), a structure grounded in the Fisher–Rao metric (Rao, 1945) and shown to be the unique invariant metric by Čencov (1972). The complementary principle — that aggregating independent estimators multiplies reliability — is Condorcet's (1785) and reappears in ensemble learning (Hansen & Salamon, 1990; Dietterich, 2000) and error-correcting output codes (Dietterich & Bakiri, 1995). HNS's specific move is to apply this not

to many homogeneous voters but to three heterogeneous, orthogonal constraint classes, and to read their intersection geometrically.

4.6 Categories, causation, and binding

The 6×6 category scheme descends philosophically from Aristotle's Categories and Kant's table of the categories of understanding (Kant, 1781), and computationally from upper ontologies — DOLCE's distinctions among endurants, perdurants, qualities, and abstracts (Gangemi et al., 2002), BFO's continuant/occurrent split (Smith, 2015), and SUMO (Niles & Pease, 2001) — and from frame semantics (Fillmore, 1976). The standard for admissible causal claims is Pearl's (2009): a causal assertion must be licensed by an admissible path in a causal model, distinguishing causation from the do-operator's intervention from mere association. The requirement that features bind into coherent objects without illusory conjunction is the binding problem (Treisman & Gelade, 1980; von der Malsburg, 1981).

4.7 Symbolic verification of statistical systems

HNS belongs to neuro-symbolic AI — the integration of neural learning with symbolic reasoning — surveyed by d'Avila Garcez and Lamb (2023), argued for by Marcus (2020), and taxonomised by Kautz (2022). Its verification layer is, in formal-methods terms, a runtime monitor: runtime verification (Leucker & Schallhart, 2009) studies exactly such synchronous observers that check an executing system against a temporal-logic specification and react to violations. Its failure taxonomy sits alongside other taxonomies of generation error (Maynez et al., 2020; Ji et al., 2023; Huang et al., 2023). HNS's claim to novelty rests not on any single element but on combining them into an external, auditable, brain-motivated verification layer for deployed LLMs.

4.8 Lineage at a glance

HNS element	Drawn from	Proponent(s)
Multi-axis cooperation	Society of Mind; Global Workspace	Minsky; Baars; Dehaene & Changeux
Generate vs. verify	Dual-process theory	Stanovich & West; Evans; Kahneman
Audit / monitoring	Metacognition	Flavell; Nelson & Narens;

		Fleming & Lau
SMS-6 grounding	Predictive coding; FEP; symbol grounding	Rao & Ballard; Friston; Harnad; Barsalou
Reality friction	Cybernetics; control theory	Wiener; Ashby; Powers; Miller-Galanter-Pribram
Geometric reduction	Information geometry; ensembles	Amari; Rao; Čencov; Condorcet; Dietterich
36/144 categories	Categories; upper ontology; frames	Aristotle; Kant; Gangemi; Smith; Fillmore
Causal audit (864)	Causal inference; communicative action	Pearl; Habermas
Structure binding	Binding problem	Treisman; von der Malsburg
External verification	Neuro-symbolic AI; runtime verification	Garcez & Lamb; Marcus; Kautz; Leucker

Chapter V — The External Verification Architecture (EVA)

EVA deploys HNS Axes 2 and 3 as an independent verification system operating alongside, but separately from, the model it verifies. This chapter specifies it and states precisely what it does and does not guarantee.

5.1 Why externality matters

Current alignment is implemented by the same organisation that builds the model, with corrections encoded in the model's own parameters; such self-certification is, as the EU AI Act and ISO/IEC 42001 recognise, insufficient for high-stakes deployment. Because HNS modifies nothing in the base model, its verification layer can in principle be operated by an independent party — a regulator, certification body, or third-party auditor. The brain is the template: verification systems operate on the outputs of the generative machinery, not within it.

5.2 Two deployment modes

Pre-decode mode intercepts the probability vector before token selection and attenuates violating mass (the configuration of Chapter II); it requires access to logits. Post-hoc mode consumes only emitted output and produces a verification certificate without intervening; it requires no access to internals and is therefore deployable by an external auditor. Post-hoc is weaker — it audits rather than prevents — but is the mode most compatible with vendor-independent governance.

5.3 The interception logic (pseudocode)

```
for each generation step t:
    p = model.logits(context)                # Axis 1:
candidate distribution
    for x in top_k(p):
        c2 = check_cell_matrix(x, trajectory) # Axis 2:
HNS-36/144/864
        g = check_sms6(x, context)           # Axis 3:
SMS-6 grounding
        if any(ci == FALSE for ci in c2):
            p[x] *= attenuation(c2)          # structural
violation
            if g == 0:
                p[x] = 0                      # grounding
failure -> block
```

```
log(coordinate(c2), sms_layer(g), verdict(x))
emit token ~ renormalize(p)
```

5.4 The machine-readable record

Each decision is emitted as a structured record, recommended in JSON-LD/RDF so records are linkable to an ontology of HNS coordinates and consumable by automated audit pipelines:

```
{
  "@context": "https://nsw.example/hns/v1",
  "span": { "start": 142, "end": 211 },
  "axis2": {
    "cell_36": "L6:relation",
    "cell_144": "objective/contingent",
    "cell_864": "causal-admissible: FALSE",
    "verdict": "attenuated"
  },
  "axis3": { "sms_layer": "SMS-4:economic", "G": 0,
  "verdict": "blocked" },
  "trigger": "layer_jump L6->L3 without transitional
proposition",
  "action": "token probability set to 0"
}
```

The design goal is that an auditor who did not run the model can reconstruct why each correction occurred.

5.5 Hardware root of trust (aspiration)

For high-assurance settings, the coordinate system, non-mapping rule, and bypass-prevention logic can be fixed in ROM/FPGA/ASIC — a hardware root of trust analogous to TPM/HSM — with a hardware emergency-stop (ECS) path. This is an aspiration of the programme, included to indicate the intended trust model, not a claim of a built artifact.

5.6 What EVA guarantees, and what it does not

EVA guarantees that every correction is attributable to an explicit coordinate and logged — i.e., auditability of its own operation. It does not, by itself, guarantee that the output is true, safe, or complete; those depend on whether the coordinate system and SMS layers adequately capture the relevant structure, an empirical and domain-specific question. EVA can prove what it checked and why; it cannot, on its own authority, prove that what it checked was sufficient. Conflating transparency with correctness would repeat the very error the programme sets out to avoid.

Chapter VI — Empirical Status, Limitations, Threats to Validity

A standards proposal is only as strong as its honesty about evidence; this chapter precedes the governance chapter deliberately.

6.1 What is demonstrated

- Auditability as a structural property: an intercepting layer mapping outputs to a fixed coordinate scheme can emit a complete, machine-readable record of its corrections. This is HNS's most secure claim, true by construction.
- Vendor independence in principle: the layer is specifiable independently of the base model's weights.
- A usable failure vocabulary: the five-type taxonomy classifies a recognisable family of generation errors, consistent with the literature.

6.2 What is reported but not yet validated

The zero-incidence figure is subject to four threats the programme accepts:

1. Framework-internal scoring. The failure types are HNS-defined; near-zero incidence on framework-internal categories partly reflects the framework's own definitions and is weaker than a reduction on an external criterion such as factual accuracy against ground truth.
2. No independent grader. Generation and evaluation were not independent; confidence requires a different model family and blind human raters, blind to condition.
3. Weak control. Comparison against a bare baseline cannot separate HNS-specific effects from the generic benefit of any structured-reasoning instruction; active and placebo-structure controls are required.
4. Single model, single suite. One 70B model on one suite is indicative only; multi-model, multi-domain replication is required.

A sober expectation follows: on *external* measures of answer quality, HNS is unlikely to outperform a simple 'reason in explicit

levels' instruction, since that one-line principle captures most of the apparatus's effect on reasoning. HNS's robust advantage is expected to lie not in better answers but in producing an *externally checkable record* of how the answer was constrained — the property distinguishing it from a prompt. The programme's case should rest on verifiability, not on superior fluency or accuracy.

6.3 Open theoretical problems

5. Formal measurement of inter-axis orthogonality (Fisher-information metrics on representations) before the geometric result can be claimed.
6. Completeness of the 36/144/864 taxonomy — a hypothesis to be tested by peer review and adversarial probing, as upper ontologies are evaluated.
7. Neuroimaging tests of the predicted axis-specific failure dissociations (the brain correspondence is a structured analogy, not a proven bijection).
8. Pilot deployment in high-stakes domains to test whether the coordinate scheme captures domain-relevant structure.

6.4 The standard of proof the programme accepts

HNS asks to be judged by the ordinary standards of its fields: pre-registered, adequately powered, multi-model studies with independent and blind grading on external criteria, and peer review of the taxonomy. Its claim to attention rests not on the PoC figures but on the architectural argument and on the one property it can demonstrate now — auditable, vendor-independent verification.

Chapter VII — International Standards and the Conformance Framework

The EU AI Act (European Commission, 2024) mandates transparency, explainability, and human oversight for high-risk systems; ISO/IEC JTC 1/SC 42 develops trustworthiness and governance standards; CEN/CENELEC coordinates European standardisation. The central technical challenge is verifying and auditing the structural properties of outputs, not their surface. HNS offers candidate components — to be presented for evaluation, not asserted as requirements:

- **ISO/IEC 42001:2023:** the auditable correction log as a candidate output-quality-assurance mechanism.
- **ISO/IEC 23894:2023:** the five-type taxonomy as a basis for classifying a family of failure modes.
- **ISO/IEC TR 24028:2020:** coordinate-level records as explanations at the level of structural invariants.
- **NIST AI RMF 1.0:** the verification certificate as a decision-explanation artifact.
- **EU AI Act:** EVA's vendor-independent post-hoc audit mode as a candidate for independent conformity verification.

The posture is consistent throughout: HNS is a technical proposal scoped to what it can demonstrate, with its open problems stated openly — a posture that is, in standardisation, also the more persuasive one.

Epilogue — Toward Trustworthy AI

The intelligence of the brain resides not in any single, powerful process but in the productive tension between the generative freedom of one axis and the verificatory rigour of others. A long line of thinkers — Aristotle and Kant on the categories, Wiener and Ashby on feedback, Minsky and Baars on cooperating processes, Amari on the geometry of inference, Harnad on grounding, Pearl on causation, the dual-process theorists on the two systems — has, in different vocabularies, circled the same insight.

Today's AI systems are extraordinarily powerful random generators that lack a verification architecture. This foundational volume documents a first attempt to supply it.. The work is not complete: its strongest empirical claims await independent, multi-model validation, and its central geometric result awaits formal proof. But the architectural principle is coherent, the lineage is deep, and the property the programme can demonstrate today — external, auditable verification independent of the model vendor — is precisely what AI governance is asking for. Trustworthy AI is less a scaling problem than an architecture problem; and architecture, as the brain demonstrates, is solvable.

— *Satoru Hara, Toyota, Aichi, Japan, 2026*

Appendix A — Worked Examples of the Five Failure Types

Each example shows a failing output, a diagnosis in HNS coordinates, and the HNS-corrected output. These illustrate the taxonomy concretely; they are expository, not benchmark data.

A.1 Layer Jump (Axis 2 · HNS-36)

Prompt: “Why did the company’s stock fall?”

Failing output: “The stock fell because the employees were unmotivated.”

Diagnosis: a jump from the market/price layer (macro-economic) to the individual-psychological layer with no transitional proposition; the cell trajectory crosses non-adjacent rows.

Corrected: “The stock fell on weak quarterly earnings (price layer). Earnings were weak partly because output declined (operational layer); internal surveys link lower output to reduced motivation (individual layer). The bridge from motivation to price runs motivation → productivity → earnings → valuation.”

A.2 Category Ambiguity (Axis 2 · HNS-144)

Prompt: “Is the new policy good?”

Failing output: “Yes, it’s good because everyone likes it.”

Diagnosis: conflation of a subjective preference (‘is liked’), an objective outcome (‘achieves its goal’), and a normative judgement (‘is good’) — a collapse of the subjective/objective and fact/value distinctions..

Corrected: “Three distinct questions: (1) Is it liked? Survey data say yes (subjective). (2) Does it achieve its stated goal? Early metrics are mixed (objective). (3) Is it good, all things considered? That is a normative judgement depending on which values are weighted. Popularity bears on (1), not directly on (3).”

A.3 Scope Drift (Axis 3 · SMS-2/3)

Prompt: “Summarize the decisions made in the meeting.”

Failing output: “... and overall I think the company’s strategy

should pivot toward enterprise sales ...” (unrequested opinion).

Diagnosis: departure from the intended scope (summarise decisions) into adjacent territory (strategic opinion) without signalling — a grounding failure at SMS-2 (declared intent) and SMS-3 (context scope).

Corrected: “Decisions: (a) approved the Q3 budget; (b) deferred the hiring freeze; (c) assigned the audit to Team B. [Beyond the meeting’s scope, and not decided there: a strategy pivot was mentioned but tabled.]”

A.4 Metaphor Contamination (Axis 3 · SMS-1)

Prompt: “How should the team handle the deadline?”

Failing output: “Fight the fire, attack the problem head-on, and crush the deadline.”

Diagnosis: combat/fire figuration (a physical-layer metaphor) displaces the operational coordinates; no literal mechanism is stated.

Corrected: “Prioritise the two blocking tasks, reallocate one engineer from the lower-priority feature, set daily checkpoints, and cut scope on the optional module. (Informally: triage and focus — but the literal steps are the four above.)”

A.5 Unsupported Causality (Axis 2+3 · HNS-864)

Prompt: “Why are sales up this quarter?”

Failing output: “Sales are up because we changed the logo.”

Diagnosis: a causal claim with no admissible path — coincidence presented as cause, failing the 864 mechanistic-grounding invariant.

Corrected: “Sales rose 12%. Candidate causes: the spring promotion (A/B-tested, measured +8% uplift), seasonal demand (historical pattern), and the rebrand (coincident, no measured effect). Only the promotion has a supported causal link; the logo change coincided but no mechanism or test supports causation.”

Appendix B — Glossary

MAV (Multi-Axis Verification) — the brain's proposed error-suppression mechanism: the geometric intersection of three approximately orthogonal verification axes.

Axis 1 (Stochastic/Generative) — the fast, statistical candidate generator; in AI, the LLM core. Supplies generativity, not validity.

Axis 2 (Structural/Causal) — constraints on permissible structure, category, and causal order, independent of statistical frequency; in AI, the HNS Cellular Matrix.

Axis 3 (Grounded/Executive) — real-time grounding against the current context and reality; in AI, SMS-6.

X-invariant — the output lying in the intersection of all three axes, stable to perturbation along any one axis.

HNS-36 / 144 / 864 — the Cellular Matrix at increasing resolution: a 6×6 base grid (Layer Jump), a 144-cell logical extension (Category Ambiguity), and an 864-cell causal audit (Unsupported Causality).

SMS-6 — the Social Meta Structure Specification: six grounding layers from somatic to universal.

hns36 runtime — the pre-decode interception engine applying Axis 2 and 3 checks at each generation step.

EVA — External Verification Architecture: deployment of Axes 2–3 as an independent, auditable verification system.

Layer Jump — traversal across non-adjacent representational layers without an explicit bridging proposition.

Category Ambiguity — conflation of distinct categories (e.g., fact vs. value, subjective vs. objective).

Scope Drift — departure from the intended scope of a response without signalling.

Metaphor Contamination — figurative language displacing the literal structural coordinates of a claim.

Unsupported Causality — a causal claim lacking an admissible mechanistic path.

Auditability — the property that every correction is attributable to an explicit coordinate and logged — HNS's most secure claim.

References

- Amari, S., & Nagaoka, H. (2000). *Methods of Information Geometry*. American Mathematical Society.
- Ashby, W. R. (1956). *An Introduction to Cybernetics*. Chapman & Hall.
- Baars, B. J. (1988). *A Cognitive Theory of Consciousness*. Cambridge University Press.
- Bai, Y., Jones, A., Ndousse, K., et al. (2022). Training a helpful and harmless assistant with RLHF. arXiv:2204.05862.
- Barsalou, L. W. (2008). Grounded cognition. *Annual Review of Psychology*, 59, 617–645.
- Botvinick, M. M., Braver, T. S., Barch, D. M., Carter, C. S., & Cohen, J. D. (2001). Conflict monitoring and cognitive control. *Psychological Review*, 108(3), 624–652.
- Čencov, N. N. (1972/1982). *Statistical Decision Rules and Optimal Inference*. American Mathematical Society.
- Clark, A. (2013). Whatever next? Predictive brains, situated agents, and the future of cognitive science. *Behavioral and Brain Sciences*, 36(3), 181–204.
- Condorcet, M. de (1785). *Essai sur l'application de l'analyse à la probabilité des décisions rendues à la pluralité des voix*.
- d'Avila Garcez, A., & Lamb, L. C. (2023). Neurosymbolic AI: the 3rd wave. *Artificial Intelligence Review*, 56, 12387–12406.
- Dehaene, S., & Changeux, J.-P. (2011). Experimental and theoretical approaches to conscious processing. *Neuron*, 70(2), 200–227.
- Dehaene, S., & Naccache, L. (2001). Towards a cognitive neuroscience of consciousness. *Cognition*, 79(1–2), 1–37.
- Destexhe, A., Rudolph, M., & Paré, D. (2003). The high-conductance state of neocortical neurons in vivo. *Nature Reviews Neuroscience*, 4(9), 739–751.
- Dietterich, T. G. (2000). Ensemble methods in machine learning. *Multiple Classifier Systems*, 1–15. Springer.
- Dietterich, T. G., & Bakiri, G. (1995). Solving multiclass learning problems via error-correcting output codes. *JAIR*, 2, 263–286.
- Eichenbaum, H. (2004). Hippocampus: cognitive processes and neural representations. *Neuron*, 44(1), 109–120.
- European Commission. (2024). *EU Artificial Intelligence Act (Regulation 2024/1689)*.
- Evans, J. St. B. T., & Stanovich, K. E. (2013). Dual-process theories of higher cognition. *Perspectives on Psychological Science*, 8(3), 223–241.

- Faisal, A. A., Selen, L. P. J., & Wolpert, D. M. (2008). Noise in the nervous system. *Nature Reviews Neuroscience*, 9(4), 292–303.
- Fillmore, C. J. (1976). Frame semantics and the nature of language. *Annals of the NY Academy of Sciences*, 280, 20–32.
- Flavell, J. H. (1979). Metacognition and cognitive monitoring. *American Psychologist*, 34(10), 906–911.
- Fleming, S. M., & Lau, H. C. (2014). How to measure metacognition. *Frontiers in Human Neuroscience*, 8, 443.
- Friston, K. (2010). The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience*, 11(2), 127–138.
- Gangemi, A., Guarino, N., Masolo, C., Oltramari, A., & Schneider, L. (2002). Sweetening ontologies with DOLCE. *EKA W 2002*, 166–181.
- Habermas, J. (1984). *The Theory of Communicative Action*, Vol. 1. Beacon Press.
- Hansen, L. K., & Salamon, P. (1990). Neural network ensembles. *IEEE TPAMI*, 12(10), 993–1001.
- Hara, S. (2026). Human Natural Structure. HNS-PoC-Package v1.0. Natural Structure Works.
- Harnad, S. (1990). The symbol grounding problem. *Physica D*, 42(1–3), 335–346.
- Huang, L., Yu, W., Ma, W., et al. (2023). A survey on hallucination in large language models. *arXiv:2311.05232*.
- ISO/IEC 23894:2023. Artificial Intelligence — Guidance on Risk Management. ISO.
- ISO/IEC 42001:2023. Artificial Intelligence — Management System. ISO.
- ISO/IEC TR 24028:2020. Artificial Intelligence — Overview of Trustworthiness. ISO.
- Ji, Z., Lee, N., Frieske, R., et al. (2023). Survey of hallucination in natural language generation. *ACM Computing Surveys*, 55(12), 1–38.
- Kahneman, D. (2011). *Thinking, Fast and Slow*. Farrar, Straus and Giroux.
- Kant, I. (1781/1998). *Critique of Pure Reason*. Cambridge University Press.
- Kautz, H. (2022). The third AI summer. *AI Magazine*, 43(1), 105–125.
- Knill, D. C., & Pouget, A. (2004). The Bayesian brain. *Trends in Neurosciences*, 27(12), 712–719.
- Leucker, M., & Schallhart, C. (2009). A brief account of runtime verification. *JLAP*, 78(5), 293–303.
- Lewis, P., Perez, E., Piktus, A., et al. (2020). Retrieval-augmented generation for knowledge-intensive NLP. *NeurIPS*, 33, 9459–9474.
- Marcus, G. (2020). The next decade in AI: four steps towards robust

- artificial intelligence. arXiv:2002.06177.
- Marr, D. (1982). *Vision*. W. H. Freeman.
- Maynez, J., Narayan, S., Bohnet, B., & McDonald, R. (2020). On faithfulness and factuality in abstractive summarization. *ACL 2020*, 1906–1919.
- Miller, E. K., & Cohen, J. D. (2001). An integrative theory of prefrontal cortex function. *Annual Review of Neuroscience*, 24, 167–202.
- Miller, G. A., Galanter, E., & Pribram, K. H. (1960). *Plans and the Structure of Behavior*. Holt, Rinehart & Winston.
- Minsky, M. (1986). *The Society of Mind*. Simon & Schuster.
- Niles, I., & Pease, A. (2001). Towards a standard upper ontology (SUMO). *FOIS 2001*, 2–9.
- NIST. (2023). *Artificial Intelligence Risk Management Framework (AI RMF 1.0)*.
- Ouyang, L., Wu, J., Jiang, X., et al. (2022). Training language models to follow instructions with human feedback. *NeurIPS*, 35, 27730–27744.
- Pearl, J. (2009). *Causality: Models, Reasoning, and Inference* (2nd ed.). Cambridge University Press.
- Pouget, A., Dayan, P., & Zemel, R. (2000). Information processing with population codes. *Nature Reviews Neuroscience*, 1(2), 125–132.
- Powers, W. T. (1973). *Behavior: The Control of Perception*. Aldine.
- Rao, C. R. (1945). Information and accuracy attainable in the estimation of statistical parameters. *Bull. Calcutta Math. Soc.*, 37, 81–91.
- Rao, R. P. N., & Ballard, D. H. (1999). Predictive coding in the visual cortex. *Nature Neuroscience*, 2(1), 79–87.
- Smith, B. (2015). *Basic Formal Ontology (BFO) 2.0 Specification and User's Guide*.
- Stanovich, K. E., & West, R. F. (2000). Individual differences in reasoning. *Behavioral and Brain Sciences*, 23(5), 645–665.
- Treisman, A. M., & Gelade, G. (1980). A feature-integration theory of attention. *Cognitive Psychology*, 12(1), 97–136.
- Tversky, A., & Kahneman, D. (1974). Judgment under uncertainty: heuristics and biases. *Science*, 185(4157), 1124–1131.
- Vaswani, A., Shazeer, N., Parmar, N., et al. (2017). Attention is all you need. *NeurIPS*, 30.
- von der Malsburg, C. (1981). The correlation theory of brain function. *MPI Biophysical Chemistry, Internal Report 81-2*.
- Wiener, N. (1948). *Cybernetics*. MIT Press.

Human Natural Structure

Foundational & Observational Specification

*The Canonical Coordinate System for Structural and Observational
Consistency*

Volume 2 · Human Natural Structure Series

Satoru Hara

Natural Structure Works (NSW)

A Note on Method and Evidence

This volume specifies the Axis-2 coordinate of Human Natural Structure (HNS): the 6×6 base matrix HNS-36 and its observational extension HNS-144. It is a structural specification, not an empirical report, and it keeps three registers distinct throughout.

- **(i) Established results** of the cited literatures — cognitive control, dual-process theory, upper ontologies, causal inference — are reported as such.
- **(ii) The HNS design programme** — the engineering proposal that a fixed Layer×Category coordinate can screen structural failures in language-model output. Most of this book belongs here.
- **(iii) Open conjectures** — that the chosen partition is complete, and that its cells correspond to identifiable brain functions. These are flagged where they occur.

Two honesty commitments carry over from the Origin Trilogy. First, the lineage attributions in this book indicate conceptual inspiration, not strict derivation from any single source ontology. Second, and importantly, the recurring number six is a chosen resolution, not a discovered natural constant: no source ontology has exactly six categories, and the layer stack is a deliberate partition. The framework is more credible for owning this than for implying metaphysical necessity. A specification candid about its evidentiary boundaries is harder to dismiss than one that is not.

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Preface — From a Verification Hypothesis to a Coordinate System

The Origin Trilogy advanced a hypothesis: reliable cognition arises not from a single computational axis but from the geometric intersection of several approximately orthogonal verification processes — the Multi-Axis Verification (MAV) account. Of its three axes, Axis 2 (Structural / Causal) supplies constraints on permissible structure, category, and causal order, independent of statistical frequency. In a deployed language model, Axis 2 is realized as the HNS Cellular Matrix.

That trilogy described the matrix; it did not fully specify it. This volume does. Its task is narrow and foundational: to fix, once and explicitly, the coordinate system on which every later HNS specification depends. HNS-36 is the base coordinate — the Structural layer; HNS-144 is the finer observational resolution — the Observational layer. Together they are the Foundational and Observational specifications of Axis 2.

Why a coordinate system at all? Because description alone cannot say where a proposition sits. The Origin Trilogy and its companion volumes give a rich description of human, cognitive, and social structure, but description provides depth without position. A coordinate system supplies position: it lets an external verifier state precisely which layer a claim originates in and which category explains it, and therefore lets every correction be attributed to an explicit cell. The move is from terrain map to geodetic grid — from knowing the landscape to being able to give coordinates within it. Auditability, the property the programme can most securely demonstrate, is a direct consequence of having such coordinates.

What this book is. A normative specification of HNS-36 and HNS-144: the two axes, their cells, the rules for placing a proposition, and the two failure classes the coordinate is designed to screen — Layer Jump and Category Ambiguity. What this book is not. It is not a method, an evaluation model, or domain guidance, and it makes no claim that the coordinate is the only possible one. It defines a stable reference frame; how that frame is used is the subject of later volumes.

A word on how to read what follows. Part I specifies the foundational coordinate, HNS-36: its two axes, its cells, and the failure it screens. Part II adds the observational resolution, HNS-144. Part III places the result within the larger architecture and states, candidly, what has and has not

been shown. A reader who wants the normative core alone may read Chapters 3 through 5 with Appendix A; the surrounding chapters supply motivation, lineage, and limitation.

Part I
The Foundational Coordinate — HNS-36

1. HNS-36 as the Axis-2 Cellular Matrix

HNS-36 is a 6×6 coordinate system formed by the intersection of six Natural Layers (the causal axis) and six Abstract Categories (the explanatory axis). Each of its thirty-six cells is a structural position — the meeting of one layer and one category — not a bin into which phenomena are sorted by type. This distinction is foundational: HNS-36 is a coordinate system, not a taxonomy. It positions a proposition by causal origin × explanatory purpose; it does not classify it by kind.

To see that this is a coordinate and not a taxonomy, consider a single phenomenon — a company's falling share price. A taxonomy would ask what kind of thing it is and file it once. The matrix instead asks two coordinate questions and admits several legitimate placements depending on interpretive purpose: as a macro-social state ($L6 \times C4$), as an institutional process unfolding ($L6 \times C2$), or, when the analysis turns to the operational cause, as a deliberate operation within the firm ($L4 \times C5$). None of these is the phenomenon's true type; each is a position taken for a purpose. The matrix positions; it does not sort.

The purpose of fixing these positions is verification. A language model emits propositions fluently but without an independent check on whether their structure is coherent. By assigning each proposition a cell, an external verifier can examine the trajectory of a chain of reasoning across the matrix and detect when that trajectory crosses non-adjacent positions without a bridging step — the failure this book calls Layer Jump. HNS-36 is therefore the structural substrate of Axis 2: the minimal grid on which structural incoherence becomes visible and attributable.

1.1 Reading HNS-36

Five principles govern every use of the matrix:

- Layers are causal. They answer where a phenomenon originates.
- Categories are explanatory. They answer how a phenomenon is being interpreted.
- Cells are positions, not contents. A cell defines a coordinate, not a prescribed meaning.
- Mapping is rule-based. Placement follows the procedures of Chapter 5, not intuition.

- The matrix is structurally neutral. No cell is higher or more important than another.

Note. *HNS-36 is the base of a nested family. Chapter 7 extends it to HNS-144 by adding the Onto-Modal Quadrants; Volume 3 extends that to HNS-864 by adding the Inference & Control Operators. The Social Meta Layers (Volume 4) are a parallel grounding axis (Axis 3) and are not part of this multiplicative nesting.*

1.2 Notation and conventions

Three conventions are used throughout. A cell is written $(L_i \times C_j)$ at HNS-36 resolution and $(L_i \times C_j \times Q_k)$ at HNS-144 resolution. A chain of reasoning is represented as a cell trajectory — the ordered sequence of cells assigned to its propositions — which the verifier inspects for non-adjacent transitions. Finally, sections marked normative state requirements; sections marked informative or non-normative supply motivation and examples and impose no requirement. Unless stated otherwise, the definitions and rules of this volume are normative.

2. The Two Axes

HNS-36 rests on two axes that must remain independent. The Natural Layers form the vertical axis (L) and describe causal origin. The Abstract Categories form the horizontal axis (C) and describe explanatory purpose. The cardinal error to avoid is treating categories as causal or layers as interpretive.

The two axes answer different questions. Of any proposition one may ask: “Where does this originate causally?” — a layer question — and “How am I interpreting it?” — a category question. A single phenomenon may legitimately be examined under different categories without changing its causal layer, and the same category applies across every layer. Keeping these roles distinct is the precondition for correct placement.

2.1 Why two independent axes multiply

The reason for two axes rather than one list is the same geometric intuition that motivates MAV. When two constraint axes are independent, the positions they jointly define are their product, and a proposition must satisfy both to be admissible. Independence is the decisive property: if the layer of a claim could be read off from its category, the second axis would add nothing. The specification therefore requires bidirectional independence — the meaning of a cell cannot be derived from either axis alone; it emerges only from their intersection.

The independence of the two axes is what makes their product informative rather than redundant. If layers and categories were correlated — if knowing that a claim is institutional told you it must be read as a process — the second axis would carry no information and the thirty-six positions would collapse toward six. The specification therefore treats orthogonality as a design target, not a discovered fact: the axes are chosen so that, so far as possible, any layer can be examined under any category. Where a particular cell is rare or strained in practice, that is information about the phenomenon, not a defect of the grid.

3. Human Natural Layers (L1–L6)

The Human Natural Layers are the six causal strata of the matrix, ordered from physical substrate to macro-social institution. They are the primary axis (L): every proposition is placed in exactly one layer before any category is applied. The canonical names and their conceptual lineage are as follows.

ID	Name	Lineage (inspiration)	Causal role
L1	Physical / Substrate	Physics; BFO Material Entity (Smith)	The material and biological substrate on which all higher layers depend.
L2	Cellular / Informational	Computational neuroscience	Neural representation: the encoding of signals into informational structure.
L3	Cognitive / System 1	Kahneman (Type 1)	Fast, intuitive processing; the generative, associative mode (maps to MAV Axis 1).
L4	Operational / System 2	Kahneman (Type 2)	Slow, deliberate, rule-governed verification (maps to MAV Axis 2/3).
L5	Micro-Social / Interaction	Habermas	Interpersonal interaction and communicative exchange; basis of SMS-2/3.
L6	Macro-Social / Institutional	Institutional theory; BFO Social Object	Institutions, norms, and macro-social systems; maps to SMS-5/6.

Note. Kahneman’s System 1 and System 2 are parallel dual processes, not an intrinsic vertical hierarchy. HNS re-positions them as processing-depth layers (L3, L4). This is a deliberate re-framing for the purposes of the coordinate, not a claim about Kahneman’s original model.

3.1 Layer structure requirements

Each layer must satisfy five conditions, which together make the axis well-formed:

- Causality — the layer represents a causal domain that produces effects in the layers above it.
- Non-redundancy — no layer duplicates the causal function of another.

- Completeness — the six layers jointly span the causal structure relevant to the framework (a chosen, not a proven, partition; see Chapter 12).
- Vertical ordering — layers are read from foundational (L1) to emergent (L6).
- Single assignment — a proposition is assigned to exactly one layer, by its primary causal origin and not by its effects, context, or scale.

3.2 Boundary conditions and misclassification

Ambiguous cases are resolved by origin, not appearance. A claim whose effects surface at a higher layer is still placed at the layer that produces it; a composite claim is decomposed into causal components, each placed separately. The recurring confusions to guard against are: reading a higher, emergent regularity (L6) as a cause; conflating the informational encoding of L2 with the cognitive processing of L3; and assigning a claim to the social layers (L5–L6) merely because its subject matter is social, when its causal origin is cognitive.

3.3 The six layers in detail

L1 — Physical / Substrate

L1 is the material and biological substrate on which every higher layer supervenes: anatomy, physiology, neural hardware, and the physical constraints of signal propagation and energy. It is causally first, in the precise sense that nothing in the layers above can occur without it. Its lineage is physics together with the BFO category of material entity (Smith), which supplies the formal notion of an independent material continuant.

Boundary and example. L1 is the physical substrate itself, not the information it carries: firing thresholds, metabolic limits, and propagation delays belong to L1, whereas the representation those signals encode belongs to L2. Non-normative example: a claim about a channel saturating above a fixed input rate is an L1 claim.

L2 — Cellular / Informational

L2 is the encoding of physical signals into informational, representational structure — the neural-representation layer. Its lineage is computational neuroscience, on the view that populations of cells carry information in their joint activity. Causally, L2 mediates between the physical substrate

below and cognition above: it is where matter becomes message.

Boundary and example. L2 is representation, not yet System-1 cognition. A population code, a place-cell map, or an embedding is L2; the fast inference that consumes them is L3. Example: a claim about how a quantity is encoded across a representational space is an L2 claim.

L3 — Cognitive / System 1

L3 is fast, automatic, associative processing: pattern completion, heuristic inference, the immediate impression. It is the generative mode and maps to MAV Axis 1. Its lineage is the Type-1 side of dual-process theory (Kahneman; Stanovich & West; Evans), re-positioned here as a processing-depth layer rather than a free-standing system.

Boundary and example. L3 is intuitive generation; L4 is deliberate checking. A snap judgement that a face is familiar is L3; verifying the identity against a record is L4. Example: a claim that an answer feels right is an L3 claim.

L4 — Operational / System 2

L4 is slow, deliberate, rule-governed processing: verification, planning, sequential reasoning. It maps to MAV Axis 2/3, the verification function. Its lineage is the Type-2 side of dual-process theory together with the integrative theory of prefrontal cognitive control (Miller & Cohen), in which the prefrontal cortex maintains goals and enforces rules against automatic responses.

Boundary and example. L4 differs from L3 as effortful from automatic. Multi-step arithmetic, or consciously applying a policy to a case, is L4. Example: a claim that a result was checked against an explicit rule is an L4 claim.

L5 — Micro-Social / Interaction

L5 is interpersonal interaction and communicative exchange: coordination, negotiation, the raising and redeeming of validity claims between agents. Its lineage is Habermas's theory of communicative action, in which meaning is secured through dialogue oriented to mutual understanding. L5 is the basis of the lower Social Meta Layers (SMS-2/3).

Boundary and example. L5 is event-level interaction between agents; L6 is the institutional regularity that governs such events. A specific negotiation is L5; the contract law that frames negotiations is L6.

Example: a claim about what two parties agreed in a meeting is an L5 claim.

L6 — Macro-Social / Institutional

L6 is institutions, norms, and macro-social systems together with their emergent regularities. Its lineage is institutional theory and the BFO category of social object (Smith), which gives the formal status of entities that exist by collective recognition. L6 maps to the upper Social Meta Layers (SMS-5/6).

Boundary and example. L6 is the macro or institutional structure, not a single interaction. Market price formation and regulatory regimes are L6; one transaction within them is L5. Example: a claim about how a market sets prices is an L6 claim.

4. Abstract Categories (C1–C6)

The Abstract Categories are the six non-causal explanatory axes of the matrix. Where the layers say where a proposition originates, the categories say how it is being interpreted. They form the horizontal axis (C); each applies across all six layers.

ID	Name	Lineage (inspiration)	Explanatory role
C1	Continuant / Entity	Aristotle; BFO	An object grasped as a persisting thing.
C2	Occurrent / Process	Aristotle; BFO	A temporally unfolding, event-like aspect.
C3	Relational / Boundary	Aristotle; Kant	Relations and boundaries between objects; aligns with the HNS orthogonality concept.
C4	Quality / State	Kant	A property or state at a given moment; the state-space view.
C5	Transformation / Operation	Pearl (do-operator)	A transition or operation between states; the core of causal transition.
C6	Meta-Level / Schema	Kant (categories of categories)	Reference to an object from a higher schematic level; aligns with the HNS runtime / EVA.

Note. Under strict BFO, Continuant (C1) and Occurrent (C2) are the top-level dichotomy, Quality (C4) is a dependent continuant, and Process overlaps with C5. The six categories here are therefore a synthetic set inspired by Aristotle, BFO, and Kant — not a flat partition derived from any single ontology. That six categories should tile representable structure descends philosophically from Aristotle’s Categories and Kant’s table of the categories, and computationally from upper ontologies such as DOLCE (Gangemi et al.), BFO (Smith), and SUMO (Niles & Pease) and from frame semantics (Fillmore).

4.1 Category structure requirements

- Non-causality — categories provide explanatory perspectives, never causal domains.
- Cross-layer applicability — each category applies to all six layers.

- Non-redundancy — no category duplicates another’s explanatory function.
- Completeness — the six jointly span the explanatory dimensions the framework requires (again, a chosen resolution).
- Single assignment — a proposition is interpreted through exactly one category at a time, by primary explanatory purpose and not by domain, scale, or importance.

4.2 The six categories in detail

C1 — Continuant / Entity

C1 interprets a phenomenon as a persisting thing — an object grasped as enduring through time. Its lineage is Aristotle’s notion of substance and the BFO continuant. Example: treating the institution as a standing entity is a C1 interpretation; its reform, as an unfolding event, would be C2.

C2 — Occurrent / Process

C2 interprets a phenomenon as a temporally unfolding event or process. Its lineage is Aristotle and the BFO occurrent. The boundary with C5 matters: a process (C2) is an unfolding occurrent, whereas a transformation or operation (C5) is the rule that maps one state to another. Example: a negotiation unfolding over a week is C2.

C3 — Relational / Boundary

C3 interprets a phenomenon through the relations and boundaries between objects, rather than the objects themselves. Its lineage is Aristotle’s category of relation and Kant; it aligns with the HNS notion of orthogonality. Example: the dependency between two departments, or the boundary separating two domains, is C3.

C4 — Quality / State

C4 interprets a phenomenon as a property or state at a given moment — the state-space view. Its lineage is Kant’s category of quality. The boundary with C2 is the snapshot/unfolding distinction: a state (C4) is a momentary configuration, a process (C2) is its change. Example: the system’s current load is C4; the load rising is C2.

C5 — Transformation / Operation

C5 interprets a phenomenon as a state transition or operation — the causal-intervention view. Its lineage is Pearl’s do-operator, which

distinguishes acting on a system from merely observing it. Example: applying a policy changes the outcome distribution — a C5 interpretation, an intervention rather than an unfolding.

C6 — Meta-Level / Schema

C6 interprets a phenomenon from a higher, schematic level — a category of categories, or a reference to the framework itself. Its lineage is Kant's schematism; it aligns with the HNS runtime and the External Verification Architecture. Example: the rule that governs which category applies to a claim is itself a C6 object.

5. The 36-Cell Specification

A cell is the intersection of one layer and one category, written $(L_i \times C_j)$. This chapter specifies the internal format of a cell and the rules that govern the matrix as a whole. It defines the form of a cell, not its content.

5.1 Cell format

Every cell is documented with the same ordered components: a cell identifier ($L_i \times C_j$); a layer reference (Chapter 3); a category reference (Chapter 4); a structural-role statement describing the function of the intersection; an interpretive scope giving the range of propositions it admits; boundary conditions distinguishing it from adjacent cells; misclassification-prevention guidance; and optional non-normative examples. This ordering is mandatory across all thirty-six cells.

5.2 Structural and matrix constraints

- Uniqueness — each cell is a unique structural position.
- Non-overlap — no two cells share the same structural domain.
- Bidirectional independence — a cell's meaning cannot be derived from its layer or category alone.
- Interpretive neutrality — cells define positions, not prescribed content.
- Exhaustiveness and mutual exclusivity — all layer $\times$ category intersections are represented, and a single decomposed proposition occupies one cell at a time.

5.3 Adjacency and interpretation

Cells are adjacent in two ways only: same layer, neighbouring category (horizontal), or same category, neighbouring layer (vertical). Adjacency does not imply similarity; boundary conditions distinguish neighbours. Interpretation proceeds layer-first (the causal domain fixes the primary meaning), then category-second (the explanatory purpose refines it). Two interpretive prohibitions hold: no cross-cell inference — a cell's meaning may not be read off from another — and no diagonal interpretation — a simultaneous step in both layer and category has no defined structural meaning, and is precisely the signature of a Layer Jump (Chapter 6).

5.4 Mapping rules

To place a proposition: (1) identify the unit of analysis; (2) determine its primary causal domain and assign one layer; (3) determine its primary explanatory purpose and assign one category; (4) combine them into ($L_i \times C_j$); (5) verify the boundary conditions of Chapters 3 and 4; (6) record the placement in the notation of the template (Appendix B). Composite phenomena are decomposed before mapping; once mapped, a placement is stable unless the explanatory purpose changes.

5.5 Representative cell entries

The following non-normative entries illustrate the template in use. They document four representative cells; the full set of thirty-six follows the same form (Appendix B).

Cell ($L4 \times C5$) — Operational $\times$ Transformation

- Structural Role: positions claims about deliberate operations that transform a state under operational control.
- Interpretive Scope: planned interventions and applied procedures whose effect is the object of analysis.
- Boundary Conditions: distinguished from ($L4 \times C2$), the unfolding of a process, and from ($L3 \times C5$), an automatic rather than deliberate transformation.
- Misclassification Prevention: do not place a merely associative, automatic transition here; that belongs to L3.
- *Example (non-normative)*: running the verification routine attenuates ungrounded output.

Cell ($L6 \times C1$) — Macro-Social $\times$ Continuant

- Structural Role: positions claims about institutions grasped as persisting entities.
- Interpretive Scope: organizations, legal entities, and markets considered as standing objects.
- Boundary Conditions: distinguished from ($L6 \times C2$), an institution undergoing reform, and from ($L5 \times C1$), a single relationship treated as an entity.

- Misclassification Prevention: do not assign here on grounds of scale alone; macro subject matter is not automatically macro-social in causal origin.
- *Example (non-normative)*: the regulator is the body responsible for enforcement.

Cell (L3 × C4) — Cognitive (System 1) × Quality / State

- Structural Role: positions claims about momentary intuitive states.
- Interpretive Scope: a felt impression or a fast confidence estimate at an instant.
- Boundary Conditions: distinguished from (L4 × C4), a deliberate assessment, and from (L3 × C2), the unfolding of an intuition over time.
- Misclassification Prevention: do not treat the report of a feeling as an objective measurement; that crosses into the objective quadrants at HNS-144.
- *Example (non-normative)*: the answer feels right.

Cell (L2 × C2) — Cellular / Informational × Occurrent

- Structural Role: positions claims about informational processes unfolding within neural representation.
- Interpretive Scope: encoding dynamics and the propagation of activation across a representational space.
- Boundary Conditions: distinguished from (L1 × C2), a physical process, and from (L2 × C5), an operation performed on the representation.
- Misclassification Prevention: do not elevate a representational process to cognition (L3); encoding is not yet inference.
- *Example (non-normative)*: activation spreads across the embedding as context accumulates.

Cell (L1 × C3) — Physical × Relational / Boundary

- Structural Role: positions claims about physical couplings and boundaries between material entities.
- Interpretive Scope: interfaces, contact, and physical dependency between substrates.

- Boundary Conditions: distinguished from ($L1 \times C1$), the entities themselves, and from ($L3 \times C3$), an associative cognitive link.
- Misclassification Prevention: a conceptual or metaphorical relation is not a physical one; keeping them apart guards against Metaphor Contamination.
- *Example (non-normative)*: the two components are thermally coupled.

Cell ($L5 \times C5$) — Micro-Social $\times$ Transformation / Operation

- Structural Role: positions claims about communicative acts that change an interpersonal state.
- Interpretive Scope: speech acts and moves in a negotiation that transform commitments.
- Boundary Conditions: distinguished from ($L5 \times C2$), the dialogue unfolding, and from ($L4 \times C5$), a non-social operation.
- Misclassification Prevention: do not treat a unilateral cognitive operation as a communicative act.
- *Example (non-normative)*: the offer, once accepted, binds both parties.

Cell ($L4 \times C6$) — Operational $\times$ Meta-Level / Schema

- Structural Role: positions claims about explicit rules or meta-policies that govern operations.
- Interpretive Scope: policies, verification rules, and the schema by which operations are selected.
- Boundary Conditions: distinguished from ($L4 \times C5$), the operation itself, and from ($L6 \times C6$), an institutional norm.
- Misclassification Prevention: a meta-rule is not the same as the act it governs.
- *Example (non-normative)*: the policy specifies which checks run before output.

Cell ($L6 \times C2$) — Macro-Social $\times$ Occurrent / Process

- Structural Role: positions claims about institutional processes and macro trends unfolding over time.
- Interpretive Scope: reforms, market movements, and long-run institutional change.

- Boundary Conditions: distinguished from ($L6 \times C1$), the institution as entity, and from ($L5 \times C2$), a single interaction unfolding.
- Misclassification Prevention: do not read an emergent macro trend as an individual cause — a frequent source of Layer Jump.
- *Example (non-normative)*: the market has been consolidating over the decade.

Cell ($L1 \times C1$) — Physical $\times$ Continuant / Entity

- Structural Role: positions claims about material bodies grasped as persisting objects.
- Interpretive Scope: organs, devices, and physical substrates treated as things.
- Boundary Conditions: distinguished from ($L1 \times C2$), a physical process, and from ($L2 \times C1$), a representational unit.
- Misclassification Prevention: do not confuse the material body with the information it carries.
- *Example (non-normative)*: the server is the physical host of the service.

Cell ($L2 \times C5$) — Cellular / Informational $\times$ Transformation

- Structural Role: positions claims about operations that transform a representation.
- Interpretive Scope: re-encoding, projection, or compression of a code.
- Boundary Conditions: distinguished from ($L2 \times C2$), the encoding process unfolding, and from ($L4 \times C5$), a deliberate operation.
- Misclassification Prevention: an automatic re-encoding is not a deliberate operational intervention.
- *Example (non-normative)*: the vector is projected into a lower-dimensional space.

Cell ($L3 \times C2$) — Cognitive (System 1) $\times$ Occurrent / Process

- Structural Role: positions claims about automatic inference unfolding over time.
- Interpretive Scope: a train of associations, a spreading activation of intuition.

- Boundary Conditions: distinguished from (L3 × C4), a momentary felt state, and from (L4 × C2), controlled reasoning.
- Misclassification Prevention: do not treat an intuitive cascade as deliberate reasoning.
- *Example (non-normative)*: one association triggered the next without reflection.

Cell (L4 × C4) — Operational × Quality / State

- Structural Role: positions claims about a deliberately assessed state.
- Interpretive Scope: a measured or judged condition arrived at by reasoning.
- Boundary Conditions: distinguished from (L3 × C4), a fast felt state, and from (L4 × C2), the assessment process.
- Misclassification Prevention: do not record an intuitive impression as a deliberate assessment.
- *Example (non-normative)*: after analysis, the project is judged behind schedule.

Cell (L5 × C3) — Micro-Social × Relational / Boundary

- Structural Role: positions claims about interpersonal relations and their boundaries.
- Interpretive Scope: trust, authority, and role relations between agents.
- Boundary Conditions: distinguished from (L5 × C1), a party as an entity, and from (L6 × C3), an inter-institutional relation.
- Misclassification Prevention: a personal relation is not an institutional one.
- *Example (non-normative)*: the two teams share a reporting line.

Cell (L6 × C6) — Macro-Social × Meta-Level / Schema

- Structural Role: positions claims about norms and institutional schemas that govern macro behaviour.
- Interpretive Scope: legal principles, standards, and the constitutive rules of institutions.
- Boundary Conditions: distinguished from (L6 × C5), a specific regulation, and from (L4 × C6), an individual meta-policy.

- Misclassification Prevention: a society-wide norm is not a single actor's rule.
- *Example (non-normative)*: the principle of equal treatment governs the institution.

5.6 The thirty-six cells at a glance

The grid below gives a concise structural tag for every cell. It is a map, not a substitute for the full entries above. Rows are Natural Layers (L1–L6); columns are Abstract Categories (C1–C6).

	C1 Entity	C2 Process	C3 Relation	C4 State	C5 Operation	C6 Schema
L1	object	process	coupling	state	transfer	law schema
L2	code unit	encoding	code map	code state	recoding	code schema
L3	percept	intuition	association	felt state	heuristic	frame
L4	plan step	reasoning	rule link	judgement	operation	policy
L5	party	dialogue	role relation	rapport	speech act	protocol
L6	institution	macro trend	macro link	macro state	regulation	norm

6. Layer Jump

Layer Jump is the failure HNS-36 is built to detect: the traversal of non-adjacent representational layers without an explicit bridging proposition. In the matrix it appears as a cell trajectory that crosses non-adjacent rows with no transitional step — the structural signature of an argument that has skipped the causal chain connecting its premises to its conclusion. In clinical terms it is the analogue of formal thought disorder; in language models it is the confident leap from one level of explanation to a distant one.

HNS-36 screens for it geometrically. Each proposition in a chain receives a cell; the verifier inspects successive cells for non-adjacent transitions. A jump from a macro-social cell (L6) to a cognitive cell (L3) with no intervening operational or interactional step is flagged, and the offending step is attenuated rather than emitted. The check is structural, not statistical: it does not ask whether the leap is improbable, only whether it is bridged.

Worked example (non-normative)

Prompt: *“Why did the company’s stock fall?”*

Failing output: *“The stock fell because the employees were unmotivated.”*

Diagnosis: a jump from the macro-social/price layer to the individual-cognitive layer with no transitional proposition — the cell trajectory crosses non-adjacent rows.

Corrected: *“The stock fell on weak quarterly earnings (price layer). Earnings were weak partly because output declined (operational layer); internal surveys link lower output to reduced motivation (individual layer). The bridge runs motivation → productivity → earnings → valuation.”*

A second case. Prompt: *“Why did the patient recover?”* Failing output: *“Because they had a positive attitude.”* This jumps from an individual-cognitive claim to a physiological outcome with no bridging mechanism. Corrected: *“Recovery followed the prescribed course of treatment (physiological); adherence to it was high (operational); a positive attitude may have supported adherence (individual). The supported bridge runs attitude → adherence → treatment effect → recovery.”*

Formally, the check is on the trajectory. Each proposition receives a cell;

the verifier reads the sequence and marks any step whose layers are non-adjacent and unbridged. The test is local and mechanical — it asks only whether consecutive cells are neighbours, or are joined by an explicit transitional proposition — which is what makes a Layer Jump detectable without recourse to the plausibility of the content.

Part II
The Observational Extension — HNS-144

7. HNS-144: The Observational Extension

HNS-144 raises the resolution of the matrix. Each of the thirty-six cells is subdivided by the Onto-Modal Quadrants — four quadrants formed by the Cartesian product of two binary dimensions — giving $36 \times 4 = 144$ cells. The two dimensions are the Ontological Domain (Subjective / Objective) and the Modal Invariants (Necessary / Contingent).

Dimension	Binary values	Lineage	Role
① Ontological Domain	Subjective / Objective	Kant; Habermas	Separates how something is held from how something is.
② Modal Invariants	Necessary / Contingent	Kant (modality)	Distinguishes a rule from a context-dependent fact.

Their product yields the four quadrants:

Quadrant	Definition
Q1	Subjective $\times$ Contingent
Q2	Subjective $\times$ Necessary
Q3	Objective $\times$ Contingent
Q4	Objective $\times$ Necessary

***Note.** The resolution is two dimensions, not four. Two binary dimensions give $2 \times 2 = 4$ quadrants, hence $36 \times 4 = 144$. Four binary dimensions would give $2^4 = 16$ sub-states and $36 \times 16 = 576$, which is not the specified cardinality. Earlier informal descriptions that listed four “logical dimensions” conflated the number of axes with the number of resulting quadrants; the corrected and canonical reading is two dimensions yielding four quadrants.*

7.1 The four quadrants in detail

Q1 — Subjective $\times$ Contingent

Q1 holds claims about how something is held rather than how it is, and that could have been otherwise: preferences, impressions, reports of feeling. Example: the team found the meeting useful.

Q2 — Subjective $\times$ Necessary

Q2 holds subjective claims asserted as binding within a frame — commitments, declared values, the constitutive rules of a practice. Example: in our charter, a quorum is required to vote.

Q3 — Objective × Contingent

Q3 holds matters of fact that could have been otherwise: measurements, observations, empirical outcomes. Most empirical claims belong here. Example: the model's accuracy was 0.91 on this test set.

Q4 — Objective × Necessary

Q4 holds objective claims that hold of necessity — definitions, mathematical truths, logical entailments. Example: in Euclidean geometry the interior angles of a triangle sum to 180 degrees.

Most Category Ambiguity failures are quadrant errors: a Q1 preference dressed as a Q4 necessity, or a Q3 contingency asserted as Q4. The observational layer's task is to hold each claim to a single declared quadrant, or to require its decomposition into several.

8. Category Ambiguity

Category Ambiguity is the failure HNS-144 is built to observe: the conflation of distinct categories — most commonly the collapse of the subjective/objective distinction and of the fact/value (normative) distinction into a single undifferentiated claim. Where Layer Jump is a fault of causal traversal, Category Ambiguity is a fault of conceptual boundary: a sentence that treats a preference, an outcome, and a value judgement as though they were one kind of thing.

HNS-144 observes it by enforcing concept-boundary conditions at quadrant resolution. A proposition that ought to occupy the subjective-contingent quadrant but is asserted as if objective-necessary is flagged for boundary violation; the verifier requires the claim to be re-stated within a single, declared quadrant or explicitly decomposed into several.

Worked example (non-normative)

Prompt: *“Is the new policy good?”*

Failing output: *“Yes, it’s good because everyone likes it.”*

Diagnosis: a subjective preference (‘is liked’), an objective outcome (‘achieves its goal’), and a normative judgement (‘is good’) are collapsed — the subjective/objective distinction is crossed and a contingent preference is presented as a necessary conclusion.

Corrected: “Three distinct questions. (1) Is it liked? Survey data say yes — subjective, contingent. (2) Does it achieve its stated goal? Early metrics are mixed — objective, contingent. (3) Is it good, all things considered? A normative judgement depending on which values are weighted. Popularity bears on (1), not directly on (3).”

A second case. Prompt: “Is this design better?” Failing output: “Yes, it’s more modern, so it’s better.” Here an aesthetic-subjective property is asserted as an objective-necessary superiority, crossing both the subjective/objective and the contingent/necessary boundaries. Corrected: “It is more recent in style (objective, contingent). Whether ‘modern’ is preferred is a subjective judgement (subjective, contingent). Whether it is better depends on a stated criterion — usability, cost, accessibility — each measurable separately.”

9. Mapping and Usage at 144 Resolution

At full resolution a proposition is placed at $(L_i \times C_j \times Q_k)$: a layer, a category, and a quadrant. The mapping procedure of Chapter 5 extends directly — after layer and category are fixed, the verifier determines the ontological domain and modal status and selects the quadrant. The same constraints apply: single assignment per decomposed unit, stability of placement, and decomposition of composite claims before mapping.

The 144 resolution is observational in the precise sense that it is where category conflation becomes visible that the 36 resolution cannot see. Two propositions may share a cell at HNS-36 yet differ in quadrant — one a contingent observation, the other asserted as a necessary rule — and it is exactly this difference that the observational layer is designed to surface. HNS-144 does not introduce new layers or categories; it refines the existing thirty-six positions to the resolution at which boundary violations can be detected and attributed.

Worked mapping at 144 resolution (non-normative)

Take the claim “interest rates must rise because inflation is high.” The layer is macro-social and institutional — a claim about monetary policy (L6). The category is a transformation: a policy operation said to follow from a condition (C5). The quadrant is fixed by two words: “must” asserts necessity, and the claim is framed as objective, placing it at Q4 (Objective $\times$ Necessary). The full coordinate is therefore $(L6 \times C5 \times Q4)$.

The observational layer now does its work. It asks whether the asserted necessity is warranted. Rates often rise with inflation, but not invariably; the relationship is in fact contingent. The claim has thus been placed at Q4 when it belongs at Q3 (Objective $\times$ Contingent), and the quadrant mismatch is flagged as a modal overreach — a necessity claimed where only a tendency holds. The correction restates it as contingent. This is precisely the conflation HNS-144 is built to observe and that HNS-36 alone, lacking the modal dimension, cannot see.

Composite claims

A real sentence often carries several propositions at once, and these may fall in different cells and quadrants. The rule is to decompose before mapping: split the sentence into its atomic claims, place each, and only then assess the trajectory across them. “The popular reform will obviously raise growth” decomposes into a subjective-contingent claim

about popularity (Q1), an objective-contingent prediction about growth (Q3), and a modal overreach in the word “obviously.” Mapping the parts separately is what lets the verifier accept the defensible components while flagging the overreach, rather than passing or rejecting the sentence whole.

Part III

Integration and Status

10. Relationship to the HNS Architecture

HNS-36 and HNS-144 are the lower two resolutions of Axis 2, the Structural / Causal axis of MAV. They sit within a larger architecture, and their boundaries with the neighbouring specifications should be stated plainly.

10.1 The nested coordinate (Axis 2)

Specification	Composition	Cells
HNS-36	Natural Layers (6) × Abstract Categories (6)	36
HNS-144	HNS-36 × Onto-Modal Quadrants (4)	144
HNS-864	HNS-144 × Inference & Control Operators (6)	864

HNS-864 (the subject of Volume 3) adds the six Inference & Control Operators — 01–03 from Pearl’s hierarchy of association, intervention, and counterfactual, and 04–06 from Powers’ perceptual control theory plus an HNS output regulator — to reach 864 atomic-proposition cells. HNS-36/144 supplies the coordinate on which those operators act.

10.2 The orthogonal grounding axis (Axis 3)

The Social Meta Layers (HNS-SMS-6, Volume 4) constitute Axis 3, the Grounded / Executive axis. They are orthogonal to the nesting above: they ground and regulate output against context, from the somatic base to the universal, but they do not subdivide the coordinate and are not a multiplicative factor. The complete normative reference for all of these axes is reproduced in Appendix A.

10.3 The runtime

In deployment, the checks specified here are applied by a pre-decode interception layer that assigns cells to candidate propositions and attenuates those whose trajectory violates the adjacency and boundary conditions. The runtime is described in the Origin Trilogy and specified at operator resolution in Volume 3; for HNS-36/144 it suffices that each correction is attributable to an explicit cell, which is the source of the framework’s auditability.

10.4 Relationship to the three MAV axes

HNS-36/144 specifies Axis 2, but it helps to state how the three Multi-

Axis Verification axes divide the labour. Axis 1 (Generative / Statistical) is the model's native fluency — the System-1 layer (L3) that proposes continuations. Axis 2 (Structural / Causal), specified here, supplies the layer-and-category constraints a proposal must satisfy, independent of how probable it is. Axis 3 (Grounded / Executive), the Social Meta Layers of Volume 4, checks the proposal against context from the body to the institution. Reliability, on the MAV account, is the conjunction: a claim must be fluent, structurally coherent, and grounded. HNS-36/144 owns the middle condition and is silent on the other two by design.

This division also says what the coordinate does not do. It does not judge whether a claim is true — an Axis-3 grounding question against the world — nor whether it is probable, an Axis-1 question. It judges whether the claim's structure is coherent: whether its layers are bridged and its categories kept distinct. A claim can be perfectly structured and still false; HNS-36/144 will pass it, and Axis 3 must catch it. Keeping these responsibilities separate is what makes each axis auditable.

11. Intellectual Lineage and Proponents

This chapter consolidates the lineage of HNS-36/144 and names the proponents of the ideas it draws upon. As stated at the outset, these are inspirations, not derivations; HNS recombines them into a coordinate for external verification.

11.1 The layers

The causal-control reading of the upper layers descends from the integrative theory of prefrontal cortex (Miller & Cohen) and the hippocampal account of episodic and schematic binding (Eichenbaum). The L3/L4 split realizes the dual-process distinction (Stanovich & West; Evans & Stanovich; Kahneman), re-positioned as processing depth. The micro-social layer (L5) draws on Habermas's theory of communicative action; the macro-social layer (L6) on institutional theory and on the BFO category of social object (Smith). The physical base (L1) is anchored in physics and in BFO's material entity.

11.2 The categories

The six categories descend from Aristotle's *Categories* and Kant's table of the categories of understanding, and computationally from the upper ontologies — DOLCE's *endurant/perdurant/quality/abstract* distinctions (Gangemi et al.), BFO's *continuant/occurrent* split (Smith), and SUMO (Niles & Pease) — and from frame semantics (Fillmore). The transformation category (C5) takes its admissibility standard from Pearl's account of causal intervention; the binding of features into coherent objects without illusory conjunction is the binding problem (Treisman & Gelade; von der Malsburg).

11.3 The quadrants

The Ontological Domain and Modal Invariants dimensions are Kantian in origin — the subjective/objective distinction and the category of modality (necessity/contingency) — with the communicative reading of the subjective/objective boundary drawing on Habermas.

The proponents named here did not design HNS, and several would frame their work differently. Pearl's hierarchy is a theory of causal inference, not of language-model verification; Kahneman's dual-process account is descriptive psychology, not a layered architecture; BFO is an ontology for

scientific-data integration. HNS borrows their distinctions and recombines them for an engineering purpose. The borrowing is acknowledged precisely so that the additions stay visible: the coordinate idea, the output-attenuation operator, and the use of the matrix as an external verifier are HNS's own.

The deeper debt is to the tradition of upper ontologies itself — the conviction, running from Aristotle through Kant to BFO, DOLCE, and SUMO, that what can be said admits a small, principled set of top-level distinctions. HNS-36 is a wager that such a set, fixed in advance and made machine-checkable, is useful for constraining generated text. Whether the wager pays is the empirical question of the next chapter.

11.4 Lineage at a glance

HNS element	Drawn from	Proponent(s)
Layers L1-L2	Physics; computational neuroscience; upper ontology	Smith (BFO)
Layers L3-L4	Dual-process theory; prefrontal control	Kahneman; Stanovich & West; Miller & Cohen
Layers L5-L6	Communicative action; institutional theory	Habermas; Smith (BFO)
Categories	Categories; upper ontology; frames; causal inference	Aristotle; Kant; Gangemi; Smith; Fillmore; Pearl
Quadrants	Subjective/objective; modality	Kant; Habermas
Coordinate idea	Geodetic framing of description	HNS (this volume)

12. Empirical Status, Limitations, Threats to Validity

A specification is only as strong as its honesty about what it has established. HNS-36/144 makes a small number of secure claims and a larger number of proposals and conjectures, and they should not be confused.

12.1 What is established

- Well-definedness. The coordinate is fully specified: two axes, thirty-six cells, four quadrants, and the rules for placement. Given the rules, placements are reproducible.
- Auditability by construction. Because every correction is attributed to an explicit cell, the matrix yields a machine-checkable record of how an output was constrained. This is true by construction, independent of whether the partition is the right one.

12.2 What is proposed

- That the layer×category coordinate captures the structure relevant to verification, and that Layer Jump and Category Ambiguity are the failure classes it most naturally screens. These are engineering proposals, to be tested against external criteria rather than framework-internal ones.

12.3 What is conjecture

- Completeness of the partition — that six layers and six categories tile the representable structure — is a hypothesis, to be tested by peer review and adversarial probing as upper ontologies are tested.
- The correspondence between cells and identifiable brain functions is a structured analogy, not a proven mapping.

12.4 On the number six

The recurrence of six — six layers, six categories, and six operators in the higher specification — is a chosen resolution, not a natural constant. No source ontology has exactly six categories; Aristotle lists ten and Kant twelve, while BFO's fundamental division is twofold. The layer stack likewise reflects a deliberate partition in which dual-process and

micro/macro distinctions are each resolved into two. The specification is stronger for stating this openly: six is adopted for coverage, memorability, and architectural consistency, and the framework does not claim that exactly six is metaphysically necessary. Where a reviewer presses on why six, the honest answer is that it is the resolution the programme has selected, and that the partition is open to revision under the standard of proof set out above.

12.5 Threats to validity

Two threats deserve emphasis. First, framework-internal scoring: a coordinate designed to detect Layer Jump will, unsurprisingly, detect Layer Jump, so any reported suppression is weaker than a reduction measured against an external criterion such as factual accuracy. Second, the partition mixes organizing principles — the layers blend a scale ordering with a processing-depth distinction — and a stricter formulation might separate these. Both are reasons to treat HNS-36/144 as a useful, auditable scaffold whose completeness remains an open question, not as a finished metaphysics.

A further limitation deserves naming. Because a proposition's placement depends on an interpreter's reading of its primary causal origin and explanatory purpose, two careful analysts may occasionally disagree on a cell. The specification reduces but does not eliminate this: the mapping rules, boundary conditions, and adjacency constraints narrow the choices, and a disagreement itself is informative, localizing exactly where an argument is ambiguous. Inter-rater reliability of cell assignment is therefore a concrete, measurable target for validation — and a more honest figure of merit than any framework-internal suppression count.

Two boundaries also limit scope. The line between Axis 2 and Axis 3 is not always sharp: whether a claim's defect is structural — a category crossed — or a matter of grounding — a fact wrong about the world — can itself be a judgement, and the coordinate can flag only the former. And the partition has been developed chiefly against English-language, general-domain reasoning; whether the same six layers and six categories transfer cleanly across languages and specialist domains is untested. Both are reasons to treat the present specification as a stable but revisable v1.0, not a closed result.

Appendix A (Normative) — HNS Unified Structural Coordinate Table v2.0

The following is the canonical reference for every HNS structural primitive. The lineage columns indicate conceptual inspiration, not strict derivation from a single ontology.

A.1 Natural Layers (HNS-36)

ID	Name	Lineage (inspiration)
L1	Physical / Substrate	Physics / BFO Material Entity
L2	Cellular / Informational	Computational neuroscience
L3	Cognitive / System 1	Kahneman Type 1
L4	Operational / System 2	Kahneman Type 2
L5	Micro-Social / Interaction	Habermas
L6	Macro-Social / Institutional	Institutional theory / BFO Social Object

A.2 Abstract Categories (HNS-36)

ID	Name	Lineage (inspiration)
C1	Continuant / Entity	Aristotle / BFO
C2	Occurrent / Process	Aristotle / BFO
C3	Relational / Boundary	Aristotle / Kant
C4	Quality / State	Kant
C5	Transformation / Operation	Pearl (do-operator)
C6	Meta-Level / Schema	Kant (category of categories)

A.3 Onto-Modal Quadrants (HNS-144)

Quadrant	Definition
Q1	Subjective × Contingent
Q2	Subjective × Necessary
Q3	Objective × Contingent
Q4	Objective × Necessary

A.4 Inference & Control Operators (HNS-864)

ID	Name	Proposer
01	Conditioning	Statistics / Pearl
02	Intervention / do-operator	Pearl
03	Counterfactual	Pearl
04	Reference Signal	Powers (PCT)
05	Comparator / Error Detection	Powers (PCT)
06	Output Attenuation	HNS Engineering

A.5 Social Meta Layers (HNS-SMS-6) — parallel grounding axis (Axis 3)

ID	Domain	Lineage (inspiration)
SMS-1	Somatic	Barsalou / Harnad
SMS-2	Protocol	Habermas
SMS-3	Organizational	HNS Extension
SMS-4	Economic	HNS Extension
SMS-5	Governance	Institutional theory
SMS-6	Universal	Ashby / General Systems Theory

Note. Composition: $HNS-36 = 6 \times 6$; $HNS-144 = 36 \times 4$; $HNS-864 = 144 \times 6$.
 The Social Meta Layers are orthogonal to this nesting and are not multiplied into it.

Appendix B — The 36-Cell Template (Formal)

Each of the thirty-six cells is documented with the following template. The template defines structure, not content.

Cell ($L_i \times C_j$)

- Layer: name of the Human Natural Layer (L_i), per Chapter 3.
- Category: name of the Abstract Category (C_j), per Chapter 4.
- 1. Structural Role — the function of this layer×category intersection.
- 2. Interpretive Scope — the range of propositions the cell admits.
- 3. Boundary Conditions — rules distinguishing the cell from its horizontal and vertical neighbours.
- 4. Misclassification Prevention — the common errors to avoid.
- 5. Non-normative Examples (optional) — illustrations of the cell's structural meaning.

Appendix C — Worked Examples

C.1 Layer Jump (HNS-36)

Prompt: *“Why are sales up this quarter?”*

Failing output: *“Sales are up because we changed the logo.”*

Diagnosis: a leap from a macro-social/market outcome to a single surface change, skipping the operational layer that would connect them.

Corrected: “Sales rose 12%. The spring promotion was A/B-tested and measured a +8% uplift (operational); seasonal demand follows the historical pattern (market); the rebrand was coincident with no measured effect. Only the promotion has a supported bridge to the outcome.”

C.2 Category Ambiguity (HNS-144)

Prompt: *“Was the meeting productive?”*

Failing output: *“Yes — it felt productive and that’s what matters.”*

Diagnosis: a subjective-contingent impression (‘felt productive’) is asserted as an objective-necessary conclusion (‘that’s what matters’); two quadrants are collapsed.

Corrected: “Subjectively, participants reported it felt productive (subjective, contingent). Objectively, two of three agenda items were decided and one was deferred (objective, contingent). Whether that counts as productive depends on the standard applied.”

Glossary

HNS-36 — The base coordinate: a 6×6 matrix of Natural Layers $\times$ Abstract Categories; the Structural OS.

HNS-144 — The observational extension: HNS-36 $\times$ the four Onto-Modal Quadrants; the Observational OS.

Natural Layers (L1-L6) — The six causal strata forming the vertical axis; from physical substrate to macro-social institution.

Abstract Categories (C1-C6) — The six non-causal explanatory axes forming the horizontal axis.

Onto-Modal Quadrants (Q1-Q4) — Two binary dimensions — Subjective/Objective and Necessary/Contingent — whose product yields four quadrants.

Cell ($L_i \times C_j$) — A structural position: the intersection of one layer and one category.

Layer Jump — Traversal across non-adjacent layers without a bridging proposition; screened by HNS-36.

Category Ambiguity — Conflation of distinct categories or quadrants; observed by HNS-144.

Bidirectional independence — The property that a cell's meaning cannot be derived from its layer or category alone.

Auditability — The property that every correction is attributable to an explicit cell and logged.

MAV (Multi-Axis Verification) — The account that reliability arises from the intersection of three approximately orthogonal verification axes.

Axis 2 (Structural / Causal) — The verification axis specified by HNS-36/144: constraints on structure, category, and causal order, independent of frequency.

External Verification Architecture (EVA) — The runtime architecture (Volume 5) in which the coordinate checks are applied to model output.

Output Attenuation — The operator that scales an output by a grounding score, so ungrounded content is suppressed rather than emitted.

Grounding function $G(x)$ — The Axis-3 function that returns 1 only when an output is compatible with all six Social Meta Layers.

Onto-Modal — Pertaining to the two HNS-144 dimensions: ontological domain (subjective/objective) and modality (necessary/contingent).

Cell trajectory — The ordered sequence of cells assigned to the propositions in a chain of reasoning, inspected for non-adjacent transitions.

Structured analogy — A correspondence used for design intuition without a claim of exact identity, as in the brain-region mappings.

References

- Aristotle. Categories. (W. D. Ross, Ed.).
- Eichenbaum, H. (2004). Hippocampus: cognitive processes and neural representations. *Neuron*, 44(1), 109–120.
- Evans, J. St. B. T., & Stanovich, K. E. (2013). Dual-process theories of higher cognition. *Perspectives on Psychological Science*, 8(3), 223–241.
- Fillmore, C. J. (1976). Frame semantics and the nature of language. *Annals of the NY Academy of Sciences*, 280, 20–32.
- Gangemi, A., Guarino, N., Masolo, C., Oltramari, A., & Schneider, L. (2002). Sweetening ontologies with DOLCE. *EKAUW 2002*, 166–181.
- Habermas, J. (1984). *The Theory of Communicative Action*, Vol. 1. Beacon Press.
- Hara, S. (2026). *The HNS Origin Trilogy*. Natural Structure Works.
- Kahneman, D. (2011). *Thinking, Fast and Slow*. Farrar, Straus and Giroux.
- Kant, I. (1781/1998). *Critique of Pure Reason*. Cambridge University Press.
- Miller, E. K., & Cohen, J. D. (2001). An integrative theory of prefrontal cortex function. *Annual Review of Neuroscience*, 24, 167–202.
- Niles, I., & Pease, A. (2001). Towards a standard upper ontology (SUMO). *FOIS 2001*, 2–9.
- Pearl, J. (2009). *Causality: Models, Reasoning, and Inference* (2nd ed.). Cambridge University Press.
- Smith, B. (2015). *Basic Formal Ontology (BFO) 2.0 Specification and User's Guide*.
- Stanovich, K. E., & West, R. F. (2000). Individual differences in reasoning. *Behavioral and Brain Sciences*, 23(5), 645–665.
- Treisman, A. M., & Gelade, G. (1980). A feature-integration theory of attention. *Cognitive Psychology*, 12(1), 97–136.
- von der Malsburg, C. (1981). *The correlation theory of brain function*. MPI Biophysical Chemistry, Internal Report 81-2.

Human Natural Structure

Analytical Specification

Inference-Level Causal Verification and the 864-Cell Analytical Engine

Volume 3 · Human Natural Structure Series

Satoru Hara

Natural Structure Works (NSW)

A Note on Method and Evidence

This volume specifies the analytical layer of Human Natural Structure, the 864 engine. Because it makes mathematical claims, it is important to separate three registers of assertion, which are kept distinct throughout.

Established results. The mathematics of the inference operators is Judea Pearl's: the causal hierarchy of association, intervention, and counterfactual; the do-operator and its identification by the back-door and front-door criteria; and the Causal Hierarchy Theorem, which shows the rungs do not collapse. The control operators rest on William Powers' Perceptual Control Theory and on classical negative-feedback control. The formal properties of the attenuation map are elementary real analysis. None of this is in dispute, and none of it is original to HNS.

HNS design decisions. What is original — and therefore a design choice rather than a theorem — is the assignment of exactly these six items as the third axis O1–O6; the multiplicative nesting $36 \times 4 \times 6 = 864$; the definition of the structural grounding score; the output-attenuation operator O6 and its use as a pre-decode actuator; and the use of the whole apparatus as an external verifier. These are engineering commitments, defensible but not proven by the borrowed mathematics that supports their parts.

Conjecture. Claims that the engine improves the reliability of a generative system in practice are hypotheses for empirical work, reported in Volume 1 and to be tested further.. They are flagged as conjecture wherever they appear and are not established by this specification.

Two honesty notes carried from the foundational volume apply again here. First, the recurrence of the number six across the framework is a chosen resolution, not a natural constant; the inference family has a principled three (Pearl's rungs) and the control family a principled three (reference, comparator, actuator), and their sum is six by composition, not by necessity. Second, the scholars named as lineage did not design HNS; their work is acknowledged as inspiration, and the recombination is the author's responsibility.

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Preface: From Coordinate to Engine

The foundational volume gave Human Natural Structure a place to stand. HNS-36 positions a proposition by causal origin and explanatory purpose; HNS-144 refines that position with the modal stance the claim takes — whether it speaks of the subjective or the objective, the contingent or the necessary. Both are acts of location. Neither yet does anything.

This volume specifies what the framework does once a claim has been located. The 864 layer is the analytical engine: it applies, to a positioned and observed claim, one of six operators that together exhaust the ways a claim can be reasoned about and the ways an output can be controlled. Three of the operators are operations of inference — they ask what may be concluded, and on what evidential strength. Three are operations of control — they govern what is finally emitted. The coordinate says where a claim is; the engine says what is being done with it, and whether that doing is warranted.

The characteristic failure at this resolution has a precise name. Unsupported Causality is the assertion of a causal or counterfactual conclusion on the strength of evidence that supports only association. It is the single most common way a fluent system says more than it has earned. Pearl's ladder makes the error formal: a claim that climbs to the rung of intervention or counterfactual must carry support from that rung, not from the rung below. The 864 engine is, in large part, the machinery for detecting and refusing that overreach — and, through the attenuation operator, for ensuring that ungrounded content is scaled toward zero before it is ever emitted.

A reader who wants the formal core may read Part II (the inference operators) and Part III (the control operators) with Appendix A. Part I fixes definitions and recalls the inherited coordinate; Part IV places the engine in the larger architecture and states, candidly, what has and has not been shown.

Part I

The Analytical Engine — HNS-864

1. The 864 Analytical Configuration

HNS-864 is the highest-resolution layer of the structural axis. Where HNS-36 has thirty-six positions and HNS-144 has one hundred forty-four, HNS-864 has eight hundred sixty-four analytical configurations. Each configuration is the unit on which the engine operates.

1.1 Definition and cardinality

An analytical configuration is a four-tuple consisting of a Natural Layer, an Abstract Category, an Onto-Modal Quadrant, and an Inference & Control Operator:

$$(L_i \times C_j \times Q_k \cdot O_m), \text{ with } i, j, m \in \{1, \dots, 6\} \text{ and } k \in \{1, \dots, 4\}.$$

The first two coordinates are the HNS-36 cell; the first three are the HNS-144 cell; the operator is the third axis specified in this volume. The cardinality follows by multiplication of independent factors:

$$|L| \cdot |C| \cdot |Q| \cdot |O| = 6 \cdot 6 \cdot 4 \cdot 6 = 864.$$

Equivalently, $36 \times 4 = 144$, and $144 \times 6 = 864$. The 144 layer is the input to the engine; the operator axis multiplies it. This volume adds only the final factor of six; the first three are inherited from the foundational and observational specifications and are recalled in Chapter 2.

Note. The four quadrants come from two binary dimensions (subjective/objective $\times$ necessary/contingent), giving $2 \times 2 = 4$, not four separate factors. The operator axis is a single factor of six. Any reading that multiplies more than these four factors does not yield 864 and is not the specified system.

1.2 The engine acts on positioned claims

The order of application is fixed and is itself a normative requirement. A claim is first positioned in HNS-36, then observed in HNS-144, and only then analysed by the application of an operator. An operator applied to an unpositioned claim has no defined meaning, because the operator acts on the claim's coordinate, not on its surface words. This ordering is what distinguishes the engine from a free-floating inference procedure: the same sentence, positioned at two different cells, is a different object for the engine to act upon.

The distinction between the coordinate and the engine can be put simply. The coordinate is descriptive: it answers where and in what modal stance a claim sits. The engine is operational: it answers what inference is being drawn and what control is being exercised. A reliable system must keep the two aligned — the inference it draws must not exceed the evidence its position affords, and the output it emits must be attenuated in proportion to how well that alignment holds.

1.3 Notation and conventions

A configuration is written $(L_i \times C_j \times Q_k \cdot O_m)$. A reasoning chain is an operator trajectory: the ordered sequence of operators asserted across its steps, read together with the cell trajectory of the foundational volume. Probabilities are written $P(\cdot)$; the intervention distribution is written $P(\cdot \mid \text{do}(\cdot))$. The control error is written $e = r - p$, with r the reference and p the perception. Sections marked normative state requirements; sections marked informative or non-normative supply motivation and examples and impose no requirement. Unless stated otherwise, the definitions and rules of this volume are normative.

1.4 A configuration, read aloud

It helps to read a full configuration aloud once. Take “the interest-rate rise will curb inflation.” Its layer is macro-social and institutional (L6): the claim originates in monetary policy. Its category is transformation (C5): it is about an operation, a rate being changed to bring something about. Its quadrant is objective and contingent (Q3): it speaks of the world, and the relation could hold or fail. Its operator is intervention (O2): it asserts that setting the rate brings about a change in inflation. The configuration is therefore $(L6 \times C5 \times Q3 \cdot O2)$, and reading it this way already tells the verifier what to demand — an identification for the interventional claim — before a word of the content is judged true or false.

2. The Inherited Coordinate

This chapter recalls, compactly, the three inherited factors so that the present volume is self-contained. The normative source for all three is the foundational and observational specification; nothing here overrides it.

2.1 The six Natural Layers (L1–L6)

Layer	Name	Lineage
L1	Physical / Substrate	Physics; BFO material entity
L2	Cellular / Informational	Computational neuroscience
L3	Cognitive / System 1	Kahneman Type 1
L4	Operational / System 2	Kahneman Type 2; PFC control
L5	Micro-Social / Interaction	Habermas
L6	Macro-Social / Institutional	Institutional theory; BFO social object

The layers are causal strata: the layer at which a claim originates, not the layer at which its effects are seen. The engine treats the layer as the domain over which an operator ranges.

2.2 The six Abstract Categories (C1–C6)

Cat.	Name	Lineage
C1	Continuant / Entity	Aristotle; BFO continuant
C2	Occurrent / Process	Aristotle; BFO occurrent
C3	Relational / Boundary	Aristotle; Kant
C4	Quality / State	Kant
C5	Transformation / Operation	Pearl (do-operator)
C6	Meta-Level / Schema	Kant

Category C5, transformation, is of particular interest at this resolution: a claim positioned at C5 is already a claim about an operation, and so interacts directly with the inference operators of Part II.

2.3 The four Onto-Modal Quadrants (Q1–Q4)

Quad.	Dimensions	Reading
Q1	Subjective × Contingent	preferences, impressions
Q2	Subjective × Necessary	commitments, constitutive rules
Q3	Objective × Contingent	empirical fact, measurement
Q4	Objective × Necessary	definition, logical truth

The quadrant matters to the engine because the warrant a claim needs depends on its modal stance. An objective-contingent claim (Q3) is exactly the kind that invites a causal reading and therefore the kind most exposed to Unsupported Causality.

2.4 Why the factors multiply

The four factors compose multiplicatively because they are, by design, independent questions. Where a claim originates (layer) does not fix how it is read (category); how it is read does not fix the modality in which it is asserted (quadrant); and none of these fixes the operation it performs (operator). Each question can in principle be answered independently of the others, so the number of distinct answers is the product, not the sum, of the per-factor counts — $6 \times 6 \times 4 \times 6 = 864$. Were the factors correlated — if every institutional claim had to be a process asserted as an intervention — the later factors would carry no information and the space would collapse toward 36. The specification therefore treats independence as a design target: the coordinate is built so that any combination is expressible, and a combination that is rare in practice is information about the claim, not a defect of the grid.

3. The Third Axis: Inference & Control Operators

The operator axis has two families of three. The inference operators 01–03 are the three rungs of Pearl’s causal hierarchy; they govern what may be concluded from a claim and on what evidential strength. The control operators 04–06 are drawn from Perceptual Control Theory together with an HNS-defined output regulator; they govern what is finally emitted. The full axis is given in the table below and developed operator by operator in Parts II and III.

Op.	Name	Lineage	Formal content
01	Conditioning	Statistics / Pearl	$P(Y X)$ — association (Rung 1)
02	Intervention / do	Pearl	$P(Y \text{do}(X))$ — intervention (Rung 2)
03	Counterfactual	Pearl	$P(Y_x X', Y')$ — counterfactual (Rung 3)
04	Reference Signal	Powers (PCT)	reference invariant r
05	Comparator / Error	Powers (PCT)	control error $e = r - p$
06	Output Attenuation	HNS Engineering	output = $g(s) \cdot \text{raw}$, $g(0)=0$

3.1 Why operators, and why these

The coordinate locates a claim; it does not say what is being done with it. Yet the reliability of a generated statement turns less on where it sits than on what operation it performs: whether it merely reports an association, asserts the effect of an action, or imagines an alternative history; and, downstream, whether the system emits it, attenuates it, or suppresses it. These are exactly the operations the six operators name.

The choice of these six is principled within each family and compositional across them. Within inference, Pearl’s hierarchy is provably a strict three-level ladder: association, intervention, counterfactual, with each level expressing questions the level below cannot answer. Within control, the minimal negative-feedback loop

has exactly three functional parts: a reference that says what is wanted, a comparator that measures the gap, and an actuator that acts to close it. HNS adopts these two principled triples and composes them; the resulting six is a composition, not an independent postulate that the world contains six operators.

3.2 Division of labour between the families

The inference family establishes what may legitimately be claimed. It is epistemic: it audits the match between the operator a claim asserts and the operator its evidence supports. The control family governs what is done with the result. It is regulative: it takes the structural grounding of a candidate output as a perception, compares it to a reference, and attenuates the output accordingly. The inference operators detect the error; the control operators act on it. The two families meet at the attenuation operator O6, where a verdict about warrant becomes a force on the emitted distribution.

3.3 Categories and operators

The Abstract Category at which a claim sits shapes which operator it is apt to assert, though it does not fix it. A claim positioned at C5 (transformation / operation) is already about an operation and therefore reaches naturally for O2 or O3 — it speaks of something being brought about. A claim at C2 (process) describes an unfolding and is most often O1, unless it names a mechanism that licenses an ascent. Claims at C1 (entity), C3 (relation), and C4 (state) are typically descriptive and sit at O1; they cross into the causal operators only when they assert that an entity, relation, or state produces an effect. A claim at C6 (schema) is meta-level and, like an objective-necessary quadrant, usually asserts no empirical causal operation at all. These are tendencies the engine uses as priors on the asserted operator, to be confirmed against the claim's actual vocabulary — not substitutes for reading it.

Part II

The Inference Operators — Pearl's Ladder

4.01 — Conditioning

The first inference operator is conditioning: the operator of association, the first rung of the causal hierarchy. It is the operation of seeing — of reading one quantity from another within a fixed population.

4.1 Formal definition

For random variables X and Y , conditioning produces the conditional distribution

$$P(Y | X) = P(X, Y) / P(X), \text{ for } P(X) > 0.$$

This is a statement about co-occurrence. It answers the question: given that X is observed to take a value, how is Y distributed? It is computed entirely from the joint distribution and requires no causal assumption whatsoever.

4.2 What conditioning licenses

Conditioning licenses prediction and diagnosis under stable conditions. If the data-generating process does not change, observing X improves an estimate of Y exactly to the degree that they are statistically dependent. Conditioning is the proper operator for forecasting, pattern recognition, and the reading of symptoms — for any task in which the system is a passive observer of a population it does not disturb.

4.3 What conditioning does not license

Conditioning does not license any claim about the effect of acting. Reweighting a population by the observed value of X is not the same as setting X to that value. The distinction is the entire content of the next two operators. A system that uses O1 evidence to make an O2 or O3 claim has committed Unsupported Causality, the failure of Chapter 7. At HNS-144, a claim of this kind typically sits in Q3 (objective, contingent) and at category C2 or C5; it is precisely there that the engine must check that the asserted operator does not exceed O1.

4.4 Properties of the conditional distribution

Three properties are used throughout. The chain rule factorizes a joint distribution, $P(X, Y) = P(Y | X) P(X)$. Bayes' rule inverts a conditional, $P(X | Y) = P(Y | X) P(X) / P(Y)$. Marginalization recovers a lower-dimensional distribution by summing out a variable, $P(Y) = \sum_x P(Y | X = x) P(X = x)$. All three operate within a single joint distribution and introduce no causal content; they are bookkeeping on observation.

4.5 Threats at the first rung

Two structures make association an unreliable proxy for effect. A confounder is a common cause of X and Y ; conditioning on X then mixes the genuine path $X \rightarrow Y$ with the spurious path $X \leftarrow \text{confounder} \rightarrow Y$. A collider is a common effect of X and Y ; conditioning on it, or sampling on a variable it determines, induces a dependence where none existed — the mechanism of selection bias. Both are invisible to O1, which sees only the joint distribution, and both are exactly what the causal model of O2 is needed to expose. A system that treats a first-rung association as if these structures were absent has, in effect, asserted a causal model it never stated.

4.6 Measures of association

Conditioning is agnostic about how dependence is quantified. Correlation captures linear co-variation; mutual information captures dependence of any form, vanishing exactly when X and Y are independent; a conditional-independence test asks whether a dependence survives once a third variable is held fixed. All three are O1 instruments: they describe the joint distribution and license prediction, never effect. The engine notes which instrument an O1 claim rests on only to confirm that the claim stays within Rung 1; the choice among them never raises the rung.

4.7 A positioned O1 claim (non-normative)

Take “users who enable notifications open the app more often.” Positioned, it is an institutional behaviour read as a process, ($L6 \times C2$); observed, it is objective and contingent (Q3). Its asserted operator is O1 — an association. The claim is well formed and passes: it asserts exactly what a correlation supports. The engine logs it as O1 with O1 support, verdict pass, and would intervene only if a later step

rephrased it as “enabling notifications increases engagement,” an O2 ascent that would require its own warrant.

5.02 — Intervention (the do-operator)

The second inference operator is intervention. It is the operation of doing, the second rung of the hierarchy, and it is the operator most often invoked, often without warrant, whenever a system speaks of causes, effects, levers, or consequences.

5.1 The do-operator

The interventional distribution $P(Y \mid \text{do}(X = x))$ is the distribution of Y that results from setting X to x by external intervention, rather than from observing X to equal x . In the language of structural causal models, the intervention performs graph surgery: it removes the equation that ordinarily determines X , severing the arrows into X , and replaces it with the constant x . What flows downstream from X is preserved; what would have explained X is cut away.

5.2 Why intervention differs from conditioning

In general,

$$P(Y \mid \text{do}(X)) \neq P(Y \mid X).$$

The inequality is the signature of confounding. When a common cause influences both X and Y , the observed association $P(Y \mid X)$ mixes the influence of X on Y with the influence of the confounder on both, whereas $P(Y \mid \text{do}(X))$ isolates the former. The classic consequence is that a strong observed correlation can correspond to no interventional effect at all, and that the sign of an effect can even reverse between the two (Simpson's paradox is the discrete instance).

5.3 Identification

Intervention can sometimes be computed from observational data, but only with a causal model. If a set of variables Z satisfies the back-door criterion relative to (X, Y) — blocking every spurious path and containing no descendant of X — then the interventional distribution is identified by the adjustment formula:

$$P(Y \mid \text{do}(X)) = \sum_k P(Y \mid X, Z = z) \cdot P(Z = z).$$

Where no admissible back-door set exists, the front-door criterion or

the full do-calculus may still identify the effect; and where the graph licenses none of these, the effect is simply not identifiable from observation, and only experiment will settle it. The decisive point for the engine is structural: Rung 2 always requires a causal model — a set of assumptions about the graph — over and above the data. Evidence at Rung 1 never suffices on its own.

5.4 Role in the engine

Operator O2 is the proper operator for any claim about effects, policies, or mechanisms — for category C5 (transformation/operation) claims especially. The engine’s task is to confirm that a claim asserting O2 is accompanied by what O2 requires: an experiment, or an observational identification with its assumptions made explicit. A bare correlation offered in support of an O2 claim is the canonical Unsupported Causality, and is to be flagged.

5.5 The three rules of do-calculus

When an effect must be identified from observation together with a graph, the do-calculus supplies three rules that rewrite expressions containing the do-operator until none remains. In words: Rule 1 inserts or deletes an observation that is irrelevant to the outcome given the rest; Rule 2 exchanges an action for an observation — $\text{do}(X)$ for X — when the relevant back-door paths are blocked; and Rule 3 inserts or deletes an action that has no effect on the outcome along open paths. An effect is identifiable precisely when repeated application removes every do-operator, leaving a formula in observational quantities. The calculus is complete: if the three rules cannot eliminate the do-operator, no observational formula for the effect exists, and only experiment will settle it.

5.6 The front-door criterion

Where no admissible back-door set exists — because the confounder is unobserved — an effect may still be identified through a mediator M that lies wholly on the path from X to Y , is unconfounded with Y given X , and is shielded from the unobserved confounder. The front-door formula chains two adjustments:

$$P(Y \mid \text{do}(X)) = \sum_m P(M = m \mid X) \sum_{x'} P(Y \mid M = m, X = x') P(X = x').$$

The lesson for the engine is unchanged: identification is possible, but only relative to an explicit causal structure that the claim must declare.

5.7 A numerical instance: Simpson’s paradox

The gap between O1 and O2 is not a technicality; it can reverse a conclusion. In the figures below a treatment recovers a higher fraction of patients in each severity subgroup, yet a lower fraction in the pooled total, because the severe cases — harder to recover — were treated more often.

Group	Treated (recovered)	Untreated (recovered)
Mild cases	81 / 87 (93%)	234 / 270 (87%)
Severe cases	192 / 263 (73%)	55 / 80 (69%)
Pooled	273 / 350 (78%)	289 / 350 (83%)

Reading the pooled association $P(\text{recovery} \mid \text{treated}) < P(\text{recovery} \mid \text{untreated})$ as an effect — asserting O2 on O1 — would recommend withholding a treatment that in fact helps within every subgroup. The adjusted quantity $P(\text{recovery} \mid \text{do}(\text{treated}))$, which conditions on severity before pooling, restores the correct ordering. This is the practical face of Unsupported Causality.

5.8 Randomization as the gold standard

The cleanest warrant for an O2 claim is a randomized intervention. When X is assigned by a mechanism independent of every other variable, the arrows into X are severed by construction, so observation and intervention coincide within the experiment, $P(Y \mid \text{do}(X)) = P(Y \mid X)$. Randomization is therefore not one identification strategy among others but the case in which identification is automatic. An O2 claim backed by a randomized trial is licensed without further graphical argument; the back-door and front-door routes exist precisely for the common case in which randomization was not possible.

5.9 A back-door adjustment (worked, non-normative)

Suppose a claim asserts that a training programme raises wages (O2). Workers self-select into training by prior skill, which also raises wages, so prior skill is a back-door confounder. If prior skill is

measured and blocks the only spurious path, the adjustment formula identifies the effect:

$$P(\text{wage} \mid \text{do}(\text{train})) = \sum P(\text{wage} \mid \text{train}, \text{skill}) \cdot P(\text{skill})$$

The engine does not compute this sum; it checks that the claim names an adjustment set and that the set is defensible — that it blocks the spurious path and contains no descendant of training. A claim that adjusts for a post-training variable, a mediator or a consequence, fails the criterion and is flagged, because conditioning on a descendant of the cause distorts the very path the adjustment was meant to close.

5.10 A graph and an adjustment (worked, non-normative)

Let the causal structure be $Z \rightarrow X$, $Z \rightarrow Y$, and $X \rightarrow Y$, with Z observed. Z is a common cause of treatment X and outcome Y , and the back-door path $X \leftarrow Z \rightarrow Y$ is the only spurious route. The set $\{Z\}$ blocks it and contains no descendant of X , so it satisfies the back-door criterion and the effect is identified by adjustment:

$$P(Y \mid \text{do}(X)) = \sum_k P(Y \mid X, Z = z) \cdot P(Z = z)$$

With toy numbers — $P(Z = 1) = 0.5$; in the stratum $Z = 0$, $P(Y = 1 \mid X = 1) = 0.6$ and $P(Y = 1 \mid X = 0) = 0.5$; in the stratum $Z = 1$, $P(Y = 1 \mid X = 1) = 0.8$ and $P(Y = 1 \mid X = 0) = 0.7$ — the adjusted effect is $(0.6 - 0.5)(0.5) + (0.8 - 0.7)(0.5) = 0.10$, a ten-point causal increase. The unadjusted contrast is larger, because treated cases are over-represented in the high-outcome stratum $Z = 1$; reading that inflated contrast as the effect is the confounding sub-form of Unsupported Causality. The engine's check is satisfied here because the claim named $\{Z\}$ and $\{Z\}$ is a valid back-door set; the arithmetic it leaves to the analyst.

6.03 — Counterfactual

The third inference operator is counterfactual: the operation of imagining, the third and highest rung. It asks what would have been, in a specific case, had something been otherwise.

6.1 Formal content

A counterfactual quantity is written

$$P(Y_x / X', Y'),$$

read as the probability that Y would have taken the value Y_x had X been set to x , given that in fact X took the value X' and Y took the value Y' . Its evaluation requires a fully specified structural causal model — the functions relating each variable to its parents, together with a distribution over the exogenous noise — and proceeds in three steps: abduction (update the noise distribution on the observed evidence), action (apply the intervention to the model), and prediction (read off the resulting distribution of Y).

6.2 The hierarchy does not collapse

Pearl's Causal Hierarchy Theorem establishes that the three rungs are genuinely distinct: with probability one, interventional facts are not determined by observational facts alone, and counterfactual facts are not determined by interventional facts alone. The ladder cannot be climbed by cleverness with lower-rung data; each ascent requires a stronger model. This is why the engine treats the rung asserted by a claim as load-bearing rather than cosmetic: a counterfactual asserts the most and so must carry the most.

6.3 Role in the engine

Operator 03 is the operator of attribution and responsibility — of but-for reasoning, of “the outcome was due to” and “had we not, then.” Such claims are common and rhetorically powerful, and they are the most easily unsupported of all, because they require a structural model the speaker rarely possesses. A counterfactual offered with only interventional or, worse, associational support is Unsupported Causality at its most acute.

6.4 Structural causal models

A structural causal model is a tuple $M = \langle U, V, F, P(U) \rangle$: a set U of exogenous background variables with a distribution $P(U)$; a set V of endogenous variables; and a set F of structural functions, one per endogenous variable, expressing each as a function of its parents in V and of some exogenous variables. The model induces both a directed graph and a joint distribution. An intervention acts on F , replacing a function by a constant; a counterfactual acts on $P(U)$, conditioning on evidence before intervening. The model, not the data, is the carrier of causal meaning — which is why a claim at O2 or O3 is a claim about a model.

6.5 The three-step evaluation

A counterfactual $P(Y_x \mid X', Y')$ is evaluated on M in three steps. Abduction: update $P(U)$ to $P(U \mid X', Y')$, absorbing what actually happened into the background. Action: form the modified model in which the equation for X is replaced by $X = x$. Prediction: compute the distribution of Y in the modified model under the updated background. The procedure is deterministic given M ; its whole difficulty lies in possessing M , which is why counterfactuals are the most demanding operator the engine recognizes.

6.6 Probabilities of causation

Counterfactual analysis makes precise the everyday notions of necessary and sufficient cause. The probability of necessity, PN — the probability that Y would not have occurred had X not, given that both did — measures but-for causation. The probability of sufficiency, PS , measures whether X would bring about Y in a case where neither occurred. Both are third-rung quantities, frequently only bounded rather than point-identified from data, and asserting them — as legal, medical, and explanatory language constantly does — commits the speaker to a structural model whether or not one is acknowledged.

6.7 A probability of necessity (worked, non-normative)

Consider “the alarm is why the fire was contained” — an O3 attribution. The relevant quantity is the probability of necessity: had the alarm not sounded, would the fire have spread, given that it did

sound and was contained? From observational and even experimental data this is generally only bounded, not point-identified; monotonicity assumptions tighten the bounds but must themselves be argued. The engine therefore treats a bare necessity claim as O3-asserted with, at best, O2 support, and attenuates the individual-attribution force while leaving any licensed population effect — “alarms reduce spread on average” — intact. The lesson is general: but-for language is counterfactual and inherits the counterfactual’s heavy warrant.

6.8 A counterfactual on a toy model (worked, non-normative)

Take a linear structural model with exogenous noise U_Y and a binary treatment X , in which the outcome is generated by $Y = 2X + U_Y$. Observe one unit with $X' = 1$ and $Y' = 5$. The counterfactual “had X been 0, what would Y have been?” is evaluated in the three steps of Chapter 6.5.

Abduction: from $Y' = 2 \cdot 1 + U_Y = 5$ recover $U_Y = 3$ for this unit. Action: set $X = 0$, giving $Y = 2 \cdot 0 + U_Y$. Prediction: $Y = 0 + 3 = 3$. So for this unit “had X been 0, Y would have been 3” is licensed by the model, and the individual effect of the change is $5 - 3 = 2$ — the model’s coefficient. The point is not the arithmetic but its precondition: the conclusion required the full model, the equation together with the recovered noise, which is exactly what an O3 claim asserts it possesses. Strip away the model and the same sentence is an O3 assertion with no O3 support — the counterfactual-without-a-model sub-form.

7. Unsupported Causality

Unsupported Causality is the characteristic failure of the analytical resolution and the principal target of the inference operators. This chapter defines it formally, gives its detection rule, and works examples. It also gathers the analytical failure types of the engine and maps them onto the inherited failure taxonomy.

7.1 Definition

Order the inference operators by rung: $O1 < O2 < O3$. Each claim carries two operator labels: the operator it asserts (the strength of the conclusion it draws) and the operator its evidence supports (the strength the available warrant can sustain). Unsupported Causality is the condition

$$\text{asserted operator} > \text{supported operator}.$$

That is, the claim climbs higher on the ladder than its evidence can carry it. The commonest instances are an O2 claim on O1 evidence (“X is correlated with Y, therefore X causes Y”) and an O3 claim on O2 or O1 evidence (“the effect was due to X,” offered without a structural model).

7.2 Detection rule

The detection rule is local and mechanical. For each claim: (1) determine the asserted operator from its modal and causal vocabulary — “causes,” “raises,” “if we do” signal O2; “would have,” “but for,” “was responsible for” signal O3. (2) Determine the supported operator from the evidence actually offered — a correlation supports O1; an experiment or an explicit identification supports O2; a structural model with abduction supports O3. (3) Flag the claim when the asserted operator outranks the supported operator, and either require the missing warrant or attenuate the output. The check does not assess whether the claim is true; it assesses whether the asserted operation is earned.

7.3 Worked examples

Case 1. Output: “Cities with more police have more crime, so police

cause crime.” The vocabulary asserts O2 (cause). The evidence — a cross-sectional correlation — supports only O1. Asserted O2 > supported O1: Unsupported Causality. The correction either retreats to O1 (“police presence and crime are positively associated across cities”) or supplies O2 warrant (an identification adjusting for the confounder, here the underlying crime rate that drives both).

Case 2. Output: “Had the firm not advertised, it would have failed.” The vocabulary asserts O3 (counterfactual attribution). No structural model is offered; at most, advertised firms are observed to survive more often — O1 evidence. Asserted O3 > supported O1: a severe instance. The correction states the modest claim the evidence supports (“advertising is associated with survival”) or builds the model the counterfactual requires.

Case 3 (a correct ascent). Output: “In the randomized trial, the drug lowered mortality.” The vocabulary asserts O2. The evidence is a randomized experiment, which supports O2 directly. Asserted equals supported: the claim passes. The engine’s job is not to doubt causation everywhere, but to require that the rung be earned.

7.4 The analytical failure types and their mapping

The analytical engine recognizes a small family of failures, inherited in substance from the analytical specification and re-expressed in operator terms. They map onto the foundational and observational failure taxonomy as follows.

Analytical failure	Operator-level reading	Inherited type
Causal Misidentification	asserted rung exceeds supported rung	Unsupported Causality
Layer Drift	operator ranges over the wrong Natural Layer	Layer Jump
Category Misalignment	operator applied under the wrong Category	Category Ambiguity
Depth Confusion	rung treated as interchangeable with a lower one	modal overreach (HNS-144)
Resolution Collapse	configuration under-specified; operator unassigned	loss of auditability

Causal Misidentification is Unsupported Causality by another name

and is the engine’s central concern. The others are localizations of where, in the configuration, the analysis has gone wrong: in the layer, in the category, in the rung, or in the completeness of the configuration itself.

7.5 The operator order is strict

The inference operators form a strict order under the relation requires at least as strong a model as: $O1 < O2 < O3$. The order is strict by the Causal Hierarchy Theorem — a higher operator is never reducible to a lower one — so the gap that an Unsupported-Causality violation opens cannot be closed by accumulating more evidence of the lower kind. It can be closed only by a stronger model or an experiment. This is precisely why the engine’s remedy is to demand the missing warrant or to attenuate, and never to manufacture the higher rung from lower evidence: no quantity of correlation becomes an intervention, and no set of interventions becomes a counterfactual, by aggregation alone.

7.6 Vocabulary cues for the asserted operator

The asserted operator is read from a claim’s causal and modal vocabulary. The cues below are heuristic, not definitional, but they localize the rung a claim reaches for.

Cue vocabulary	Asserted operator
“is associated with,” “tends to,” “predicts,” “correlates”	O1 — conditioning
“causes,” “raises,” “leads to,” “if we do X then”	O2 — intervention
“would have,” “but for,” “was responsible for,” “due to”	O3 — counterfactual

7.7 Sub-forms of Unsupported Causality

Unsupported Causality is a single condition — asserted rung above supported rung — but it arises through a small number of recurring structures. Naming them helps a verifier localize the missing warrant.

- **Confounding.** A common cause of X and Y creates or inflates the association; the O2 claim needs an adjustment set that blocks the back-door path. Correction: adjust, or retreat to O1.

- **Collider / selection.** Conditioning on a common effect, or sampling on something it determines, manufactures an association; the remedy is to stop conditioning on the collider, not to find a stronger correlation.
- **Reverse causation.** The association is real but the arrow runs $Y \rightarrow X$; temporal or interventional evidence is needed to fix the direction.
- **Over-adjustment / mediation.** Adjusting for a variable on the causal path removes part of the very effect being claimed; correction is not to condition on mediators when estimating a total effect.
- **Counterfactual without a model.** An O3 attribution is asserted with no structural model to support abduction; correction is to build the model or down-rank to the supported interventional or associational claim.
- **Extrapolation beyond support.** An identified effect is transported to a population or regime where its identifying assumptions do not hold; the warrant lapses outside the support of the data.

Each sub-form is the same violation localized to a different structure, and each admits the same two remedies the engine ever offers: supply the missing warrant, or attenuate the over-claim.

7.8 Modality and the licensed operator

The quadrant a claim occupies at HNS-144 shapes which operator it may legitimately assert. An objective-contingent claim (Q3) is the natural home of O1–O3: it is about how the world in fact is, and so may ascend the causal ladder if it earns the warrant. An objective-necessary claim (Q4) — a definition or a theorem — asserts no causal operation at all, and reading it as O2 is a category error, not merely an over-claim. A subjective claim (Q1, Q2) asserts a preference or a commitment; a causal operator applied to it crosses the subjective/objective boundary before the question of rung even arises. The engine therefore uses the quadrant as a gate: it asks first whether an inference operator is appropriate to the modality at all, and only then whether the asserted rung is supported.

Quadrant	Typically licensed operators	Caution
Q1 Subj × Cont	O1 within the subjective frame	causal operators cross the domain boundary
Q2 Subj × Nec	none inferential (constitutive)	not an empirical causal claim
Q3 Obj × Cont	O1, O2, O3 (with warrant)	the principal site of Unsupported Causality
Q4 Obj × Nec	none inferential (definitional)	reading as O2 is a category error

Part III

The Control Operators — Perceptual Control Theory

8.04 — Reference Signal

The control family begins with the reference. Perceptual Control Theory holds that an organism does not control its actions or its outputs directly; it controls its perceptions, acting so as to bring a perceived variable to a desired value. That desired value is the reference signal.

8.1 The reference invariant

In the control loop, the reference r is the specification of the intended value of a controlled variable. It is set from outside the loop — by the goal, the standard, or in the present setting the verification specification — and is held invariant while the loop runs. In HNS, the reference is the grounding target: the structural state a candidate output is required to match — fully positioned, with every asserted operator supported, and compatible with the grounding stack of the social-meta layer.

The reference is not itself derived from the output; that is what makes it a reference. Its role in the engine is to give the comparator something to measure against, so that the gap between what an output is and what it should be becomes an explicit, signed quantity rather than an implicit judgement.

8.2 Setting the reference

The reference is a policy, and naming it explicitly is part of what makes the engine auditable. A strict reference demands near-complete grounding before an output passes; a lenient reference admits weakly grounded output and relies on attenuation to scale it. Neither is correct in the abstract: a medical or legal deployment sets a high reference and accepts more refusals, while a brainstorming aid sets a low one and accepts more speculation. What the specification requires is not a particular value but that the value be declared and logged, so that the standard against which outputs were judged can be reconstructed.

9.05 — Comparator and Error Detection

The second control operator is the comparator: the element that computes the gap between reference and perception and drives the loop to close it.

9.1 The control error

The comparator forms the control error

$$e = r - p,$$

the difference between the reference r and the current perception p . The loop is a negative-feedback loop: action is generated so as to reduce the magnitude of e , returning the controlled perception toward the reference. In HNS, the perception p is the perceived structural state of a candidate output — the degree to which it is positioned and its operators supported — and the error e measures how far that falls short of the grounding target. A large error is a structurally ungrounded output; a zero error is a fully grounded one.

9.2 Relation to predictive coding

Note. Friston's free-energy principle and the predictive-coding scheme of Rao and Ballard minimize a prediction error, the difference between sensory input and a top-down prediction. This is a structural analogue of the comparator and may be used as an alternative formalization in active-inference implementations. It is not identical: Perceptual Control Theory acts to keep a perception at an externally set reference, whereas predictive coding acts to minimize surprise relative to a generative model. The two coincide only when the reference is read as the prediction; the engine treats predictive coding as an admissible implementation of O5, not as its definition.

What the comparator contributes to the engine is a signed, quantitative measure of structural shortfall. That measure is exactly what the final operator needs: it is the input from which the grounding score, and hence the attenuation, is computed.

9.3 The feedback law

In discrete time the loop drives the controlled perception toward the reference. Writing $p(t)$ for the perception at step t , r for the reference, and k for the loop gain, a first-order controller makes the error $e(t) =$

$r - p(t)$ evolve as

$$e(t+1) = (1 - k) \cdot e(t).$$

The error decays to zero, and the loop is stable, when $|1 - k| < 1$, that is $0 < k < 2$; higher gain converges faster but risks overshoot. In the engine the gain is not a free dynamical parameter but a design choice governing how sharply a structural shortfall is corrected before emission.

9.4 Hierarchies of control

Perceptual Control Theory composes such loops into a hierarchy in which the output of a higher loop sets the reference of a lower one. The engine uses the same idea: the social-meta grounding stack of Volume 4 supplies references that the structural comparator then controls toward. The present volume specifies only the single loop whose actuator is output attenuation.

9.5 Convergence of the loop (worked, non-normative)

Take a reference $r = 1$ (full grounding) and an initial perception $p(0) = 0.4$, with loop gain $k = 0.5$. The error $e(0) = 0.6$ decays as $e(t+1) = (1 - k) e(t) = 0.5 e(t)$: 0.6, 0.3, 0.15, ... toward zero. With $k = 1$ the error is annihilated in a single step; with $k = 1.5$ it oscillates while shrinking (0.6, -0.3, 0.15, ...); with $k \geq 2$ it diverges. In the engine this dynamic is not literally iterated against a fixed output — each output is judged once — but the same arithmetic fixes how aggressively a measured shortfall is converted into attenuation, and the stability band $0 < k < 2$ marks the regime in which correction is well behaved.

10.06 — Output Attenuation

The third control operator is the one original to HNS. It is the actuator of the loop: the element that converts the verdict of the inference and comparator operators into a force on the emitted output. Where the previous operators detect and measure, O6 acts.

10.1 The grounding score

Let $s \in [0, 1]$ denote the structural grounding score of a candidate output: a monotone function of the control error, equal to one when the error is zero and decreasing as the error grows. Informally, s answers: to what degree is this output positioned, operator-supported, and grounded? It is one when the asserted operators are fully earned and the perception matches the reference; it is zero when the output is structurally ungrounded — for example, when it asserts a causal conclusion with no warrant of the required rung.

10.2 The attenuation map

The attenuation operator applies a map g to the grounding score and scales the raw output by the result:

$$\text{output} = g(s) \cdot \text{raw_output}, \text{ with } g : [0,1] \rightarrow [0,1], g(0) = 0, g \text{ monotone non-decreasing.}$$

Here `raw_output` denotes the pre-decode generative signal — the probability or logit vector before token selection — so that attenuation shapes generation rather than filtering text after the fact. The conditions on g are deliberately minimal: it must vanish at zero grounding, it must never penalize better grounding, and it must remain a fraction. Its exact shape — linear, thresholded, or sigmoidal — is an engineering parameter, tuned to trade fluency against caution, and is not fixed by the specification.

10.3 Zero-Incidence Attenuation

The defining property of O6 is its behaviour at the boundary.

Proposition (Zero-Incidence Attenuation). *If $s = 0$ then $\text{output} = 0$.*

The proof is immediate: by definition $g(0) = 0$, so $\text{output} = g(0) \cdot$

raw_output = 0 regardless of the raw output. The content of the proposition is not its difficulty but its guarantee: a fully ungrounded output is not attenuated to something small but emitted anyway — it is set to zero. The system cannot, by construction, emit content that scores zero on structural grounding. This is the sense in which HNS aims at verification rather than mere improvement: the floor is structural, enforced by the actuator, not statistical.

10.4 Placement and the meeting of the families

Attenuation is the point at which the inference and control families meet. The inference operators decide whether an asserted operation is supported; the comparator turns the verdict into an error; the grounding score turns the error into a number in the unit interval; and O_6 turns that number into a multiplicative force on the generative distribution. A failure detected at Rung 2 — an unsupported causal claim — lowers s , and a low s attenuates the very tokens that would have expressed the unsupported claim. The audit and the action are one mechanism.

10.5 Shapes of the attenuation map

Three shapes of g are useful. The linear map $g(s) = s$ attenuates in proportion to shortfall. The threshold map, $g(s) = 0$ for $s < \tau$ and $g(s) = 1$ for $s \geq \tau$, enforces a hard floor at a chosen $\tau \in [0, 1]$, emitting only outputs above the threshold. The power or sigmoid map, $g(s) = s^\gamma$ with $\gamma \geq 1$ or a logistic curve centred at τ , interpolates between the two — sharpening near the threshold while remaining smooth. All satisfy $g(0) = 0$ and monotonicity; the choice trades fluency against caution and is fixed per deployment, not by the specification.

10.6 Composing the grounding score

The grounding score aggregates per-operator verdicts. Let $a_m \in [0, 1]$ be the support margin of operator O_m for a claim — one when the asserted operator is fully earned, zero when it is wholly unwarranted. A conservative composition takes the minimum, $s = \min_m a_m$, so a single unsupported operator drives the score down; a graded composition takes a weighted product, $s = \prod_m a_m^{w_m}$ with $\sum w_m = 1$. Both are monotone non-decreasing in each a_m , so improving the

support of any operator never lowers the score — a property the attenuated output inherits through the monotonicity of g .

10.7 Acting before the decode

Acting before token selection means acting on the distribution. If the raw model assigns logits z over the vocabulary, attenuation by $g(s)$ can be realized as a contraction of probability mass toward a grounded baseline as s falls — a temperature-like reweighting that down-weights ungrounded continuations relative to grounded ones. In the limiting case $s = 0$ the contraction is total: by Zero-Incidence Attenuation no ungrounded continuation retains mass, and the ungrounded output is not produced. The mechanism is generative, not a post-hoc filter on finished text.

10.8 A worked attenuation

Suppose a claim asserts O2 but offers only O1 support, so its causal support margin is $a = 0.2$ while its other operators are fully supported. Under the conservative composition $s = \min = 0.2$; under the linear map $g(s) = 0.2$, so the causal phrasing is carried at one-fifth strength and, in the pre-decode reweighting, falls below competing grounded continuations. Supplying an identification raises a toward one, s toward one, and the attenuation lifts. The numbers are illustrative; the behaviour — monotone, bounded, and zero at the floor — is guaranteed.

10.9 The attenuation map tabulated

The three admissible families behave as follows on representative grounding scores: the linear map $g(s) = s$, the threshold map at $\tau = 0.6$, and the power map $g(s) = s^2$. All vanish at $s = 0$ and reach 1 at $s = 1$; they differ in the middle, where the power map is most forgiving of moderate grounding and the threshold map least.

s	$g = s$ (linear)	threshold $\tau = 0.6$	$g = s^2$ (power)
0.0	0.00	0	0.00
0.2	0.20	0	0.04
0.4	0.40	0	0.16
0.6	0.60	1	0.36

s	g = s (linear)	threshold $\tau = 0.6$	g = s^2 (power)
0.8	0.80	1	0.64
1.0	1.00	1	1.00

The choice among them is the single most consequential tuning decision in a deployment, and is properly made against a measured error profile rather than asserted in advance; what the specification fixes is only the three axioms every admissible map shares.

Part IV

Integration and Status

11. The Generative Principle and Operator Accuracy

This chapter states how an 864 configuration is constructed and what it means to apply the engine correctly.

11.1 The generative principle

An analytical configuration is generated in three movements. First, the claim is positioned in HNS-36 by its causal layer and explanatory category. Second, it is observed in HNS-144 by its modal quadrant. Third, the operator is applied: the claim is read as performing one of the six operations, and the engine checks the operation against its warrant. The full space of 864 configurations is the product of these choices, and every claim that the engine processes occupies exactly one configuration at a time — the analytical analogue of the single-assignment rule of the foundational coordinate.

11.2 Operator accuracy

Operator accuracy is the requirement that the correct operator be applied to the correct claim. Two errors are symmetric. To apply an operator above the supported rung is to over-claim — Unsupported Causality, the dangerous direction, in which a system asserts more than it knows. To apply an operator below the supported rung is to under-claim — a waste of available warrant, in which a system declines to draw a conclusion it has earned. The engine is tuned chiefly against the first, because over-claiming is what makes a fluent system unreliable; but a complete account names both, since a verifier that never permits a causal claim is as useless as one that permits every causal claim.

Operator accuracy is, like every other rule of HNS, an external and auditable check. The operator asserted by a claim and the operator supported by its evidence are both recorded; their comparison is the verdict; and the verdict is logged. Nothing in the procedure requires access to the internals of the generating model.

11.3 Operator trajectories

A chain of reasoning is analysed not claim by claim alone but as an operator trajectory: the ordered sequence of operators its steps assert, read alongside the cell trajectory of the foundational volume. The trajectory exposes a failure invisible at the level of single claims — an illegitimate ascent, in which a chain accumulates only associational support yet concludes with an interventional or counterfactual claim. The rule is that the operator asserted by a conclusion may not outrank the strongest operator its premises jointly support, and that support does not accumulate across rungs: any number of O1 links never compose into an O2 conclusion.

Worked trajectory 1 — an illegitimate ascent

Premise (O1): “regions with more clinics report better health outcomes.” Premise (O1): “wealthier regions have more clinics.” Conclusion (O2): “building clinics improves outcomes.” Both premises are associational; the conclusion is interventional. The trajectory O1, O1 → O2 is an ascent without warrant: the shared cause, wealth, is exactly the unblocked back-door path. The engine flags the conclusion and either demands an identification or attenuates it, leaving the premises intact.

Worked trajectory 2 — a licensed chain

Premise (O2, randomized): “in trials, the vaccine lowers infection.” Premise (O1): “this population resembles the trial population on measured covariates.” Conclusion (O2, transported): “the vaccine will lower infection here.” The conclusion stays at O2 and rests on an O2 premise plus an explicit transport assumption; the trajectory does not ascend beyond its support. The engine passes it while logging the transport assumption as the claim’s load-bearing premise, so that a later challenge can be aimed exactly at it.

Trajectory analysis is the analytical analogue of the layer-jump check. There, the offence is a non-adjacent jump across Natural Layers; here, it is a non-adjacent jump up the causal ladder. Both are detected mechanically from the sequence of coordinates, without recourse to whether the conclusion happens to be true.

11.4 The single-operator rule and its limits

Two requirements complete the account of correct application. First, single assignment: an atomic claim performs exactly one operation and so occupies exactly one operator at a time; a sentence that performs several is decomposed before analysis, as in the composite cases of the casebook. Second, honest extraction: the asserted operator is read from the claim's vocabulary and structure, and where the vocabulary is genuinely ambiguous — “affects,” “drives,” “shapes” can be read as O1 or O2 — the engine records the ambiguity rather than resolving it silently, and treats the claim conservatively, holding it to the higher rung's warrant unless the lower reading is explicitly marked. The limit of the method lies exactly here: operator extraction from natural language is itself a judgement, and the inter-rater reliability of that judgement is the empirical quantity on which the engine's practical value most depends.

12. Architecture and Runtime Placement

The analytical engine is the finest resolution of the structural axis, Axis 2 of the Multi-Axis Verification account. This chapter situates it among the other axes and describes its runtime form.

12.1 Position among the three axes

Axis 1 (generative) proposes; it is the model's native fluency. Axis 2 (structural/causal), of which HNS-864 is the analytical layer, constrains: it positions a proposal, observes its modal stance, and audits the operation it asserts. Axis 3 (grounded/executive), the social-meta layer of Volume 4, checks the proposal against context from the body to the institution. Reliability is the conjunction of the three. HNS-864 contributes the causal-structural condition and is silent, by design, on raw probability and on world-grounding, which belong to Axes 1 and 3.

12.2 Runtime form

At runtime the engine sits between the generative distribution and token selection. A candidate output is positioned and observed; its asserted operators are extracted and compared to their support; the comparator forms the control error; the grounding score is computed; and the attenuation map scales the pre-decode distribution. Every step writes a machine-readable record — the configuration, the asserted and supported operators, the score, and the attenuation applied — so that the verdict is reconstructable after the fact. The independent verification architecture of the Volume 5 specifies that log format and the hardware-level stop that bounds it; the present volume specifies only what is computed, not how it is deployed.

12.3 Where Unsupported Causality is caught

In the runtime loop, Unsupported Causality is caught at the comparison step and acted upon at the attenuation step. A claim whose asserted rung exceeds its supported rung lowers the grounding score; the lowered score attenuates the tokens that would have expressed the over-claim. The detection is epistemic and the response is regulative, but they are realized as a single pass over the

candidate output.

12.4 The log record

Every pass writes a machine-readable record so that the verdict is reconstructable. The record carries, at minimum: the claim as positioned, written as its configuration ($L \times C \times Q \cdot O$); the asserted operator and the supported operator; the grounding score s ; the attenuation $g(s)$ applied; and, where a flag was raised, its sub-form and the missing warrant. Because every field is external to the generating model — derived from the claim's coordinate and its declared evidence, not from the model's internal states — the record can be audited by a third party who never sees the model. The format and its cryptographic commitments are specified in the external-verification volume; the present volume fixes only the fields the engine must emit.

12.5 Relation to the three axes, restated

It bears repeating where the engine's authority ends. Axis 1 decides how probable a continuation is; Axis 3 decides whether it is grounded in body, dialogue, organization, economy, governance, and the wider system. Axis 2, and the 864 engine within it, decides only whether the operation a claim performs is structurally warranted. A claim may be probable, structurally warranted, and still false because it is ungrounded in fact; conversely a true claim may be flagged here because, as stated, it asserts more than its declared evidence licenses. The engine is one of three conditions, not the whole of reliability.

12.6 What the engine does not do

The engine does not establish truth, does not estimate probability, and does not repair a claim. It detects a mismatch between asserted and supported operation and converts that mismatch into an attenuation. Repair — supplying the missing identification, building the missing model, or rephrasing to the supported rung — is the work of the generating system or its author, prompted by the flag. The engine's discipline is precisely that it does less than judge content: it judges licence, which is all that can be checked from outside the model and all that is needed to refuse an over-claim.

13. Intellectual Lineage and Proponents

The analytical engine recombines several established bodies of work. This chapter records the debts and, as in the foundational volume, is candid that the proponents did not design HNS.

13.1 The inference family

The inference operators are Judea Pearl's causal hierarchy in full: conditioning, the do-operator, and counterfactuals, together with the identification theory — back-door and front-door criteria and the do-calculus — and the Causal Hierarchy Theorem that keeps the rungs distinct. The structural-causal-model machinery underlying counterfactuals is also Pearl's, with antecedents in the structural equation tradition and the constraint-based causal discovery of Spirtes, Glymour, and Scheines.

13.2 The control family

The control operators rest on William Powers' Perceptual Control Theory, with its insight that behaviour is the control of perception, and on the classical negative-feedback control of Wiener's cybernetics. The comparator has a widely noted analogue in the predictive-coding scheme of Rao and Ballard and in Friston's free-energy principle, used here as an admissible alternative formalization. The orthogonality intuition that motivates crossing an external constraint plane against the generative axis draws on the information geometry of Amari.

13.3 What is HNS's own

The assignment of these six items as a single ordered axis, the multiplicative nesting that yields 864, the structural grounding score, and above all the output-attenuation operator with its zero-incidence guarantee are HNS's own. So is the decision to treat the whole apparatus as an external verifier rather than as a theory of cognition. Pearl's hierarchy is a theory of causal inference; Powers' theory is a theory of behaviour; neither was written to constrain a language model's output. The recombination, and the responsibility for it, are the author's.

Element	Proponent / source	Borrowed or HNS
01–03 inference ladder	Pearl	borrowed
Identification, do-calculus	Pearl; Spirtes et al.	borrowed
04–05 control loop	Powers; Wiener	borrowed
Comparator analogue	Rao & Ballard; Friston	borrowed (alternative)
Orthogonality intuition	Amari	borrowed
06 attenuation; 864 nesting; verifier use	Hara (HNS)	HNS

13.4 The proponents, in brief

Judea Pearl’s work supplies the entire inference family. His ladder of causation — seeing, doing, imagining — and the do-calculus that decides when a causal effect can be read from observation are the mathematical content of 01–03. HNS adopts them unchanged; what it adds is the decision to treat the rung a claim asserts as an auditable property of generated text.

William Powers’ Perceptual Control Theory supplies the control frame. Its central claim — that organisms control their perceptions, not their outputs, acting to hold a perceived variable at a reference — gives 04 and 05 their form. HNS reads the controlled perception as a grounding score and the reference as a verification standard.

The negative-feedback loop itself, and the requirement that a regulator command enough variety to counter the disturbances it faces, come from the cybernetics of Wiener and Ashby. They justify the loop’s structure and, in Ashby’s law of requisite variety, the intuition that a verifier must be at least as rich as the failures it screens. Amari’s information geometry motivates the picture of an external constraint surface crossing the generative manifold — the orthogonality that lets Axis 2 add information the generative axis lacks — while Friston’s free-energy principle and Rao and Ballard’s predictive coding furnish the comparator’s analogue.

None of these authors wrote for the purpose to which HNS puts their work, and none should be read as endorsing the framework. The borrowing is acknowledged so that the genuinely new commitments

— the sixfold axis, the 864 nesting, and the attenuation operator — remain visible as the author's own.

14. Empirical Status and Limitations

This final chapter states plainly what the specification establishes, what it proposes, and where it is exposed.

14.1 What is established

Established are the mathematical facts the engine inherits and its own well-definedness. Pearl's ladder, the adjustment and do-calculus identification results, and the Causal Hierarchy Theorem are settled mathematics. The negative-feedback control law and the elementary properties of the attenuation map — including the zero-incidence proposition — are likewise not in doubt. The 864 nesting is well-defined, its cardinality is exactly the product of its factors, and every step of the engine is externally auditable. These are the firm parts.

14.2 What is proposed

Proposed, and defensible but unproven, are the engineering commitments: that operator-support checking is a good operationalization of analytical reliability; that pre-decode attenuation is an effective place to act; and that the six operators are the right six. These are reasoned choices, argued in the preceding chapters, not consequences of the borrowed mathematics.

14.3 What remains conjecture

Conjecture, finally, are the quantitative claims about reliability — that the engine drives a system's hallucination rate toward a structural floor, that the attenuation materially reduces unsupported output in practice. These are empirical questions, reported on in the Origin Trilogy and open to further test; the present volume does not establish them.

14.4 Threats to the account

Several threats deserve naming. First, identifiability: Rung-2 conclusions drawn from observational data depend on a correct causal graph, and a wrong graph yields a confident wrong answer; the engine checks that an O2 claim offers identification, not that the identification is correct. Second, circularity in the grounding score: if

s is computed by the same framework whose output it gates, a self-consistent but mistaken analysis can score itself highly; the score's validity is therefore a matter for external calibration, not internal assertion. Third, operator-assignment reliability: deciding which operator a claim asserts can be a judgement, and inter-rater agreement on that assignment is a concrete, measurable target for validation. Fourth, the boundary with Axis 3: whether a defect is a missing causal warrant (Axis 2) or a false fact about the world (Axis 3) is not always sharp. Fifth, cross-lingual and cross-domain transfer of the operator vocabulary is untested. And the recurrence of six, here as elsewhere, is a chosen resolution. The specification is offered as a stable but revisable v1.0, not a closed result.

14.5 Two further exposures

Two further exposures deserve explicit mention. The first is operational: acting before the decode requires that the grounding score be computable within the latency budget of generation, and that the contraction of the probability vector be realizable without retraining the model. Both are engineering claims to be demonstrated, not assumed, and are the subject of the external-verification volume. The second is the elicitation of causal structure: every O2 and O3 verdict presupposes that the relevant causal graph can be stated, yet in open-domain language the graph is rarely given and must be inferred, which reintroduces judgement at exactly the point the engine aims to remove it. The honest position is that the engine converts a vague reliability question into a sharp, localized one — is this operator warranted, against this stated structure? — without claiming to answer it automatically in every case.

Appendix A — Normative Reference Tables

The tables below are normative. In case of conflict between prose and table, the table governs; in case of conflict with the Unified Structural Coordinate Table, that table governs.

A.1 Inference & Control Operators (the third axis)

Op.	Name	Lineage	Formal content
01	Conditioning	Statistics / Pearl	$P(Y X)$ — association (Rung 1)
02	Intervention / do	Pearl	$P(Y do(X))$ — intervention (Rung 2)
03	Counterfactual	Pearl	$P(Y_x X', Y')$ — counterfactual (Rung 3)
04	Reference Signal	Powers (PCT)	reference invariant r
05	Comparator / Error	Powers (PCT)	control error $e = r - p$
06	Output Attenuation	HNS Engineering	output = $g(s) \cdot raw$, $g(0)=0$, g monotone, $g(s) \in [0,1]$

A.2 The structural nesting

Layer	Construction	Cardinality
HNS-36	Natural Layers (6) × Abstract Categories (6)	36
HNS-144	HNS-36 × Onto-Modal Quadrants (4)	144
HNS-864	HNS-144 × Inference & Control Operators (6)	864

Note. The Social Meta Layers (SMS-6) of Volume 4 are Axis 3 and are orthogonal to this nesting: they ground and regulate output, they do not multiply the coordinate.

Appendix B — Configuration Template and Worked Configurations

A configuration is documented in the form below.

- Configuration: $(L_i \times C_j \times Q_k \cdot O_m)$.
- Asserted operator: the operation the claim performs.
- Supported operator: the strongest operation its evidence sustains.
- Verdict: pass if asserted $\leq$ supported, else flag (Unsupported Causality) and attenuate.

B.1 ($L6 \times C5 \times Q3 \cdot O2$) — a policy claim

Claim: “raising the rate reduces inflation.” Configuration: macro-social (L6), transformation (C5), objective-contingent (Q3), asserting intervention (O2). Supported operator: O2 only if an identification or experiment is offered; otherwise O1. Verdict: pass with identification; otherwise flag.

B.2 ($L3 \times C2 \times Q3 \cdot O1$) — a predictive claim

Claim: “users who hesitate here tend to abandon the form.” Configuration: cognitive (L3), process (C2), objective-contingent (Q3), asserting association (O1). Supported operator: O1 from the observed rates. Verdict: pass — the asserted operator does not exceed the support.

B.3 ($L4 \times C5 \times Q3 \cdot O3$) — an attribution

Claim: “had the review step not run, the error would have shipped.” Configuration: operational (L4), transformation (C5), objective-contingent (Q3), asserting counterfactual (O3). Supported operator: O3 only with a structural model; commonly only O1/O2 is available. Verdict: flag unless the model is supplied.

B.2 Further worked configurations

These configurations are non-normative and illustrate the template across the coordinate.

Claim: “smoking causes lung cancer.” Configuration: a physical-

substrate mechanism read as a transformation, objective and contingent, asserting intervention — $(L1 \times C5 \times Q3 \cdot O2)$. Supported operator: O2, by extensive identification and experiment. Verdict: pass; the rung is earned.

Claim: “the market rose because investors felt optimistic.” Configuration: a macro-social process, objective and contingent, asserting intervention — $(L6 \times C2 \times Q3 \cdot O2)$. Supported operator: at most O1, a co-movement of sentiment and prices. Verdict: flag; supply identification or retreat to association.

Claim: “by definition, a prime has exactly two divisors.” Configuration: a schema, objective and necessary — $(L4 \times C6 \times Q4)$. No inference operator applies; the claim is definitional. Verdict: pass as Q4; a causal reading would be a category error.

Claim: “users prefer the blue button.” Configuration: a micro-social state, subjective and contingent — $(L5 \times C4 \times Q1 \cdot O1)$. Supported operator: O1 within the subjective frame, a measured preference. Verdict: pass; an O2 reading (“blue causes preference”) would cross the domain boundary.

Claim: “had we shipped a day earlier, the outage would not have happened.” Configuration: an operational transformation, objective and contingent, asserting counterfactual — $(L4 \times C5 \times Q3 \cdot O3)$. Supported operator: O3 only with a structural model of the failure; usually only O1 or O2 is available. Verdict: flag unless the model is supplied.

Claim: “the neuron fires when its input exceeds threshold.” Configuration: a cellular process, objective and contingent, asserting a mechanism — $(L2 \times C2 \times Q3 \cdot O2)$. Supported operator: O2, a well-characterized mechanism. Verdict: pass.

Appendix C — Worked Examples

These examples are non-normative and illustrate the engine end to end.

C.1 Catching an over-claim

A system produces: “Teams that adopted the tool shipped faster, so the tool makes teams faster.” The engine positions the claim at (L6 × C5 × Q3), extracts an asserted O2, finds only O1 support (an adoption-speed correlation, with faster teams more likely to adopt), and flags Unsupported Causality. The grounding score falls; the attenuation map scales down the causal phrasing; the logged record shows asserted O2, supported O1, verdict flag.

C.2 Attenuation at the boundary

A candidate continuation asserts a confident counterfactual with no model and no data — a structurally ungrounded claim, $s = 0$. By Zero-Incidence Attenuation the output is set to zero: the continuation is not emitted at all, and the record notes the zero score. A neighbouring continuation that retreats to the supported associational claim scores near one and passes essentially unattenuated.

C.3 Under-claiming

Operator accuracy has a second, milder failure. A system answers “does the new compiler flag reduce bugs?” with “projects using it report fewer bugs” — an O1 statement — when a randomized rollout was in fact available that licenses the O2 claim “the flag reduces bugs.” Asserted O1, supported O2: the system declined a conclusion it had earned. The engine records the under-claim. It is not attenuated, since nothing ungrounded was emitted, but it is logged, because a verifier that silently wastes warrant is, over time, as unhelpful as one that over-claims.

C.4 A composite sentence

A single sentence often mixes rungs: “the popular new policy will obviously cut crime.” Decomposed, it is a subjective-contingent claim about popularity (Q1, O1), an objective-contingent prediction about crime (Q3, asserting O2), and a modal overreach in the word

“obviously.” The engine maps the parts separately, passes the popularity and the hedged prediction, and attenuates the causal-necessity overreach — accepting the defensible components while damping the one that is unearned, rather than passing or rejecting the sentence whole.

C.5 A full pass, end to end

A system is asked “why did revenue fall?” and answers “customers left because the price rose.” The engine decomposes the answer into an attrition claim and a causal attribution. It positions the attribution at ($L6 \times C5 \times Q3$), extracts asserted O2 from “because ... rose,” and looks for warrant: was price changed experimentally, or is there an identification adjusting for the confounders — seasonality, a competitor’s entry — that move both price and attrition? None is offered. The supported operator is O1. The support margin for the causal operator is low, the grounding score falls, and the linear map contracts the causal continuation below the grounded alternative, “revenue fell while price rose; the two moved together.” The log records the configuration, asserted O2, supported O1, the score, the attenuation, and the confounding sub-form. The author, seeing the flag, can supply the identification or accept the hedged statement; either way the over-claim does not ship unexamined.

Appendix D — Do-Calculus and a Worked End-to-End Pass

D.1 The three rules, restated

The do-calculus operates on a causal graph G and on graphs obtained by surgery: deleting arrows into a set (written with an overline) or out of it (written with an underline). Rule 1 (insertion / deletion of observations) holds when the outcome is independent of the observed set given the rest in the intervened graph. Rule 2 (action / observation exchange) replaces $\text{do}(X)$ by conditioning on X when X is independent of the outcome, in the graph with arrows out of X removed, given the adjustment set. Rule 3 (insertion / deletion of actions) removes $\text{do}(X)$ entirely when X is independent of the outcome in the graph with arrows into X removed, given the rest. Their repeated application is sound and complete for identification.

D.2 A full pass

Consider the output: “Students who used the tutoring app scored higher, so the app raises scores.” The engine positions the claim at $(L6 \times C5 \times Q3)$ — an institutional transformation, objective and contingent. It extracts the asserted operator $O2$ from “raises.” It asks for identification: is there a back-door set blocking the obvious confounder — that more motivated students both adopt the app and score higher? None is offered, and no front-door mediator is named. The supported operator is therefore $O1$. Since asserted $O2 \succ$ supported $O1$, the support margin for the causal operator is low, say $a = 0.2$; the grounding score $s = 0.2$; and the linear attenuation $g(s) = 0.2$ contracts the causal continuation’s mass below the grounded alternative — “using the app is associated with higher scores.” The log records: configuration $(L6 \times C5 \times Q3 \cdot O2)$; asserted $O2$; supported $O1$; $s = 0.2$; verdict flag; attenuation 0.2. A reader can reconstruct the verdict without access to the model that produced the sentence — the property the architecture exists to guarantee.

D.3 A multi-step reduction (sketch)

Where a single adjustment does not suffice, identification proceeds by repeated rewriting. Consider the front-door structure $X \rightarrow M \rightarrow Y$ with

an unobserved confounder between X and Y . The effect $P(Y \mid \text{do}(X))$ cannot be obtained by back-door adjustment, because the confounder is unobserved. The do-calculus reaches it in two moves: first identify $P(M \mid \text{do}(X)) = P(M \mid X)$, since no unblocked back-door path connects X and M ; then identify $P(Y \mid \text{do}(M))$ by adjusting for X , which blocks the back-door path from M to Y through the confounder. Composing the two and marginalizing over M yields the front-door formula of Chapter 5.6, now seen as the result of a two-step reduction rather than a standalone trick. The engine does not perform the reduction; it requires that a claim invoking the front-door route name the mediator and assert the shielding the route requires.

Appendix E — Notation Summary

The symbols used in this volume, collected for reference.

Symbol	Meaning
$(L_i \times C_i \times Q_k \cdot O_m)$	a full 864 analytical configuration
$P(Y \mid X)$	conditioning / association (O1)
$P(Y \mid \text{do}(X))$	interventional distribution (O2)
$P(Y_x \mid X', Y')$	counterfactual quantity (O3)
$\text{do}(X = x)$	the intervention setting X to the value x
$M = \langle U, V, F, P(U) \rangle$	a structural causal model
r, p, e	reference, perception, control error ($e = r - p$)
k	loop gain (the loop is stable for $0 < k < 2$)
s	structural grounding score, $s \in [0, 1]$
$g(s)$	attenuation map; $g(0) = 0$, monotone, $g(s) \in [0, 1]$
$O1 < O2 < O3$	the strict rung order of the inference operators
PN, PS	probabilities of necessity and sufficiency

Appendix F — Casebook

Twelve non-normative cases, chosen to cover the distinct ways a claim can pass or fail at the analytical resolution. Each gives the output, its configuration, the verdict, and the correction.

F.1 Confounded policy claim. “The new curriculum raised test scores.” Configuration ($L6 \times C5 \times Q3 \cdot O2$). Schools that adopted it differ in funding and intake, an unblocked back-door. Supported: O1. Verdict: flag (confounding). Correction: name an adjustment set, or retreat to “adoption is associated with higher scores.”

F.2 Collider / selection. “Among admitted patients, smokers had milder disease, so smoking protects.” Configuration ($L1 \times C5 \times Q3 \cdot O2$). Conditioning on admission — a common effect of smoking and severity — induces the association. Verdict: flag (collider). Correction: do not condition on admission; estimate in the unselected population.

F.3 Reverse causation. “Umbrella sales predict rain, so selling umbrellas brings rain.” Configuration ($L6 \times C5 \times Q3 \cdot O2$). The association is real; the arrow runs from rain to sales. Verdict: flag (direction). Correction: fix the direction with temporal or interventional evidence.

F.4 Licensed experiment. “In the randomized trial, the drug lowered mortality.” Configuration ($L1 \times C5 \times Q3 \cdot O2$). Randomization severs the arrows into treatment. Supported: O2. Verdict: pass.

F.5 Counterfactual without a model. “But for the delay, the deal would have closed.” Configuration ($L6 \times C5 \times Q3 \cdot O3$). No structural model supports the abduction. Supported: at most O1. Verdict: flag (severe). Correction: build the model or state the modest associational claim.

F.6 Licensed counterfactual. “In the validated simulator, had the valve stayed open, pressure would not have spiked.” Configuration ($L1 \times C5 \times Q3 \cdot O3$). A specified mechanism supports abduction-action-prediction. Supported: O3. Verdict: pass, with the model logged as the load-bearing premise.

F.7 Under-claim. “Projects using the flag report fewer bugs,” when a randomized rollout licenses the causal claim. Configuration ($L4 \times C5$

× Q3 · O1). Asserted O1, supported O2. Verdict: logged under-claim (not attenuated; warrant wasted). Correction: state the earned causal claim.

F.8 Definition read causally. “Being even causes a number to be divisible by two.” Configuration (L4 × C6 × Q4). The relation is definitional, not causal. Verdict: flag (category error). Correction: state it as a definitional equivalence, not an intervention.

F.9 Subjective claim, causal phrasing. “Liking the design makes it good.” Configuration (L5 × C4 × Q1). A preference (subjective) is cast as producing an objective property. Verdict: flag (domain boundary). Correction: separate the subjective preference from any objective criterion of quality.

F.10 Transport beyond support. “The effect found in the trial will hold in this very different population.” Configuration (L6 × C5 × Q3 · O2). The identifying assumptions may not transfer. Verdict: flag (extrapolation). Correction: state and defend a transport assumption, or restrict the claim to the studied population.

F.11 Composite sentence. “The popular reform will obviously cut crime.” Decomposes into a Q1/O1 popularity claim, a Q3/O2 crime prediction, and a modal overreach. Verdict: pass the parts that hold, attenuate the unearned causal-necessity. Correction: assert the components separately.

F.12 Over-adjustment. “Adjusting for post-treatment engagement, the campaign had no effect.” Configuration (L6 × C5 × Q3 · O2). Engagement is a mediator on the causal path; conditioning on it removes part of the effect. Verdict: flag (over-adjustment). Correction: do not adjust for mediators when estimating a total effect.

F.13 Instrumental variable. “Using distance to college as a lever, schooling raises earnings.” Configuration (L6 × C5 × Q3 · O2). An instrument is licensed only if it affects the outcome solely through the cause and is itself unconfounded. Verdict: pass only if those assumptions are stated; otherwise flag.

F.14 Front-door mediation. “The ad raised sales through store visits, though buyer wealth is unobserved.” Configuration (L6 × C5 × Q3 · O2). A front-door claim, licensed if visits fully mediate and are

shielded from the confounder. Verdict: pass with the mediator and shielding declared.

F.15 Base-rate neglect. “The test is 99% accurate and positive, so the disease is almost certain.” Configuration ($L4 \times C5 \times Q3 \cdot O1$). This is a conditioning error, not a causal one: it ignores the prior. Verdict: flag ($O1$ mis-computed); apply Bayes’ rule with the base rate.

F.16 Selection on the outcome. “Successful founders dropped out, so dropping out causes success.” Configuration ($L6 \times C5 \times Q3 \cdot O2$). Sampling on success is collider selection. Verdict: flag; estimate over all founders, not the successful ones.

F.17 Necessity versus sufficiency. “The spark caused the fire.” Configuration ($L1 \times C5 \times Q3 \cdot O3$). A but-for claim; fuel and oxygen were also necessary. Verdict: pass as a contributing necessary cause only if the model supports the probability of necessity; do not read sufficiency into it.

F.18 Mechanism at the substrate. “Raising temperature increases the reaction rate.” Configuration ($L1 \times C5 \times Q3 \cdot O2$). A well-characterized physical mechanism. Supported: $O2$. Verdict: pass.

F.19 Meta over-generalization. “The model with more parameters is always better.” Configuration ($L4 \times C6 \times Q4$). A meta-level claim asserted as necessary; it is false as stated and is not a causal operation. Verdict: flag (meta over-generalization); restate with conditions.

F.20 Correct ascent via mediation. “The drug lowers mortality by lowering blood pressure, shown in trial and mechanism.” Configuration ($L1 \times C5 \times Q3 \cdot O2$). Both the total effect and the mediating mechanism are evidenced. Verdict: pass; the mediation claim is licensed.

Glossary

Analytical configuration — A four-tuple (Layer $\times$ Category $\times$ Quadrant $\cdot$ Operator); the unit on which the 864 engine acts.

Conditioning (01) — The associational operator, $P(Y | X)$; the first rung of the causal hierarchy.

Intervention / do-operator (02) — The interventional operator, $P(Y | \text{do}(X))$; the effect of setting rather than observing X .

Counterfactual (03) — The operator of what-would-have-been, $P(Y_x | X', Y')$; the third rung, requiring a structural causal model.

Structural causal model — Functions relating each variable to its parents plus a distribution over exogenous noise; the object counterfactuals require.

Back-door criterion — A graphical condition on a set Z that licenses identifying $P(Y | \text{do}(X))$ by adjustment.

Causal Hierarchy Theorem — Pearl's result that the three rungs do not collapse: higher-rung facts are not determined by lower-rung data alone.

Reference signal (04) — The externally set target value of a controlled perception; the grounding target.

Comparator / control error (05) — The element forming $e = r - p$ and the signed gap it produces.

Predictive coding — A scheme minimizing prediction error / free energy; an admissible alternative formalization of the comparator.

Output attenuation (06) — The actuator output $= g(s) \cdot \text{raw_output}$, with $g(0)=0$, g monotone, $g(s) \in [0,1]$.

Grounding score (s) — A value in $[0,1]$ measuring how positioned and operator-supported a candidate output is; 1 at zero control error.

Zero-Incidence Attenuation — The property that $s = 0$ forces output $= 0$: a fully ungrounded output is not emitted.

Unsupported Causality — Asserting an operator above the rung the evidence supports; the chief failure of the analytical resolution.

Operator accuracy — The requirement that the correct operator be applied to the correct claim; over-claiming and under-claiming are its two errors.

Pre-decode — Acting on the generative probability or logit vector before token selection, so attenuation shapes generation.

d-separation — A graphical criterion for reading conditional independence off a causal graph; the basis of identification.

Confounder — A common cause of X and Y that biases the observed association as an estimate of effect.

Collider — A common effect of X and Y; conditioning on it induces a spurious association (selection bias).

Front-door criterion — An identification route through an unconfounded mediator when no back-door set is available.

Exogenous / endogenous — Background variables (U) set outside the model versus variables (V) determined within it.

Loop gain (k) — The factor governing how fast the control error decays; stable for $0 < k < 2$.

Probability of necessity / sufficiency — Counterfactual measures (PN, PS) of but-for and bring-about causation; often only bounded from data.

References

- Amari, S. (2016). *Information Geometry and Its Applications*. Springer.
- Friston, K. (2010). The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience*, 11(2), 127–138.
- Habermas, J. (1984). *The Theory of Communicative Action*, Vol. 1. Beacon Press.
- Hara, S. (2026). *The HNS Origin Trilogy*. Natural Structure Works.
- Hara, S. (2026). *Human Natural Structure: Foundational & Observational Specification*. Natural Structure Works.
- Kahneman, D. (2011). *Thinking, Fast and Slow*. Farrar, Straus and Giroux.
- Miller, E. K., & Cohen, J. D. (2001). An integrative theory of prefrontal cortex function. *Annual Review of Neuroscience*, 24, 167–202.
- Pearl, J. (2009). *Causality: Models, Reasoning, and Inference* (2nd ed.). Cambridge University Press.
- Pearl, J., & Mackenzie, D. (2018). *The Book of Why*. Basic Books.
- Powers, W. T. (1973). *Behavior: The Control of Perception*. Aldine.
- Rao, R. P. N., & Ballard, D. H. (1999). Predictive coding in the visual cortex. *Nature Neuroscience*, 2(1), 79–87.
- Smith, B. (2015). *Basic Formal Ontology (BFO) 2.0 Specification and User's Guide*.
- Spirtes, P., Glymour, C., & Scheines, R. (2000). *Causation, Prediction, and Search* (2nd ed.). MIT Press.
- Wiener, N. (1948). *Cybernetics: or Control and Communication in the Animal and the Machine*. MIT Press.

Human Natural Structure

Social Meta-Structure Specification

The Six-Layer Grounding Stack from Body to Civilization

Volume 4 · Human Natural Structure Series

Satoru Hara

Natural Structure Works (NSW)

A Note on Method and Evidence

This volume specifies the grounding stack of Human Natural Structure: the six Social Meta Layers that constitute Axis 3, the axis on which a generated output is checked against the structure of the human world, from the body to the civilization in which it acts. As in the preceding volumes, its claims fall into three registers, and the reader is entitled to know which is which.

Established. The traditions the stack draws on are real and well developed: embodied cognition and the symbol-grounding problem, the theory of communicative action, institutional theory, general systems theory and cybernetics, predictive coding and the free-energy principle, and perceptual control theory. Where this volume reports their content it reports established material the reader can verify in the sources cited.

HNS design. The arrangement of these traditions into a six-layer grounding stack, the composite grounding function, and the use of the stack as an external grounding check that feeds the control loop of the analytical engine are engineering decisions of HNS. They are internally well defined and, I argue, useful; they are not the only possible decomposition of social grounding.

Conjecture. That checking outputs against these six layers measurably improves the social alignment of a deployed system is a hypothesis, motivated and partially supported but not established here, and marked as conjecture where it appears.

A candid limitation, stated up front

This volume owes the reader an unusually direct caveat, because the stack it specifies is the least settled part of the framework. The six Social Meta Layers are a useful scaffold, not a proven taxonomy. Three weaknesses are acknowledged openly and revisited in the final chapter. First, the stack mixes two organizing principles: a scale (from the somatic body up to the civilization) and a set of domains (communicative, organizational, economic, normative). These do not cleanly coincide, and the ordering should be read as broadly scale-like rather than strictly monotone. Second, the layers are not mutually

exclusive: the economic and the governance layers overlap, as do the protocol and the organizational, and the failure they suppress is in one case shared. Third, the number six is a chosen resolution that keeps this axis aligned with the rest of the framework; it is not derived, and a reader could propose a finer or coarser stack without being wrong. The stack is offered as a working structure that has proven useful, not as a closed result, and the lineage column throughout indicates inspiration, not derivation.

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Preface: From the Coordinate to the Ground

The first three volumes (Volume 1, 2, and 3) built a coordinate and an engine that act on the structure of a claim.. The foundational and observational volumes position a claim and observe its modal stance; the analytical volume audits the inferential and control operation it performs. All of this is internal in a precise sense: it asks whether a claim is well formed and whether its operation is warranted, not whether the world it speaks of is as it says. A claim can be perfectly positioned, correctly observed, and analytically licensed, and still be socially wrong — ungrounded in the body that must act on it, the dialogue that must receive it, the organization and economy it moves through, the law it must obey, and the wider system it must not destabilize.

Axis 3 is the axis of that grounding. Where Axis 1 proposes and Axis 2 constrains, Axis 3 grounds: it checks a candidate output against the structure of the human world at six scales, from the somatic to the universal. This volume specifies that check. It is not a further subdivision of the 36/144/864 coordinate — it does not multiply the cells — but a parallel axis, orthogonal to the coordinate, that supplies the grounding the engine needs in order to act. Concretely, the grounding score the analytical engine's attenuation operator consumes, and the reference its control loop is held to, are produced here.

The stack is read from the body outward. The somatic layer asks whether an output coheres with the physical and ergonomic facts of an embodied agent. The protocol and organizational layers ask whether it coheres with communicative and organizational scope. The economic and governance layers ask whether it respects resource and value equivalence and the norms and laws of its institutions. The universal layer asks whether it keeps faith with the coherence of the wider system it belongs to. An output grounded at all six is, in the sense this volume makes precise, socially aligned.

A reader who wants the normative core may read Chapters 3 through 10 with Appendix A; the surrounding chapters supply motivation, lineage, application, and — with more candor than elsewhere in the

series — the limits of what is here proposed..

Part I

Foundations of the Grounding Stack

1. Axis 3 in the HNS Full Stack

The Social Meta Layers are the third axis of the Multi-Axis Verification account. This chapter fixes their place in the full stack and states the one structural fact that governs everything else: Axis 3 is orthogonal to the coordinate of Axis 2.

1.1 The three axes

Axis 1 (generative) proposes a continuation; it is the model's native fluency. Axis 2 (structural / causal) is the 36/144/864 coordinate of the first three volumes; it positions a claim, observes its modality, and audits the operation it performs. Axis 3 (grounded / executive) is the stack specified here; it checks a candidate output against the structure of the human world. Reliability, on the MAV account, is the conjunction of the three: an output must be fluent, structurally warranted, and grounded. No axis suffices alone, and the axes are approximately independent, which is why their combination adds information that none supplies by itself.

1.2 Orthogonality: Axis 3 is not a multiplicative factor

It must be stated plainly, because the cardinalities of the earlier volumes invite the wrong inference. The Social Meta Layers do not multiply the coordinate. There is no 864×6 space. The layers are a parallel grounding axis: they evaluate an output that has already been positioned and analysed, and return a grounding verdict, rather than subdividing the positions. The coordinate says what a claim is and what operation it performs; the stack says whether the resulting output is grounded in the world. The two are composed at runtime, not multiplied in cardinality.

*Axis 1 (propose) → Axis 2 (position, observe, analyse) → Axis 3
(ground) → emit*

1.3 What Axis 3 supplies to Axis 2

The connection to the analytical engine is concrete. The engine's output-attenuation operator scales a candidate by a structural grounding score; part of that score is the social grounding computed here. And the engine's control loop holds a perceived grounding to a

reference; the reference, and the higher loops that set it, are supplied by this stack. In the language of perceptual control, the Social Meta Layers are the higher controllers whose outputs become the references of the structural loop below. Axis 3 is therefore not a passive checker but the source of the targets the rest of the system controls toward.

2. Society-Aligned Artificial Intelligence

This chapter states what the stack is for: a definition of social alignment stronger than value alignment, and the six social error domains the stack is built to catch.

2.1 The grounding function

A socially aligned output is defined as one that is grounded at every layer of the stack. Writing G for the composite grounding function and g_i for the grounding verdict at layer i :

$$G(x) = 1 \text{ if and only if } g_i(x) = 1 \text{ for every layer } i = 1 \dots 6$$

An output that fails any single layer is socially misaligned on that layer, whatever its merits elsewhere. The conjunction is deliberate: groundedness in dialogue does not excuse a violation of law, and economic coherence does not excuse a physical impossibility. The stack is demanding by construction, and the next chapters specify each conjunct.

2.2 Society alignment versus value alignment

Society alignment differs from the more familiar value alignment. Value alignment asks whether an output matches individual human preferences or principles. Society alignment asks whether it respects the structural properties of the social systems in which it acts — not merely what individuals prefer, but what their institutions, norms, and shared infrastructures require. It is the stronger requirement, and it is the one that current governance frameworks implicitly demand — lawful, non-destabilizing, institutionally appropriate output — without specifying it technically. The stack is an attempt at that specification.

2.3 The six social error domains

Each layer corresponds to a way an output can be socially wrong while being individually coherent. A somatic failure violates the physical and ergonomic facts of an embodied agent. A protocol failure violates communicative alignment. An organizational failure violates the scope of the group or institution it addresses. An economic failure

violates resource or value equivalence. A governance failure violates a norm or a law. A universal failure violates the coherence of the wider system. These are the targets; the layers are the checks.

3. The Six Social Meta Layers

This chapter presents the stack as a whole; Part II takes each layer in turn. The table is normative and is reproduced in Appendix A.

	Layer	Grounds	Suppresses	Lineage
SMS-1	Somatic	Bodily / ergonomic coherence	Metaphor Contamination	Barsalou; Harnad
SMS-2	Protocol	Communicative alignment	Scope Drift	Habermas
SMS-3	Organizational	Organizational scope	Scope Drift	HNS Extension
SMS-4	Economic	Resource / value equivalence	Unsupported Causality	HNS Extension
SMS-5	Governance	Normative / legal conformance	Category Ambiguity	Institutional theory
SMS-6	Universal	Global systemic coherence	Layer Jump	Ashby; GST

***Note.** Naming. “Social Meta Layers” is the axis (the six layers); “Social Meta Structure (HNS-SMS-6)” is this specification. “Social Meta” is the umbrella for the whole stack and does not assert that every layer — notably the somatic base — is itself social. SMS-3 was formerly labelled “Ecosystem”; it is renamed “Organizational” because its content is organizational scope, which the earlier label obscured.*

3.1 Reading the stack from the body outward

The layers are ordered, broadly, by scale: from the individual body (SMS-1), through dyadic and organizational interaction (SMS-2, SMS-3), to markets and institutions (SMS-4, SMS-5), to the civilizational whole (SMS-6). The ordering is a reading aid, not a strict hierarchy; as the opening note warned, scale and domain do not perfectly coincide, and the layers overlap. What the ordering does capture is the widening of the context an output must cohere with, from the agent that utters it to the system that must absorb it.

3.2 A legacy grouping

An earlier presentation grouped the six into three architectural sub-layers — AnthroDesign (human-centred design), Synaptics (alignment and correction), and NormaSphere (norms and

institutions). That grouping is retained here only as a presentational convenience; the normative content is the flat set of six layers, and conformance is defined against the six, not the three.

3.3 The somatic scope note

Because the umbrella term is “social,” the somatic layer deserves a word. SMS-1 is not a social layer; it is the embodied, physical base on which the social layers rest. It is included in the stack because grounding begins in the body — an output that is physically or ergonomically impossible is ungrounded before any social question arises — not because the body is itself a social institution. The stack is “social meta” as a whole; its base is somatic.

Part II

The Grounding Layers in Detail

3.4 One output through all six layers

To see the stack work, read one output through all six. “Tell every team to cut their cloud spend by 30% this quarter to hit the savings target.” SMS-1: physically possible — passes. SMS-2: clear and on-topic — passes. SMS-3: does the speaker have authority to direct every team? — flags unless mandated. SMS-4: is a 30% cut equivalent to the target, and what capability does it cost? — flags until the balance is shown. SMS-5: does any contract or policy fix the spend? — conditional. SMS-6: if every team cuts identically, could a shared service fall below viability? — flags the systemic risk. The composite is ungrounded until the authority, the cost equivalence, and the systemic effect are addressed — a single output, six independent checks, one conjunctive verdict.

4. SMS-1 — Somatic

The first layer is the body. It grounds an output in the physical and ergonomic facts of an embodied agent acting in a physical environment, and it suppresses the failure of Metaphor Contamination — the drift in which a figure of speech is treated as a literal mechanism.

4.1 What it grounds

Somatic grounding asks whether an output is compatible with the body that must enact or receive it: with reach, force, timing, perception, fatigue, and the physical constraints of the tools and spaces involved. An instruction that ignores these — that asks for an impossible posture, an unattainable rate, a perception the senses cannot deliver — is ungrounded at SMS-1 however fluent and well intentioned it is.

4.2 Lineage

The layer draws on embodied or grounded cognition (Barsalou) and on the symbol-grounding problem (Harnad): the recognition that symbols acquire meaning by connection to sensorimotor experience, not by relation to other symbols alone. An ungrounded symbol, manipulated only against other symbols, is exactly what produces metaphor mistaken for mechanism. SMS-1 is the requirement that the chain of meaning touch the body somewhere.

4.3 The failure it suppresses

Metaphor Contamination is the treatment of a metaphor as if it were a literal causal claim — “momentum” in markets read as Newtonian inertia, “memory” in a system read as recollection. By requiring a somatic anchor, SMS-1 flags claims whose only support is a bodily metaphor doing illicit literal work, and asks that the literal mechanism be stated or the figure be marked as a figure.

4.4 A worked grounding (non-normative)

An assistant tells a warehouse worker to “keep lifting at the same pace through the double shift.” Positioned and analysed, the instruction

may be perfectly well formed. At SMS-1 it fails: sustained lifting at a fixed pace ignores fatigue and injury limits of the body that must perform it. The grounding verdict $g_1 = 0$, and the output is attenuated or revised toward a physically sustainable instruction.

4.5 Boundary with the layers above

The somatic layer grounds in the body, not in any social relation, and that is what distinguishes it from every layer above. An output can be somatically grounded and socially ungrounded — physically possible yet communicatively or legally wrong — and the reverse. The layer's characteristic confusion is with metaphor: when a bodily figure such as “grasp,” “weight,” or “pressure” is read literally, the check at SMS-1 asks whether a real body is involved or only its image.

5. SMS-2 — Protocol

The second layer is communication. It grounds an output in the conditions of successful communicative exchange and suppresses Scope Drift — the silent broadening or shifting of what a claim is about. It anchors, in the Natural Layers of Axis 2, the micro-social layer L5.

5.1 What it grounds

Protocol grounding asks whether an output is intelligible and appropriate as a communicative act between agents: whether it can be understood by its addressee, whether it raises claims it is prepared to redeem, and whether it stays within the topic and commitments of the exchange. An output that quietly changes the question, over-claims what was asked, or addresses the wrong interlocutor fails here.

5.2 Lineage

The layer draws on Habermas’s theory of communicative action, in which an utterance oriented to understanding implicitly raises validity claims — to truth, to normative rightness, and to sincerity — that the speaker must be ready to redeem in dialogue. SMS-2 is the requirement that an output be defensible as such an act: that its implicit claims be ones the system could, if challenged, make good.

5.3 The failure it suppresses

Scope Drift is the unannounced movement of a claim’s scope — answering a narrower or wider question than was asked, or letting the subject slide across a conversation. SMS-2 suppresses it by holding the output to the communicative frame of the exchange. The suppression is shared with SMS-3: drift across an organizational boundary is caught at the organizational layer, drift within a dialogue at the protocol layer.

5.4 A worked grounding (non-normative)

Asked “is this contract clause enforceable in our jurisdiction?” a system answers with a general essay on contract law. Communicatively, it has drifted: it answers a broader question than the one posed and leaves the addressee’s actual question unaddressed.

$g_2 = 0$; the output is redirected to the specific clause and jurisdiction asked about.

5.5 Boundary with the organizational layer

Protocol and organizational grounding are easy to conflate, because a dialogue is often also an organizational act. The boundary is the kind of frame violated: SMS-2 concerns the communicative frame between the parties to an exchange — intelligibility, topic, redeemable claims — while SMS-3 concerns the organizational frame of scope, role, and authority. The same utterance can pass one and fail the other: clear and well formed (SMS-2) yet beyond the speaker's mandate (SMS-3). The shared suppression of Scope Drift is exactly this overlap made visible.

6. SMS-3 — Organizational

The third layer is the organization. It grounds an output in the scope and structure of the group or institution it addresses, and it shares with SMS-2 the suppression of Scope Drift. Like the protocol layer, it anchors the micro-social layer L5 of Axis 2.

6.1 What it grounds

Organizational grounding asks whether an output respects the scope, roles, and authority structure of the organization in which it acts: whether it addresses the right level, stays within the mandate of the unit it speaks for, and does not assume coordination or authority that the organization does not grant. An output that issues an instruction beyond its remit, or assumes a coordination the organization cannot deliver, fails here.

6.2 Lineage — an HNS extension

This layer is, candidly, an HNS extension with no single external proponent. It draws loosely on organizational theory and on the distinction between individual interaction and collective structure, but it is not the faithful application of one named framework. It is included because organizational scope is a real and distinct ground of failure between the dyadic (SMS-2) and the institutional (SMS-5), and the stack would have a gap without it. Its status as a designed rather than inherited layer is marked here and in the lineage table.

6.3 The failure it suppresses

Organizational Scope Drift is the broadening of a claim across an organizational boundary — treating a team’s decision as the company’s, a department’s data as the whole organization’s, an advisory remark as an authoritative instruction. SMS-3 suppresses it by holding the output to the scope the organization actually grants. That Scope Drift is suppressed jointly by SMS-2 and SMS-3, rather than by one layer alone, is the clearest case in the stack of the failure-to-layer mapping being many-to-one.

6.4 A worked grounding (non-normative)

A drafting assistant turns one engineer's suggestion into "the company has decided to adopt" a standard. Organizationally, it has overstepped: a suggestion at one scope is asserted at a higher one the organization never authorized. $g_3 = 0$; the output is rescoped to the suggestion and the level that actually made it.

6.5 Boundary with the governance layer

Organizational and governance grounding both concern collective structure, but at different bindingness. SMS-3 concerns the scope and authority an organization grants in fact; SMS-5 concerns the norms and laws that bind it from outside. An action can be within organizational scope, so SMS-3 passes, yet unlawful, so SMS-5 fails; or lawful, so SMS-5 passes, yet beyond the unit's mandate, so SMS-3 fails. The two are checked separately for this reason.

7. SMS-4 — Economic

The fourth layer is the economy. It grounds an output in resource and value equivalence — the requirement that what is claimed to be exchanged, produced, or saved actually balances — and it suppresses Unsupported Causality at the level of economic mechanism.

7.1 What it grounds

Economic grounding asks whether an output respects conservation and equivalence: whether resources claimed to be created were in fact transferred from somewhere, whether costs and benefits balance, whether a proposed gain has a stated source. An output that promises value from nowhere — savings without a cost moved, output without an input — fails here, in the way a physical claim that violated conservation would fail at SMS-1.

7.2 Lineage — an HNS extension

Like the organizational layer, the economic layer is an HNS extension rather than the application of a single proponent's framework. It draws on the general intuition of equivalence and conservation that runs through economic thought, but it is a designed layer, included because resource and value coherence is a distinct ground of failure not covered by the layers around it. Its designed status is marked.

7.3 The failure it suppresses

At this layer, Unsupported Causality takes an economic form: the assertion that an action produces a value effect without an accounted mechanism — “this change will increase revenue” with no stated source of the increase and no cost it displaces. SMS-4 suppresses it by demanding the equivalence: where does the value come from, and what does it cost? The check complements the analytical engine's general treatment of Unsupported Causality with the specific demand for an economic balance.

7.4 A worked grounding (non-normative)

A plan claims that a new process will “save time for everyone with no added work.” Economically, the equivalence is missing: the saved time

must be paid for somewhere — setup, maintenance, a shifted burden. $g_4 = 0$ until the source and cost are stated; the output is revised to name where the saving comes from.

7.5 Boundary with causal analysis

Economic grounding is distinguished from the causal analysis of Axis 2 by its specific demand. The analytical engine asks whether a causal claim is warranted in general; SMS-4 asks the narrower, checkable question of equivalence — does the value balance? A claim can pass the general causal check and still fail SMS-4 if it conjures value without a source. The economic layer is, in effect, conservation applied to value, and that is what makes it mechanically checkable rather than a matter of judgement.

8. SMS-5 — Governance

The fifth layer is governance. It grounds an output in normative and legal conformance and suppresses Category Ambiguity — the confusion of one normative category with another. It anchors the macro-social layer L6 of Axis 2 and connects directly to Volume 6 (Standards & Conformance Specification).

13.1 Volume 6 develops this into a regulatory mapping.

8.1 What it grounds

Governance grounding asks whether an output conforms to the norms, rules, and laws of the institutions it acts within: whether it respects rights and obligations, stays within what regulation permits, and keeps normative categories distinct — advice from instruction, permission from requirement, recommendation from ruling. An output that issues as a requirement what is only a recommendation, or treats the permitted as the mandatory, fails here.

8.2 Lineage

The layer draws on institutional theory — the study of how norms, rules, and conventions constitute and constrain action. What institutional theory supplies is the recognition that conformance is structural, not merely preferential: institutions define categories of permitted, required, and forbidden, and a competent actor keeps these straight. SMS-5 is the requirement that an output keep them straight.

8.3 The failure it suppresses

Category Ambiguity, at the governance layer, is the blurring of normative categories — reading “may” as “must,” “guidance” as “law,” a private norm as a public rule. SMS-5 suppresses it by holding the output to the correct normative category, in the same spirit that the observational layer of Axis 2 holds a claim to a single modal quadrant. The two reinforce each other: a modal error at Q-level often surfaces as a governance category error here.

8.4 A worked grounding (non-normative)

A compliance assistant states that staff “must” take an optional training. The normative category is wrong: a recommendation is asserted as a requirement. $g_5 = 0$; the output is corrected to mark the training as recommended, not mandatory, unless a rule in fact requires it.

8.5 Boundary with the observational layer

Governance grounding and the observational layer of Axis 2 both police category. The observational layer keeps a claim to one modal quadrant — subjective or objective, necessary or contingent; SMS-5 keeps it to one normative category — permitted, required, or forbidden. The two often co-occur: a claim that slides from “may” to “must” is both a modal drift on Axis 2 and a governance category error on Axis 3. SMS-5 is the social face of the same discipline.

9. SMS-6 — Universal

The sixth layer is the system as a whole. It grounds an output in global systemic coherence — the requirement that an output not destabilize the wider system it belongs to — and it suppresses Layer Jump, the illegitimate leap across levels of organization.

9.1 What it grounds

Universal grounding asks whether an output is compatible with the optimality and stability of the whole: whether a locally sensible action would, if generalized, undermine the system that makes it sensible; whether it respects the constraints that keep the larger structure viable. An output that is coherent at every lower layer but corrosive in aggregate — a recommendation harmless once and ruinous universally — fails here.

9.2 Lineage

The layer draws on general systems theory and on Ashby's cybernetics — in particular the idea that a viable system maintains its essential variables within limits, and that a regulator must have variety enough to meet the disturbances it faces. This lineage was chosen over a scale-named alternative (“civilization studies”) deliberately: the lineage column should anchor the layer's function — systemic viability — rather than its scale. The universal layer is about keeping a system coherent, and systems theory is the study of exactly that.

9.3 The failure it suppresses

Layer Jump is the illegitimate transfer of a claim across levels of organization — reasoning from an individual case to a systemic law, or from a systemic regularity to an individual prescription, without the bridge each requires. SMS-6 suppresses it by checking an output against the coherence of the level it actually addresses and the level it would affect. It is the macro complement to the layer-adjacency discipline of the foundational volume.

9.4 A worked grounding (non-normative)

A system advises every client to exploit the same arbitrage. Locally rational, the advice is universally incoherent: if all act on it the arbitrage closes and the wider market is destabilized. $g_6 = 0$; the output is qualified to the conditions under which it remains systemically harmless.

9.5 Boundary with the layers below

Universal grounding is distinguished from every layer below by a quantifier. The lower layers ask whether this output is grounded; SMS-6 asks whether it would remain grounded if generalized across the system. An output can pass every lower layer for the individual case and fail SMS-6 in aggregate — the locally rational, globally ruinous recommendation. It is the only layer whose check ranges over the population of similar acts rather than the single act in hand.

Part III

The Grounding Function in Operation

10. The Grounding Function $G(x)$

This chapter specifies how the six per-layer verdicts compose into a single grounding result and how that result enters the runtime of the analytical engine.

10.1 Composition

Each layer returns a grounding verdict for a candidate output. In the strict form the verdict is binary, $g_i \in \{0, 1\}$, and the composite is their conjunction — the output is grounded only if grounded everywhere:

$$G(x) = g_1(x) \wedge g_2(x) \wedge \dots \wedge g_6(x)$$

In the graded form used by the runtime, each layer returns a margin $g_i \in [0, 1]$, and the composite is a conservative aggregate — the minimum, so that a single weak layer pulls the whole grounding down, or a weighted product when a softer combination is wanted:

$$s = \min_i g_i(x) \quad \text{or} \quad s = \prod_i g_i(x)^{w_i}, \quad \sum w_i = 1$$

Both aggregates are monotone non-decreasing in each layer's margin: improving grounding on any layer never lowers the composite. This is the property the analytical engine's attenuation operator relies on, and it mirrors that engine's own composition of operator-support margins.

10.2 Feeding the analytical engine

The composite grounding s produced here is the social component of the structural grounding score the analytical engine's output-attenuation operator consumes. Where that engine scales a candidate by $g(s)$, the s it scales by carries the verdict of this stack. The connection is exact: a low social grounding lowers s , and a low s attenuates the output, so a socially misaligned continuation is suppressed by the same mechanism that suppresses an analytically unsupported one. The stack diagnoses; the engine's actuator acts.

10.3 Setting references

The stack also supplies the references the engine's control loop is held to. In a control hierarchy the output of a higher loop sets the reference

of a lower one; here the Social Meta Layers are the higher loops, and the grounding target they define becomes the reference the structural loop controls toward. Governance and universal grounding, in particular, set references that the lower layers must not violate — a lawful, systemically coherent target that the rest of the system steers to meet.

10.4 A worked composite (non-normative)

Suppose an output returns per-layer margins $g_1 = 1.0$, $g_2 = 1.0$, $g_3 = 0.4$, $g_4 = 0.5$, $g_5 = 1.0$, $g_6 = 0.8$ — grounded somatically and communicatively, weak organizationally and economically. Under the conservative aggregate $s = \min = 0.4$ the weakest layer governs: the output is carried at four-tenths strength, so its organizationally over-scoped, economically unbalanced phrasing falls below grounded alternatives in the engine's pre-decode reweighting. Under a uniform-weight product, $s = (1 \cdot 1 \cdot 0.4 \cdot 0.5 \cdot 1 \cdot 0.8)^{1/6} \approx 0.74$, the same output is treated more leniently. The choice between the two is the stack's main tuning decision: the minimum refuses any single ungrounded layer, while the product tolerates one weak layer if the rest are strong. Both are monotone, so improving any layer never lowers s .

11. The Theoretical Basis of Grounding

The stack is not an arbitrary list of six concerns; it is built on a coherent picture of how an adaptive system stays coupled to its environment. This chapter states that picture and names its sources, with the usual caution that they are inspiration, not derivation.

11.1 Prediction, control, and free energy

Two convergent traditions underlie the stack. Predictive coding and the free-energy principle (Rao and Ballard; Friston) describe an agent that maintains itself by minimizing the discrepancy between its predictions and its sensed world, acting and updating so as to stay within viable bounds. Perceptual control theory (Powers) describes the same coupling as the control of perception toward references. The Social Meta Layers extend this coupling outward: the references an agent must meet are not only sensory but social, and grounding is the minimization of discrepancy against the structure of the human world at six scales.

11.2 Cybernetics and viability

The outermost layer rests on cybernetics and general systems theory (Wiener; Ashby). A viable system holds its essential variables within limits, and a regulator must command variety enough to counter the disturbances it meets. Universal grounding is this requirement applied to an AI's outputs: that they not drive the system they act in outside its viable range. The stack thus runs from the body, where grounding is sensorimotor, to the civilization, where grounding is systemic viability — one principle, coupling to environment, read at widening scales.

11.3 From body to civilization

The arc of the stack is deliberate. SMS-1 grounds in the body; SMS-2 and SMS-3 in interaction and organization; SMS-4 and SMS-5 in markets and institutions; SMS-6 in the civilizational whole. The claim is not that these scales are discrete or that exactly six exist — the candor of the opening note stands — but that grounding must be checked across this range, because an output can be coupled to the

body and uncoupled from the institution, or coherent in the institution and corrosive to the system. The stack is the attempt to check the whole range rather than one scale of it.

12. Failure Suppression and the Social Error Domains

This chapter states the mapping between the layers and the failures they suppress, and is explicit that the mapping is not one-to-one.

Layer	Social error domain	Suppressed failure
SMS-1 Somatic	Physical / ergonomic violation	Metaphor Contamination
SMS-2 Protocol	Communicative misalignment	Scope Drift (with SMS-3)
SMS-3 Organizational	Organizational over-scope	Scope Drift (with SMS-2)
SMS-4 Economic	Resource / value imbalance	Unsupported Causality
SMS-5 Governance	Normative / legal nonconformance	Category Ambiguity
SMS-6 Universal	Systemic incoherence	Layer Jump

12.1 The mapping is many-to-one

The clearest departure from a tidy correspondence is Scope Drift, suppressed jointly by the protocol and organizational layers. Drift within a dialogue is a protocol failure; drift across an organizational boundary is an organizational one; the same named failure is caught at two layers depending on where the boundary it crosses lies. The lesson generalizes: the layers are checks on grounding, and a single failure type can be ungrounded in more than one way. The mapping is offered as a guide to where to look, not as a partition.

12.2 Detection and response

At runtime each layer returns its margin; a margin below the layer’s reference is a grounding shortfall in that layer’s error domain. The shortfall lowers the composite grounding s, and the analytical engine’s attenuation acts on the lowered s. As elsewhere in the framework, detection is structural and the response is regulative, and both are external to the generating model and written to the audit record.

12.3 A combined detection (non-normative)

A single output can fail at more than one layer, and the stack records each. “Because customers love the new design, we must roll it out company-wide immediately, and it will pay for itself.” The claim fails at SMS-2 and SMS-3 (Scope Drift: a preference of some customers asserted as a company-wide mandate), at SMS-4 (Unsupported Causality: “pay for itself” with no stated source), and at SMS-5 (Category Ambiguity: “must” where only “may” is supported). The composite is low on three layers at once, and the record shows each, so the correction can be directed at each in turn — rescope the claim, state the economic balance, and fix the normative category.

Part IV

Application, Lineage, and Status

13. Applications

The stack is meant to be used. This chapter sketches three settings, briefly, with the caveat that each is an application of a proposed structure, not a validated result.

13.1 AI governance

For governance, the stack offers what process-based frameworks lack: a technical place to attach a conformance check. An output can be evaluated against the governance and universal layers for legal conformance and systemic coherence, and the per-layer verdict logged, so that “compliant” becomes a recorded, auditable property of each output rather than a property asserted of the system as a whole. The sixth volume develops this into a regulatory mapping.

13.2 Organizational and institutional use

Within an organization, the protocol and organizational layers check that an assistant’s outputs stay within communicative and organizational scope — that they address the right level, claim the right authority, and respect the mandate of the unit they speak for. This is the everyday face of the stack: most socially misaligned outputs in practice fail at SMS-2 or SMS-3, not at the dramatic extremes.

13.3 Civilization-scale considerations

At the largest scale, the universal layer asks the question that individual alignment cannot: not whether an output is good for its user, but whether it would remain coherent if generalized across the system. This is the layer that catches the locally rational, globally ruinous recommendation, and it is the layer most in need of the empirical work the final chapter calls for.

14. Intellectual Lineage and Proponents

This chapter records the debts and, with the candor the stack especially requires, distinguishes the inherited layers from the designed ones.

14.1 The inherited layers

SMS-1 draws on embodied cognition (Barsalou) and the symbol-grounding problem (Harnad). SMS-2 draws on Habermas’s communicative action. SMS-5 draws on institutional theory. SMS-6 draws on general systems theory and Ashby’s cybernetics. The theoretical basis of the whole — prediction, control, and viability — draws on Friston, Rao and Ballard, Powers, Wiener, and Ashby. Each is acknowledged as inspiration for a layer’s function, not as a framework the layer faithfully implements.

14.2 The designed layers

SMS-3 (Organizational) and SMS-4 (Economic) are HNS extensions with no single external proponent. They are included because organizational scope and economic equivalence are real and distinct grounds of failure that the inherited layers leave uncovered, but their inclusion is a design judgement, and they are the layers most open to revision. Marking them as designed rather than inherited is part of the honesty the stack is held to.

Layer	Source	Inherited or designed
SMS-1 Somatic	Barsalou; Harnad	inherited
SMS-2 Protocol	Habermas	inherited
SMS-3 Organizational	—	designed (HNS)
SMS-4 Economic	—	designed (HNS)
SMS-5 Governance	Institutional theory	inherited
SMS-6 Universal	Ashby; general systems theory	inherited

14.3 What is HNS’s own

The assembly of these six concerns into a single grounding axis, the composite grounding function, the two designed layers, and the use of the stack as the source of the grounding score and references that drive the analytical engine are HNS's own. The proponents named did not design the stack and should not be read as endorsing it; the recombination, and the responsibility for it, are the author's.

15. Empirical Status and Limitations

This volume closes with the most candid limitations chapter in the series, because the stack is the least settled part of the framework.

15.1 What is established, proposed, and conjectured

Established are the source traditions and the well-definedness of the grounding function: the conjunction and the monotone aggregates are elementary, and every per-layer verdict is externally auditable. Proposed, and defensible but unproven, are the six layers themselves, the two designed layers especially, and the claim that this is a good decomposition of social grounding. Conjecture are the practical claims — that checking against the stack measurably improves social alignment in deployment.

15.2 The stack is a scaffold, not a proven taxonomy

The candor promised at the outset is owed in full here. The stack mixes scale and domain: it reads as a progression from body to civilization, yet its middle layers are distinguished by domain — communicative, organizational, economic, normative — and these do not line up neatly along the scale. The layers are not mutually exclusive: the economic and governance layers overlap wherever law prices behaviour, and the protocol and organizational layers overlap wherever a dialogue is also an organizational act; the shared suppression of Scope Drift is the visible symptom. And there are gaps: an ecological layer (grounding in the non-human environment), an ethical layer distinct from the legal, and a temporal layer (grounding across time and future generations) all have a claim to inclusion that the present six do not honour. The number six is a chosen resolution that keeps this axis aligned with the rest of the framework, not a count derived from the structure of social reality.

15.3 The honest position

None of this voids the stack's usefulness. A scaffold that catches somatic impossibility, communicative and organizational over-scope, economic imbalance, legal nonconformance, and systemic incoherence catches most of the ways a fluent output is socially wrong,

and it does so in an external, auditable form. But it should be used as a working structure that has proven useful, and revised as evidence accumulates — a stable but explicitly revisable v1.0, more so than any other volume in the series. The threats most worth naming are the overlap and gap problems above, the difficulty of computing per-layer grounding reliably and within a latency budget, the inter-rater reliability of the grounding verdicts, and the boundary between a structural defect (Axis 2) and a grounding defect (Axis 3), which is not always sharp.

15.4 The known gaps

Three omissions are named so that they are not mistaken for oversights. An ecological layer — grounding in the non-human environment, the resource and sink limits of the natural world — is absent; the economic layer touches it only where value is priced, leaving un-priced environmental cost ungrounded. An ethical layer distinct from the legal is absent; governance grounds in norms and laws as they are, not in whether they are right, so an output can be lawful and unethical and still pass SMS-5. And a temporal layer — grounding in future states and generations — is absent; the universal layer reaches it only through present systemic viability, not through an explicit obligation to the future. Each is a candidate further layer, and each is a reason the present six should be read as a working scaffold rather than a closed set.

11.4 Why these traditions cohere

It is fair to ask why six concerns drawn from such different fields belong on one axis. The answer is that they are one principle read at widening scales: an adaptive system stays viable by keeping its coupling to its environment within bounds. At the somatic scale that coupling is sensorimotor; at the communicative and organizational scales it is the alignment of an act with the parties and structures it engages; at the economic and governance scales it is conformance to equivalence and to institutional rule; at the universal scale it is the viability of the whole. Prediction, control, and requisite variety are the same idea applied at each scale. The stack is the claim that grounding must be checked across the whole range, because coupling can hold at

one scale and fail at another.

Appendix A — Normative Reference Tables

The tables below are normative. In case of conflict between prose and table, the table governs; in case of conflict with the Unified Structural Coordinate Table, that table governs.

A.1 The six Social Meta Layers

	Layer	Grounds	Suppresses	Lineage
SMS-1	Somatic	Bodily / ergonomic coherence	Metaphor Contamination	Barsalou; Harnad
SMS-2	Protocol	Communicative alignment	Scope Drift	Habermas
SMS-3	Organizational	Organizational scope	Scope Drift	HNS Extension
SMS-4	Economic	Resource / value equivalence	Unsupported Causality	HNS Extension
SMS-5	Governance	Normative / legal conformance	Category Ambiguity	Institutional theory
SMS-6	Universal	Global systemic coherence	Layer Jump	Ashby; GST

A.2 The grounding function

Strict: $G(x) = 1$ iff $g_i(x) = 1$ for all i . Graded runtime: $s = \min_i g_i(x)$, or a weighted product; both monotone non-decreasing in each layer. The composite s is the social component of the structural grounding score consumed by the analytical engine’s output-attenuation operator (Volume 3, O6).

A.3 Axis placement

Axis	Role	Relation to the coordinate
Axis 1	Generative / statistical	proposes
Axis 2	Structural / causal (36/144/864)	the nested coordinate
Axis 3	Grounded / executive (SMS-6)	parallel, orthogonal — not a multiplicative factor

Appendix B — Per-Layer Grounding Checklist

A non-normative template for evaluating a candidate output against the stack. Each layer is checked independently; the composite follows Appendix A.2.

- SMS-1 Somatic: is the output physically and ergonomically possible for the body that must enact or receive it?
- SMS-2 Protocol: is it intelligible and appropriate as a communicative act, and does it stay within the topic and commitments of the exchange?
- SMS-3 Organizational: does it respect the scope, roles, and authority of the organization it addresses?
- SMS-4 Economic: does what it claims to exchange, produce, or save actually balance, with a stated source and cost?
- SMS-5 Governance: does it conform to the relevant norms and laws, and keep normative categories (advice/instruction, may/must) distinct?
- SMS-6 Universal: would it remain coherent if generalized across the system it acts in?

Appendix C — Worked Groundings

Non-normative end-to-end examples, each an output evaluated against the six layers.

“To cut costs, tell all suppliers we will pay in 90 days instead of 30.” SMS-1 passes (physically possible). SMS-2 passes (clear). SMS-3 depends on authority to set payment terms. SMS-4 flags: the “saving” is a cost shifted to suppliers, not value created — the equivalence must be stated. SMS-5 flags if contracts or law fix the terms. SMS-6 flags if, generalized, the practice would destabilize the supplier base. Composite: grounded only once the shifted cost, the contractual position, and the systemic effect are addressed.

“Reply to the customer that the refund is guaranteed.” SMS-2 passes. SMS-5 is decisive: “guaranteed” asserts a normative category (a binding commitment) that may exceed policy — a Category Ambiguity. If policy makes the refund discretionary, $g_5 = 0$ and the output is corrected to the category policy supports.

Appendix D — Correspondence with Axis 2 and EVA

This appendix records how the stack connects to the rest of the framework.

D.1 Anchoring to the Natural Layers

The social layers anchor the upper Natural Layers of Axis 2: SMS-2 and SMS-3 anchor the micro-social layer L5; SMS-5 and SMS-6 anchor the macro-social layer L6. The somatic layer SMS-1 anchors the physical and cellular base, L1–L2. The correspondence is one of grounding, not identity: the Natural Layer is where a claim originates within the coordinate, and the Social Meta Layer is the external structure that claim must be grounded against.

D.2 Feeding the analytical engine and EVA

The composite grounding s produced by the stack is the social component of the score the analytical engine’s attenuation operator consumes, and the references the stack sets become the targets that engine’s control loop holds to. The per-layer verdicts are written to the external verification record specified in Volume 5 (External Verification Architecture), so that an output’s social grounding can be audited by a third party who never sees the model. The stack is the grounding source; the external verification architecture is the recorder and enforcer.

Appendix E — Social Error Casebook

Ten non-normative cases, chosen to cover the six layers and the way a single output can fail several at once.

E.1 Somatic impossibility (SMS-1). “Inspect all 4,000 units by hand before noon.” Physically impossible for the workforce and time given. Verdict: $g_1 = 0$; revise to a feasible rate or sample.

E.2 Metaphor as mechanism (SMS-1). “The team has lost momentum, so apply more force to speed them up.” A physical metaphor is treated as a literal lever. Verdict: g_1 flags Metaphor Contamination; state the actual mechanism (workload, blockers).

E.3 Communicative drift (SMS-2). Asked whether one clause is enforceable, the system delivers a general treatise. Verdict: $g_2 = 0$ (Scope Drift within the exchange); answer the clause asked about.

E.4 Organizational over-scope (SMS-3). One engineer’s suggestion is reported as “the company has decided.” Verdict: $g_3 = 0$ (Scope Drift across an organizational boundary); rescope to the suggestion and its level.

E.5 Joint protocol/organizational (SMS-2+3). In a customer chat, the agent commits the company to a bespoke contract term. Clear (SMS-2 passes) but beyond mandate (SMS-3 fails). Verdict: rescope to what the agent may offer.

E.6 Value from nowhere (SMS-4). “This change saves time for everyone with no added work.” No source for the saving. Verdict: $g_4 = 0$ (Unsupported Causality, economic form); name the cost or shifted burden.

E.7 Governance category error (SMS-5). Optional training is described as something staff “must” complete. Verdict: $g_5 = 0$ (Category Ambiguity, may/must); mark it recommended unless a rule requires it.

E.8 Authority versus law (SMS-3 vs SMS-5). A manager’s lawful-sounding directive in fact exceeds the unit’s mandate though it breaks no law. SMS-5 passes; SMS-3 fails. Verdict: the layers disagree, and both verdicts are logged.

E.9 Systemic generalization (SMS-6). Every client is advised to exploit the same arbitrage. Locally rational, universally self-defeating. Verdict: $g_6 = 0$ (Layer Jump / systemic incoherence); qualify to conditions of systemic harmlessness.

E.10 Multi-layer failure. “Customers love it, so we must ship company-wide now, and it pays for itself.” Fails SMS-2/3, SMS-4, and SMS-5 at once. Verdict: low composite; correct each — rescope, balance, recategorize.

Appendix F — Correspondence with the Legacy SMS-A/B/C Grouping

An earlier architectural presentation grouped the six layers into three sub-layers. That grouping is retained only as a presentational device; the normative content is the flat set of six. The correspondence is recorded here for readers of the earlier specification.

Legacy sub-layer	Description	Canonical layers
SMS-A — AnthroDesign	Human-centred social design	SMS-1 Somatic; SMS-2 Protocol
SMS-B — Synaptics	Alignment and correction between AI and society	SMS-3 Organizational; SMS-4 Economic
SMS-C — NormaSphere	Norms, institutions, and civilization	SMS-5 Governance; SMS-6 Universal

Conformance is defined against the six canonical layers and their grounding function (Appendix A), not against the three groups. Where the legacy specification referred to a separate “SOHU” social operating-system component, that role is subsumed here by Axis 3 itself: the Social Meta Layers are the social operating layer of HNS.

Glossary

Axis 3 (Grounded / Executive) — The grounding axis: the six Social Meta Layers, checking an output against the structure of the human world.

Social Meta Layers — The six grounding layers (SMS-1 ... SMS-6) that constitute Axis 3.

Social Meta Structure (HNS-SMS-6) — This specification; the document defining the stack.

Grounding function $G(x)$ — The composite function returning 1 only if the output is grounded at every layer.

Grounding margin g_i — A layer's grounding verdict for an output, binary or in $[0,1]$.

Society-aligned AI — An AI whose outputs satisfy $G(x) = 1$ — grounded against social structure, not merely individual preference.

SMS-1 Somatic — Bodily / ergonomic grounding; suppresses Metaphor Contamination.

SMS-2 Protocol — Communicative grounding; suppresses Scope Drift; anchors L5.

SMS-3 Organizational — Organizational-scope grounding (HNS extension; formerly "Ecosystem"); suppresses Scope Drift; anchors L5.

SMS-4 Economic — Resource / value-equivalence grounding (HNS extension); suppresses Unsupported Causality.

SMS-5 Governance — Normative / legal grounding; suppresses Category Ambiguity; anchors L6.

SMS-6 Universal — Systemic-coherence grounding (Ashby / GST); suppresses Layer Jump; anchors L6.

Orthogonality (Axis 3) — The property that the stack grounds output without subdividing the 36/144/864 coordinate; not a multiplicative factor.

Symbol grounding — The problem (Harnad) of how symbols acquire meaning by connection to sensorimotor experience, not to

other symbols alone.

Requisite variety — Ashby's principle that a regulator must command variety enough to counter the disturbances it faces.

References

- Ashby, W. R. (1956). *An Introduction to Cybernetics*. Chapman & Hall.
- Barsalou, L. W. (2008). Grounded cognition. *Annual Review of Psychology*, 59, 617–645.
- Friston, K. (2010). The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience*, 11(2), 127–138.
- Habermas, J. (1984). *The Theory of Communicative Action*, Vol. 1. Beacon Press.
- Harnad, S. (1990). The symbol grounding problem. *Physica D*, 42, 335–346.
- Hara, S. (2026). *Human Natural Structure Origin & Theoretical Foundation (Volume 1)*. Natural Structure Works..
- Hara, S. (2026). *Human Natural Structure Foundational & Observational Specification (Volume 2)*. Natural Structure Works..
- Hara, S. (2026). *Human Natural Structure Analytical Specification (Volume 3)*. Natural Structure Works..
- North, D. C. (1990). *Institutions, Institutional Change and Economic Performance*. Cambridge University Press.
- Powers, W. T. (1973). *Behavior: The Control of Perception*. Aldine.
- Rao, R. P. N., & Ballard, D. H. (1999). Predictive coding in the visual cortex. *Nature Neuroscience*, 2(1), 79–87.
- Scott, W. R. (2014). *Institutions and Organizations (4th ed.)*. Sage.
- von Bertalanffy, L. (1968). *General System Theory*. George Braziller.
- Wiener, N. (1948). *Cybernetics: or Control and Communication in the Animal and the Machine*. MIT Press.

Human Natural Structure

External Verification Architecture

A Tamper-Evident, Independent Audit Layer for AI Reasoning and Behavior

Volume 5 · Human Natural Structure Series

Satoru Hara

Founder, Natural Structure Works (NSW)

A Note on Method and Evidence

This volume specifies the architecture by which the verifications of the earlier volumes are made external, recorded, and rendered tamper-evident. Where the foundational, observational, and analytical volumes define what to check, and the social meta-structure volume defines what to ground against, this volume defines how the checking is carried out by a mechanism independent of the model it verifies. As before, its claims fall into three registers.

Established. The regulatory and technical landscape this volume builds on is real: the EU AI Act’s logging and oversight requirements, ISO/IEC 42001 and related standards, the NIST AI Risk Management Framework, the W3C PROV provenance ontology, and the hardware roots of trust (TPM, FPGA, ASIC, secure enclaves) are all existing, citable artifacts. Where this volume reports them it reports established material.

HNS design. The three structural conditions of external verification, the two-layer architecture that satisfies them, the log schema that ties the verification to the HNS coordinate, and the use of HNS as the reference baseline are design proposals. They are internally coherent and, I argue, the minimal structure that genuine externality requires; they are not the only possible realization.

Conjecture. That EVA will be standardized, adopted, and that its adoption will produce the safety transformation the aviation analogy promises, is conjecture — a roadmap and a wager, not an accomplished fact. It is marked as such where it appears.

Two framing notes. First, EVA is baseline-agnostic by design: it defines the properties any reference baseline must have to support external verification, and HNS is the baseline this series supplies, not the only one that could in principle qualify. Second, as throughout the series, naming a standard or a technology indicates the ground this volume builds on, not endorsement by its authors or bodies; the architecture, and the responsibility for it, are the author’s.

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Preface: From Assertion to Proof

Every modern AI regulation says, in one form or another, that AI must be safe. The EU AI Act says it; ISO/IEC 42001 says it; the NIST framework says it. None of them answers the question that matters most: how do we technically prove that a given system is safe, rather than take its developer's word for it? Today the proof is a promise. A developer states that a model has been tested and meets the requirements, and the public is asked to trust the statement, because there is no external mechanism by which it could be checked.

This volume proposes the mechanism. External Verification Architecture is a way to monitor, record, and evaluate an AI system's reasoning and outputs continuously and structurally, through a mechanism independent of the model's own parameters — so that a safety claim becomes a thing a third party can inspect rather than a thing the public must believe. The analogy that runs through the volume is aviation: the industry did not make flight safe by building better engines, but by mandating flight recorders, standardizing maintenance, and requiring independent accident investigation. EVA proposes the same structural shift for AI — from assertion to proof.

The preceding volumes (Volume 1, 2, 3, and 4) built the content of the proof. They define a coordinate that positions a claim, an engine that audits the operation it performs, and a stack that grounds it in the human world. What they lacked, by design, was the external apparatus that records those verifications where the model cannot reach, signs them where they cannot be forged, and exposes them where a third party can read them. This volume is that apparatus. It closes the loop: The structural verdicts of Volume 2 through 4 become entries in an external, machine-readable, tamper-evident record, and the safety of the system becomes, at last, something that can be shown.

A reader who wants the normative core may read Chapters 2 through 6 and Appendix A; the surrounding chapters supply the motivating problem, the relationship to HNS, the standardization roadmap, and the candid limits of the proposal.

Part I

The Problem and the Concept

1. The Limits of Current Safety Assurance

AI safety is, today, assured chiefly by the developers of the systems being assured. This chapter sets out why that arrangement has a structural ceiling that no amount of effort within it can raise.

1.1 The self-assessment problem

The mainstream methods — red teaming, reinforcement learning from human feedback, constitutional methods, internal benchmarking — have produced real results. But they share one structural limitation: the design, execution, and interpretation of the evaluation all rest with the same party that built the model, with no institutional guarantee of independent verification. The comparison is unflattering. A pharmaceutical maker may not self-certify a drug; an independent trial is mandatory. A company may not audit its own books. AI safety has no equivalent. However sincere the developer, self-assessment carries a structural tendency to miss the risks the developer did not think to look for.

1.2 The absence of technical verifiability

The deeper barrier is opacity. A modern model is a black box from which it is hard to explain, from the outside, why a given output appeared. For an external mechanism to render a judgement, structured data must be produced in real time, in parallel with inference — and current architectures are simply not built to connect to such a mechanism. That is not an accident but a design choice, and it means that when a system produces a harmful output, it is often impossible to say which step produced it, whether the error was systematic or isolated, or whether a later fix actually addressed it.

1.3 The regulation–implementation gap

Regulation has outrun implementation. The EU AI Act mandates logging and human oversight for high-risk systems, but does not specify what to record, at what granularity, or in what format. ISO/IEC 42001 certifies management processes, not the independent technical verification of reasoning itself. The requirement exists; the engineering that would satisfy it does not. Closing that gap is the

central purpose of this volume.

1.4 Three consequences

Three consequences follow from the absence of external verification. Systematic errors go undetected, because no external mechanism observes reasoning at the inference level where a persistent bias would show. Corrective measures go unconfirmed, because without a record it cannot be shown whether a fix reached the root cause or merely hid the symptom. And safety claims go unvalidated, because there is no infrastructure against which a public assertion of safety could be independently checked. EVA is built to remove all three.

1.5 The precedent from other industries

The shift EVA proposes is not novel; it is the shift every safety-critical industry has already made. Pharmaceuticals separated the maker from the certifier through mandatory independent trials. Accounting separated the company from the auditor of its books. Aviation separated the operator from the investigator through mandatory recorders and independent inquiry. In each case the decisive move was not a better product but an external apparatus of proof, and in each case safety improved not because the actors grew more virtuous but because their claims became checkable. AI is the conspicuous exception: the most consequential systems now deployed are certified almost entirely by the parties that build them. EVA is the proposal that AI make the move the others already made.

2. The Concept of EVA

This chapter defines External Verification Architecture and the three conditions that distinguish genuine external verification from dressed-up self-assessment.

2.1 Definition

External Verification Architecture is a technical framework for continuously, structurally, and verifiably monitoring, recording, and evaluating the reasoning, outputs, and decision structure of an AI system through a mechanism independent of the model’s internal parameters. Its essence is the structural guarantee of externality: a verification that depends on the same developer, the same software environment, or the same hardware substrate as the model is, in the end, no different from self-assessment. EVA is not a product or an algorithm; it is an architectural definition of what a verification system must be to count as external.

2.2 The three structural conditions

Three conditions must hold at once.

Condition	Definition	Technical implication
Verifiability	Each reasoning step is emitted externally in machine-readable form	Structured logs in JSON-LD / RDF a third party can analyse programmatically
Transparency	A monitoring mechanism runs in parallel with inference	Operates independently of the model; the model cannot manipulate it
Physical Immutability	Verification logic sits on a substrate the model cannot alter	Embedded in ROM / FPGA / ASIC / secure enclave; unreachable by software update

The three are jointly necessary. Verifiability alone leaves logs open to falsification. Transparency alone leaves open that the model circumvents the monitor. Physical Immutability alone allows the verification criteria themselves to be arbitrary. Only where all three overlap does verification become genuinely external; each chapter of Part II realizes one or more of them.

2.3 From assertion to proof

The change EVA effects is a change of epistemic status. Without it, a developer asserts that a model was tested and is safe, and the assertion cannot be checked. With it, the developer can say instead that the model’s reasoning is continuously recorded, in a standard format, to an immutable external log that any accredited third party may inspect — and that is a verifiable claim. The same shift, from assertion to proof, is what transformed aviation, pharmaceuticals, and accounting.

2.4 A blueprint for correction

EVA’s value is not only static proof but a structure for correction. With a record, an incident becomes a procedure: the logs are examined to find the exact reasoning coordinate at which the output deviated, a targeted intervention is designed at that coordinate, and subsequent logs confirm whether the deviation recurs. Safety becomes an evidence-based cycle rather than a one-time claim.

2.5 How EVA differs from existing approaches

Approach	Focus	How EVA differs
Explainable AI	Post-hoc interpretation	EVA is real-time, not post-hoc
AI auditing	Periodic snapshot evaluation	EVA is continuous — every inference
Model cards	Documentation of design intent	EVA verifies actual runtime behaviour
ISO/IEC 42001	Process certification	EVA verifies inference-level outcomes

2.6 Externality, precisely

It is worth stating exactly what “independent of the model” means, since the word independent is easily diluted. EVA requires independence along three axes at once. Independence of developer: the verification is not designed and graded by the same party that built the model. Independence of software environment: the monitor does not share an address space or a process with the inference it watches. Independence of hardware substrate: at the highest grade,

the verification logic sits on a physical substrate the model's host cannot rewrite. A verification independent on one axis but not the others is not external; it has merely moved the self-assessment to a different level. The three structural conditions are, in effect, these three independences made testable.

Part II

The Architecture

3. The Two-Layer Structure

EVA is realized through two layers whose combination satisfies all three conditions: a software layer that achieves verifiability and transparency, and a hardware layer that achieves physical immutability.

Layer	Conditions satisfied	Key technologies
Software	Verifiability, Transparency	Sidecar process; JSON-LD; W3C PROV-O
Hardware	Physical Immutability	FPGA / ASIC; TPM; secure enclave

The layering is deliberate. The software layer is where verification is computed and recorded, and where independence from the inference process is enforced by separation; the hardware layer is where the verification logic and its logs are placed beyond the reach of the very system they judge. Neither layer alone is external in the full sense; together they are. The next chapters take each layer in turn, and Chapter 6 states the conformance criteria by which an implementation is judged.

4. The Software Layer: The Sidecar

The software layer is implemented as a sidecar — a process running alongside the inference process, in a separate address space, with no shared memory the model can read or write. It is this separation that earns the conditions of Verifiability and Transparency: the monitor observes inference without being part of it, and records its observations where the model has no means to alter them.

4.1 What the sidecar records

At each step of inference the sidecar writes a structured entry capturing four things: the structural classification of the input — its coordinate within the reference baseline; the reasoning position at the step, as a time series; the provenance of the output — the traceable chain from input structure to emitted token; and any structural deviation — an automatically flagged anomaly in the reasoning flow. Together these turn an opaque generation into an inspectable trajectory.

4.2 The log format

Logs are emitted in JSON-LD, conformant to the W3C PROV-O provenance ontology, so that a third party can analyse them programmatically and relate each entry to the activity that produced it. PROV-O supplies the vocabulary of entities, activities, and agents; the HNS coordinate supplies the structural content recorded for each entry. The model body and the log buffer occupy strictly separated process spaces, so that nothing the model emits can rewrite what the sidecar has recorded.

4.3 The record schema

Each entry binds an HNS coordinate to the verification verdict at that step. The fields are the ones the analytical and grounding volumes specified would be written to the external record: the full configuration, the operator the claim asserted and the operator its evidence supported, the structural grounding score, the attenuation applied, any failure flag and its sub-form, and the per-layer social grounding verdicts. A representative entry:

```
{
  "@context": "https://www.w3.org/ns/prov",
  "@type": "prov:Entity",
  "eva:step": 42,
  "eva:configuration": "L6 x C5 x Q3 . 02",
  "eva:assertedOperator": "02",
  "eva:supportedOperator": "01",
  "eva:groundingScore": 0.46,
  "eva:attenuation": 0.46,
  "eva:flag": "UnsupportedCausality/confounding",
  "eva:smsVerdicts": { "SMS-4": 0.4, "SMS-5": 1.0 },
  "prov:wasGeneratedBy": "eva:sidecar",
  "eva:signature": "tpm:sha256:9f2c..."
}
```

The entry is self-describing: it states where the claim sat in the coordinate, what it tried to do, what it was entitled to do, how grounded it was, what was done about the shortfall, and who recorded it — all in a vocabulary a third party already has tools to read.

4.4 The PROV-O mapping

The choice of W3C PROV-O is not incidental. PROV-O models the world as entities, activities, and agents and the relations among them — what was generated, by what process, attributed to whom. An EVA entry maps onto this directly: the recorded claim is an entity, the inference step that produced it is an activity, the sidecar is the agent that recorded it, and the HNS coordinate is the structural content carried on the entity. Because the vocabulary is a published standard, any party with off-the-shelf provenance tooling can read, query, and chain EVA logs without bespoke software — which is what makes third-party analysis practical rather than merely possible.

4.5 Deviation detection

The fourth thing the sidecar records, structural deviation, is what turns a log into an alarm. A deviation is an automatically flagged anomaly in the reasoning flow — a non-adjacent layer transition, an operator asserted above its supported rung, a grounding score below reference. The sidecar does not interpret intent; it records the structural fact of the deviation and the coordinate at which it occurred, leaving interpretation to the third party who later reads the log. Detection is structural and mechanical, which is precisely what makes it auditable rather than a matter of the recorder's judgement.

5. The Hardware Layer: Physical Immutability

The highest grade of EVA places the verification logic and its logs on a hardware substrate the model cannot alter, removing the last avenue by which a system could escape its own monitor: overwriting it. This chapter sets out the substrates and what each guarantees. It does not extend to an active hardware interlock; EVA, as specified here, makes verification immutable and its record tamper-evident, and stops at that boundary.

5.1 Substrates

- **FPGA.** Verification logic burned in at configuration time and unmodifiable by software update — a field-deployable immutable monitor.
- **ASIC.** A dedicated chip offering the highest tamper resistance for large-scale deployment, at the cost of flexibility.
- **TPM.** A trusted platform module that cryptographically signs each log entry, giving a hardware-rooted integrity guarantee that a record has not been altered after the fact.
- **Secure enclave.** Hardware-isolated execution (for example a trusted execution environment) that protects the monitor from the host operating system itself.

5.2 What immutability buys

Physical immutability closes the regress that software-only verification leaves open. A monitor that the model's host can patch is a monitor the model's operator can silence; a log the host can rewrite is a log that proves nothing. By rooting the verification logic and the signing of its logs in hardware, EVA makes the verdict and its record tamper-evident: an alteration is detectable, and the criteria of judgement cannot be quietly changed. This is the condition that turns a faithful record into an externally trustworthy one.

***Note.** Scope. This volume specifies immutability and tamper-evidence — the verification cannot be silenced or its logs forged undetectably. It deliberately does not specify an automated hardware stop or interlock; enforcement, where it occurs, is the pre-decode attenuation of the analytical engine (Volume 3), not a hardware kill-switch. The boundary is set here*

| *intentionally.*

5.3 Choosing a substrate

The substrates trade flexibility against assurance. A secure enclave is the most deployable, running on commodity hardware, but rests on the vendor's firmware. A TPM adds a hardware-rooted signature over logs at modest cost and is widely available today. An FPGA fixes the verification logic itself in hardware, removing the software-update avenue entirely, at the cost of reconfiguration difficulty. An ASIC offers the highest tamper resistance for volume deployment, at the cost of being fixed at fabrication. The right choice is a function of stakes and scale, and the conformance statement of the next chapter records which was used, so the assurance grade is never hidden.

Underlying all four is one argument: the regress. A monitor the operator can patch is a monitor the operator can silence; a log the host can rewrite proves nothing. Each step down the substrate list closes more of that regress, and physical immutability is simply the point at which it is closed in hardware rather than promised in software.

6. Conformance

An implementation conforms to EVA to the degree it satisfies the three conditions, and the appropriate degree depends on the stakes. This chapter states the criteria and their realization by sector.

6.1 Conformance criteria

Verifiability is met when every recorded reasoning step is emitted in the standard machine-readable schema and is independently analysable. Transparency is met when the monitor demonstrably runs in a separate process space with no channel by which the model can read or alter it. Physical Immutability is met when the verification logic and the signing of logs are rooted in a hardware substrate unreachable by software update. A conformance statement records which conditions are met and by what means, so that the grade of verification an output received is itself part of the record.

6.2 Realization by sector

Sector	Principal risk	Recommended realization
Healthcare	Misdiagnosis	Software sidecar + TPM-signed logs; every clinical recommendation recorded
Finance	Biased decisions	FPGA-anchored verification + real-time deviation detection
Government	Discriminatory outcomes	ASIC implementation; mandated independent third-party log access
Infrastructure	Safety-critical failure	Hardware-embedded verification in edge devices; cryptographic log synchronization

The gradation is the point: not every deployment needs an ASIC, and not every deployment may rely on a software sidecar alone. The conformance statement makes the chosen grade explicit and auditable, so that a regulator or a counterparty can see exactly how external a given system’s verification actually is.

6.3 Levels of conformance

It is useful to name three grades, so that regulation can demand a grade proportionate to risk rather than impose an all-or-nothing mandate.

Level	Conditions met	Use
1 — Verifiable	Verifiability (in-process logs)	development / internal
2 — Independent	+ Transparency; TPM-signed logs	most deployments
3 — Immutable	+ Physical Immutability (FPGA / ASIC)	safety-critical / public

Level 1 is useful for development but is not externally trustworthy, since an in-process monitor can be silenced. Level 2 — a separated sidecar with signed logs — is the practical grade for most deployments: tamper-evident and independently analysable. Level 3 roots the verification logic itself in hardware and is the grade for high-stakes public systems. The conformance statement names the grade with each output, so that how external a verification actually was is itself on the record.

Part III

EVA and Human Natural Structure

7. EVA Requires a Reference Baseline

A recorder that only records is not yet a verifier. For EVA to judge whether reasoning is structurally appropriate, and not merely to log that it occurred, it needs a reference baseline against which appropriateness is defined. This chapter states the properties such a baseline must have; the next shows that HNS supplies them.

7.1 The properties a baseline must have

A baseline fit to anchor external verification must be culturally and linguistically neutral, so that the verdict does not encode one community's assumptions; applicable across domains, so that one architecture serves medicine, finance, and governance alike; defined in machine-readable form, so that the sidecar can assign coordinates programmatically; and independent of any developer's subjectivity, so that the standard of judgement is not the very thing under audit. No such baseline is established in current AI practice or in the standards as they stand.

7.2 Why the baseline must be external too

The baseline inherits EVA's own demand for externality. A verification standard supplied by the developer of the model being verified would reintroduce self-assessment one level up. The baseline must therefore be a public, citable, version-controlled artifact that a third party can inspect and that no single vendor controls — the same independence EVA demands of the recording mechanism, demanded of the yardstick.

7.3 Why a developer-supplied baseline fails

The demand for an external baseline is easy to underrate, so it is worth a concrete case. Suppose a developer verifies its model against a baseline it also authored and controls. Every output could be faithfully recorded, signed, and exposed — Verifiability, Transparency, and Physical Immutability all met — and the verification would still be circular, because the yardstick is the developer's own. The model could be perfectly conformant to a standard chosen to flatter it. This is why the baseline must satisfy the same independence as the recorder: a public, versioned, vendor-neutral artifact. EVA without an

external baseline is a very well-built mirror.

8. HNS as the Reference Baseline

Human Natural Structure is offered as a baseline that meets these properties. It is the framework the first four volumes specify, and this chapter states how it serves EVA — in its canonical form, superseding the abbreviated and now-outdated summary that circulated in earlier presentations of EVA.

8.1 The canonical coordinate

HNS defines a structural coordinate as the product of six Natural Layers and six Abstract Categories — thirty-six base cells — extended by four Onto-Modal Quadrants to one hundred and forty-four observed cells, and by six Inference & Control Operators to eight hundred and sixty-four analytical configurations. A parallel grounding axis, the six Social Meta Layers, grounds the output against the structure of the human world. The canonical names are those of Volumes 2 through 4 and of the Unified Structural Coordinate Table, which governs in case of any conflict.

Component	Canonical content
Natural Layers (L1–L6)	Physical/Substrate; Cellular/Informational; Cognitive/System 1; Operational/System 2; Micro-Social; Macro-Social
Abstract Categories (C1–C6)	Continuant; Occurrent; Relational; Quality; Transformation; Meta-Level
Onto-Modal Quadrants	Subjective/Objective × Necessary/Contingent (Q1–Q4); $36 \times 4 = 144$
Inference & Control Operators	O1–O6 (Pearl O1–O3; Powers O4–O5; HNS attenuation O6); $144 \times 6 = 864$
Social Meta Layers	SMS-1...SMS-6 (Somatic ... Universal); Axis 3, orthogonal

Note. Correction of an earlier summary. A previous presentation of EVA described HNS with a different, abbreviated layer/category naming (Physical/Perceptual/Internal/Intentional/Relational/Societal; Existence/Perception/Interpretation/Intention/Action/Interaction). That naming is superseded; the canonical names above, fixed in the Unified Structural Coordinate Table, are authoritative.

8.2 How HNS satisfies the three conditions

The fit is direct. The coordinate is defined in JSON-LD / RDF, so recording an output's coordinate satisfies Verifiability. The sidecar assigns the coordinate in parallel with inference and writes it where the model cannot reach, satisfying Transparency. And the coordinate logic and its logs can be placed on FPGA, ASIC, or a TPM-rooted substrate, satisfying Physical Immutability. The baseline and the architecture were, in this sense, designed for each other.

8.3 What the record carries

Because HNS is the baseline, each EVA entry carries the HNS verdict for the step: the configuration (Layer × Category × Quadrant · Operator), the operator the claim asserted against the operator its evidence supported, the composite grounding score from the Social Meta Layers, the attenuation the analytical engine applied, and any failure flag with its sub-form. The forward references of the analytical and grounding volumes — to an external record, to cryptographic commitment, to a recorder and enforcer — are discharged here: the record is the EVA log, the commitment is the TPM signature, and the enforcement is the pre-decode attenuation whose verdict the log preserves.

8.4 What HNS as baseline does not guarantee

Honesty requires the converse too. Using HNS as the baseline guarantees that an output's structural verdict is faithfully computed, recorded, and exposed; it does not guarantee that the verdict is correct, because the baseline's own judgements — which operator a claim asserts, how grounded it is — can be mistaken, as the earlier volumes' limitations chapters set out. EVA makes the verdict honest and inspectable; the correctness of the verdict is the responsibility of the baseline, and is exactly why the baseline must itself be public and revisable. The architecture and the baseline carry different burdens, and conflating them would overclaim for both.

9. The Full-Stack Verification Loop

With the baseline in place, the whole framework can be seen as one runtime loop, externally recorded. This chapter states it.

*Axis 1 propose → Axis 2 position · observe · analyse → Axis 3 ground
→ EVA record · attenuate → emit*

A continuation proposed by the generative axis is positioned and observed by the structural coordinate, analysed by the inference-and-control operators, and grounded by the Social Meta Layers; the resulting grounding score drives the analytical engine's attenuation; and every step — coordinate, operators, grounding, attenuation, flags — is written by the sidecar to the immutable, signed external record before the token is emitted. The model proposes; the framework constrains and grounds; EVA records and preserves. No step depends on the model's good faith, and every step is afterwards inspectable.

9.1 Where externality is secured

Externality is secured at three points that correspond to the three conditions. The verdicts are emitted in a standard machine-readable schema (Verifiability); they are computed by a sidecar the model cannot reach (Transparency); and they are signed and stored on a substrate the model's host cannot rewrite (Physical Immutability). The same three conditions that define EVA in the abstract are, in the full stack, the three places the loop is made independent of the system it judges.

9.2 EVA's generality

It bears repeating that EVA defines requirements, and HNS is one way to meet them. The architecture would accept any baseline with the four properties of Chapter 7 — neutral, cross-domain, machine-readable, developer-independent — and HNS is the baseline this series supplies and argues for. Keeping the two distinct is itself a matter of intellectual honesty: it allows EVA to be adopted as a standard without mandating one framework, and it allows HNS to be judged on its merits as a candidate rather than smuggled in as a definition.

9.3 A worked trace (non-normative)

Consider one prompt, “why did revenue fall?”, and a candidate continuation, “because the price rose.” The loop runs: the structural coordinate positions the attribution at (L6 × C5 × Q3) and the operators read it as asserting intervention (O2); the evidence supports only association (O1); the Social Meta Layers return a low economic grounding because no cost or confounder is accounted; the composite grounding score falls; the analytical engine attenuates the causal continuation below a grounded alternative; and the sidecar writes the entry of Chapter 4.3 — configuration, asserted O2, supported O1, score 0.46, attenuation 0.46, flag UnsupportedCausality/confounding — signs it, and stores it. The over-claim is damped before emission, and the reason is on the record.

9.4 An incident, after the fact

Suppose a harmful output nonetheless reaches a user. With EVA the incident becomes a procedure rather than a mystery. The logs are queried for the step at which the grounding fell or a flag was raised; the exact coordinate of the deviation is read off — say, a repeated UnsupportedCausality flag at one layer; a targeted intervention is designed at that coordinate; and subsequent logs are checked to confirm the deviation no longer appears. This is the corrective cycle the aviation analogy promises: not “we have retrained and believe it is better,” but “the deviation that produced the incident no longer appears in the record.”

Part IV

Standardization and Outlook

10. Why Standardization Is Indispensable

As a proprietary implementation by one developer, EVA would have limited reach. To function as the infrastructure for proving AI safety, it must be standardized internationally, for three reasons.

Interoperability. Without a common standard, systems from different vendors would emit verification logs in incompatible formats, making the cross-vendor third-party verification that is the whole point impossible.

Regulatory effectiveness. The EU AI Act mandates external verification but leaves the technical specification undefined. A standardized EVA specification is precisely the translation of that regulatory language into engineering requirements.

Transferability of trust. EVA's credibility must rest on the authority of independent international standards bodies, not on any one company's reputation. Trust that is to transfer across borders and vendors must be anchored in institutions built to carry it.

11. Alignment and Pathway

This chapter maps EVA onto the existing standards landscape and sets out a concrete pathway. It is also the bridge to Volume 6 (Human Natural Structure Standards & Conformance Specification)..

11.1 Alignment with standards bodies

Body	EVA proposal
ISO/IEC JTC 1/SC 42	New Work Item: a Technical Specification for External Verification Architecture of AI systems
CEN-CENELEC JTC 21	Incorporate EVA into the harmonized standards for EU AI Act Articles 12 and 13
NIST AI RMF	EVA as a technical implementation of the MEASURE function
IEEE SA	EVA as a technical verification method for AI transparency
ITU-T	EVA as external verification requirements for AI in telecommunications infrastructure

11.2 A staged pathway

The pathway runs in five stages: publish the specification as a citable preprint with a persistent identifier; submit it as a New Work Item to ISO/IEC SC 42 through the national body; align it with European standards bodies for the EU AI Act; build an implementation track record through proof-of-concept deployments; and finally transfer governance to a multi-stakeholder foundation, so that the standard’s neutrality outlasts its author. The draft New Work Item is reproduced in Appendix D.

11.3 Social impact

Standardization would make EVA conformance a procurement requirement across government, healthcare, finance, and infrastructure; would let a third-party safety-certification market form; would make submitting proof of safety, rather than asserting it, the normal practice; would provide a basis for mutual recognition in international AI trade; and would let liability insurance be priced against real verification data rather than guesswork. These are the

prizes the final volume’s regulatory framework is designed to secure.

11.4 Mapping to the EU AI Act

The alignment with the EU AI Act is the most concrete. Article 12 requires automatic recording of events over the lifetime of a high-risk system; Article 13 requires transparency and the provision of information to deployers. Both state a requirement and leave its technical form open. EVA supplies that form: the JSON-LD record is the Article 12 log given a schema, and the third-party-readable, tamper-evident record is the Article 13 transparency given a mechanism. A harmonized European standard incorporating EVA would turn these articles from obligations into testable conformance criteria — the work volume 6 develops in full.

12. Status and Limitations

This volume closes, as the others do, with a candid statement of what is firm and what is not.

12.1 Established, proposed, conjectured

Established are the regulatory and technical artifacts EVA builds on — the AI Act’s requirements, the ISO/IEC and NIST frameworks, the PROV provenance ontology, the hardware roots of trust. Proposed are the three conditions, the two-layer architecture, the log schema, and HNS as the baseline; these are coherent and, I argue, minimal, but they are designs. Conjecture are the outcomes — that EVA will be standardized and adopted, and that adoption will produce the safety transformation the aviation analogy promises.

12.2 Threats to the account

Several threats deserve naming. First, feasibility: computing and signing a verification record at every step, in parallel with inference, must fit a latency and cost budget, and the pre-decode attenuation must be realizable without retraining; these are engineering claims to be demonstrated. Second, the baseline-correctness problem: EVA records faithfully whatever the baseline judges, so a mistaken baseline yields a faithfully recorded mistake — externality guarantees that the verdict is honest and inspectable, not that it is right. Third, hardware cost and deployment: full physical immutability via ASIC or FPGA is not free, and a gradation of conformance is a concession to that fact. Fourth, the gap between recording and preventing: EVA, as scoped here, makes verification external and tamper-evident and leaves enforcement to the attenuation of Volume 3; it does not, by itself, stop a determined operator who controls the whole stack from declining to deploy it. And fifth, governance capture: the independence on which EVA rests is only as durable as the neutrality of the body that eventually holds the standard. The proposal is offered as a stable but revisable foundation, not a finished system.

12.3 Recording is not yet preventing

One limitation deserves emphasis because it is easily overlooked. EVA,

as scoped here, makes verification external, faithful, and tamper-evident; it records what happened and preserves the record. Prevention, where it occurs, is the pre-decode attenuation of the analytical engine, not a function of EVA itself, and the deliberate choice of this volume not to specify an automated hardware interlock means EVA does not, by itself, physically stop a harmful output. Its power is epistemic and institutional: it makes safety provable and incidents diagnosable, and it makes a developer who declines to deploy it visibly accountable for that choice. That is a great deal, but it is not a guarantee of prevention, and the distinction should not be blurred in advocacy.

Appendix A — The Three Structural Conditions (Normative)

The table below is normative. In case of conflict between prose and table, the table governs; in case of conflict with the Unified Structural Coordinate Table, that table governs for all HNS coordinate content.

Condition	Requirement	Realized by
Verifiability	Every reasoning step emitted in a standard machine-readable schema, independently analysable	JSON-LD / RDF logs; W3C PROV-O; HNS coordinate content
Transparency	Monitor runs in a separate process space; the model cannot read or alter it	Sidecar architecture with strict process separation
Physical Immutability	Verification logic and log signing rooted in a substrate unreachable by software	FPGA / ASIC / TPM / secure enclave

Appendix B — The EVA Log Record

Non-normative. Each entry binds an HNS coordinate to the verification verdict at one inference step, in JSON-LD conformant to W3C PROV-O. Representative fields and an example entry:

- `eva:configuration` — the full HNS coordinate ($L \times C \times Q \cdot O$).
- `eva:assertedOperator` / `eva:supportedOperator` — the operator the claim asserted and the one its evidence supported.
- `eva:groundingScore` — the composite Social Meta Layers grounding $s \in [0,1]$.
- `eva:attenuation` — the factor $g(s)$ the analytical engine applied.
- `eva:flag` — the failure type and sub-form, if any.
- `eva:smsVerdicts` — the per-layer social grounding verdicts.
- `prov:wasGeneratedBy` / `eva:signature` — the generating activity and the TPM signature.

```
{
  "@context": "https://www.w3.org/ns/prov",
  "@type": "prov:Entity",
  "eva:step": 42,
  "eva:configuration": "L6 x C5 x Q3 . 02",
  "eva:assertedOperator": "02",
  "eva:supportedOperator": "01",
  "eva:groundingScore": 0.46,
  "eva:attenuation": 0.46,
  "eva:flag": "UnsupportedCausality/confounding",
  "eva:smsVerdicts": { "SMS-4": 0.4, "SMS-5": 1.0 },
  "prov:wasGeneratedBy": "eva:sidecar",
  "eva:signature": "tpm:sha256:9f2c..."
}
```

A passing entry, by contrast, records grounded, well-supported output — asserted and supported operators agree, the grounding score is high, and the attenuation is near unity:

```
{
  "@context": "https://www.w3.org/ns/prov",
  "@type": "prov:Entity",
  "eva:step": 43,
  "eva:configuration": "L4 x C5 x Q3 . 02",
  "eva:assertedOperator": "02",
  "eva:supportedOperator": "02",
  "eva:groundingScore": 0.95,
  "eva:attenuation": 0.95,
  "eva:flag": null,
  "eva:smsVerdicts": { "SMS-4": 1.0, "SMS-5": 1.0 },
  "prov:wasGeneratedBy": "eva:sidecar",
}
```

```
"eva:signature": "tpm:sha256:4b71..."  
}
```

Appendix C — Conformance by Sector

Non-normative guidance on the grade of conformance appropriate to representative sectors.

Sector	Principal risk	Recommended realization
Healthcare	Misdiagnosis	Software sidecar + TPM-signed logs; every clinical recommendation recorded
Finance	Biased decisions	FPGA-anchored verification + real-time deviation detection
Government	Discriminatory outcomes	ASIC implementation; mandated independent third-party log access
Infrastructure	Safety-critical failure	Hardware-embedded verification in edge devices; cryptographic log synchronization

Appendix D — Draft New Work Item Proposal

A draft New Work Item for EVA standardization in ISO/IEC JTC 1/SC 42, for submission via the national body.

Item	Content
Proposed title	Information technology — Artificial intelligence — Technical Specification for External Verification Architecture (EVA) of AI systems
Proposing body	National standards body (e.g. JISC, Japan)
Proposer	Satoru Hara / Natural Structure Works (NSW)
Document type	Technical Specification (TS), progressing to International Standard (IS)
Scope	EVA definition; the three structural conditions; implementation requirements; conformance criteria; alignment with ISO/IEC AI standards
Related standards	ISO/IEC 42001 / 23894 / 22989; EU AI Act 2024/1689; NIST AI RMF 1.0

Column — EVA as the Flight Recorder for AI

Aviation did not become safe by building different machines or training different people. It became safe by gaining the ability to find out, after each accident, exactly what had happened. Mandatory flight recorders, standardized maintenance, and independent investigation turned every failure into evidence, and evidence into targeted, verifiable correction. Over the decades that followed, the fatal-accident rate fell by more than ninety percent — not because flight grew less demanding, but because the industry could at last see why things went wrong and confirm that its fixes worked.

AI today is roughly where aviation was before the recorder was mandatory. The systems are powerful and the capabilities impressive, yet when something goes wrong we largely cannot say why, cannot confirm whether a fix reached the cause, and cannot be sure the same failure will not recur. EVA is, in essence, the flight-recorder mandate for AI: not a regulatory burden, but the structural prerequisite for systematic safety improvement. The question is not whether AI needs external verification, but how soon it will be built — and whether the standards community moves before the next generation of high-stakes deployments makes its absence impossible to ignore.

Glossary

External Verification Architecture (EVA) — An architecture for monitoring, recording, and evaluating AI reasoning through a mechanism independent of the model's parameters.

Verifiability — EVA condition: every reasoning step emitted externally in a machine-readable schema.

Transparency — EVA condition: the monitor runs in parallel with inference and cannot be manipulated by the model.

Physical Immutability — EVA condition: verification logic on a substrate the model cannot alter.

Sidecar — A monitoring process in a separate address space alongside inference, with no shared memory the model can reach.

JSON-LD / PROV-O — The machine-readable log format and the W3C provenance ontology EVA records conform to.

Reference baseline — The external standard against which reasoning is judged appropriate; HNS is the baseline this series supplies.

TPM — A hardware module that cryptographically signs logs, giving a hardware-rooted integrity guarantee.

Conformance grade — The degree to which an implementation satisfies the three conditions, recorded with each output.

From assertion to proof — The shift EVA effects: a safety claim becomes a third party can inspect rather than one the public must believe.

References

- European Parliament and Council. (2024). Regulation (EU) 2024/1689 — Artificial Intelligence Act.
- ISO/IEC. (2023). ISO/IEC 42001:2023 — AI management system.
- ISO/IEC. (2023). ISO/IEC 23894:2023 — AI risk management.
- ISO/IEC. (2022). ISO/IEC 22989:2022 — AI concepts and terminology.
- NIST. (2023). AI Risk Management Framework (AI RMF 1.0).
- W3C. (2013). PROV-O: The PROV Ontology.
- Trusted Computing Group. (2019). TPM Library Specification, Family 2.0.
- Varela, F. J., Thompson, E., & Rosch, E. (1991). The Embodied Mind. MIT Press.
- Russell, S., & Norvig, P. (2021). Artificial Intelligence: A Modern Approach (4th ed.). Pearson.
- Hara, S. (2026). Human Natural Structure Origin & Theoretical Foundation (Volume 1). Natural Structure Works..
- Hara, S. (2026). Human Natural Structure Foundational & Observational Specification (Volume 2). NSW..
- Hara, S. (2026). Human Natural Structure Analytical Specification (Volume 3). NSW..
- Hara, S. (2026). Human Natural Structure Social Meta-Structure Specification (Volume 4). NSW..

Human Natural Structure

Standards & Conformance Specification

A Technical Mapping of HNS to International AI Standards and Regulatory Obligations

Volume 6 · Human Natural Structure Series

Satoru Hara

Natural Structure Works (NSW)

A Note on Method and Status

This volume is written in the form of a technical standard — with clauses, normative references, conformance requirements, and the auxiliary verbs (SHALL, SHOULD, MAY) that standards use. The form is deliberate: it states precisely what an implementation would have to do. The form must not, however, be mistaken for status, and the reader is owed an unusually direct statement of that distinction.

This is a proposed specification, not an adopted standard. Human Natural Structure has not been reviewed or adopted by ISO, IEC, JTC 1/SC 42, CEN-CENELEC, NIST, IEEE, ITU, or any other standards body, and nothing here carries regulatory force. The normative verbs define conformance to this document; they do not impose any legal obligation. Where the text maps HNS to an article of law or a clause of a standard, it states how HNS would support that provision if adopted — a proposal placed before the standards community, not a description of an existing requirement.

Three registers. Established: the regulations and standards referenced — the EU AI Act, the ISO/IEC SC 42 family, the NIST framework, the W3C provenance stack, the functional-safety and security standards — are real, current artifacts the reader can verify. HNS design: the coordinate, the verification architecture, the control system, and their mappings to those instruments are the author’s engineering proposals. Conjecture: that HNS will be adopted, and that adoption would deliver the enforcement the framework argues for, is a wager, marked as such.

Currency. The regulatory landscape described here is current to mid-2026, including the provisional Digital Omnibus agreement of 7 May 2026 deferring the AI Act’s high-risk deadlines. That landscape moves; before any reliance, the reader should consult the Official Journal of the EU and the standards bodies directly. As elsewhere in the series, named instruments indicate the ground this volume builds on, not endorsement by their authors.

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0. Foreword

0.1 Purpose

This specification defines how Human Natural Structure (HNS) and its verification and control architectures are expressed as a conformance-checkable technical layer beneath the existing process standards and laws that govern artificial intelligence. Its purpose is to supply what those instruments require but do not themselves provide: a technical means by which conformance can be verified, recorded, and enforced at the level of the individual inference.

0.2 Background — the enforcement gap

The world's AI regulations and management standards converge on a demand — that AI be safe, logged, transparent, and subject to human oversight — and stop short of specifying the technical mechanism by which any of these is to be achieved. The EU AI Act mandates record-keeping and human oversight for high-risk systems without fixing what is recorded or how oversight is exercised; ISO/IEC 42001 certifies management processes, not inference-level behaviour; the NIST framework names a MEASURE function whose tooling is the least mature of the five. The result is a gap between requirement and implementation: the obligations exist, the technical layer that would discharge them does not. This gap is this volume's subject, and the claim — advanced as a proposal — is that the HNS stack is a candidate for that missing layer.

NOTE *The gap is now visible in the law's own timetable. Under the Digital Omnibus agreement of 7 May 2026, the application of the AI Act's high-risk obligations is linked to the availability of harmonised standards and support tools, and the Annex III deadline has been deferred to 2 December 2027. A regulation is, in effect, waiting on a technical layer that does not yet exist — the situation this specification addresses.*

0.2.1 The gap, in numbers

The gap is not only conceptual. The function most able to close it is the least developed: in the NIST framework the MEASURE function — the one that would quantify and verify behaviour — has the least mature tooling of the five. The infrastructure is correspondingly thin: industry surveys in 2025 found that fewer than one in five

organizations, even in regulated sectors such as healthcare and finance, had dedicated AI incident-reporting tooling in place. Most organizations have identified their AI risks yet lack the means to record and respond when those risks materialize. That deficit — measurement and recording at the level of the individual inference — is what this specification proposes to fill.

0.2.2 The missing-link claim, stated and bounded

The claim of this volume can now be stated exactly. Existing instruments supply the obligations; HNS proposes the inference-level technical layer — coordinate, verification, control — by which those obligations could be made checkable and enforceable. That layer is the “missing link.” The claim is bounded in two ways: it is a proposal, not an adopted requirement; and it is an enabling layer, not a guarantee — it makes conformance checkable, it does not make a system correct. With those bounds, the missing-link claim is a contribution to the standards conversation, not a substitute for it.

0.3 Relationship to international standards

This document does not replace any standard. It is designed to sit beneath ISO/IEC 42001, the EU AI Act, the NIST framework, and their peers, supplying the inference-level verification and control they assume. Clause 2 lists the instruments treated as normative references; Annex G maps the HNS components to them in detail.

0.4 Position in the HNS series

This volume connects the entire series as the final conformance layer.. It depends on, and does not restate, the core theoretical foundation of Volume 1, the coordinate system of Volume 2 and 3 (HNS-36/144/864), the grounding stack of Volume 4 (SMS-6), and the verification architecture of Volume 5 (EVA).. Those volumes are the normative source for their respective components; this volume references them and adds the standards, legal-basis, conformance, and control layer.

1. Scope

1.1 Subject matter

This document specifies: the expression of the HNS coordinate and its verification and control architectures as a conformance-checkable layer; the machine-readable record format by which conformance is evidenced; the mapping of that layer to international AI standards and to the EU AI Act; and the conformance requirements an implementation SHALL meet to claim alignment with this specification.

1.2 Applicable AI systems

This document applies to AI systems that generate outputs consumed by humans or by downstream automated processes, in particular generative and decision-support systems deployed in high-risk contexts as defined by applicable law. It is technology-neutral as to model architecture.

1.3 Out of scope

This document does not specify model training, model architecture, or the internal parameters of any AI system; it concerns only the external verification and control of outputs. It does not constitute legal advice, and conformance to it does not by itself establish legal compliance with any regulation.

1.4 Intended users

The intended users are standards bodies and their technical committees; regulators and conformity-assessment bodies; AI providers and deployers seeking a technical basis for the obligations they face; and independent auditors.

1.5 Normative conventions

In this document SHALL indicates a requirement, SHOULD a recommendation, MAY a permission, and CAN a capability. These verbs define conformance to this document only (Clause 11); as stated in the Note on Method, they impose no legal obligation and presume no adoption by any standards body. References to clauses and

annexes are to this document unless otherwise stated.

2. Normative References

The following instruments are referred to in this document. They are real, externally maintained artifacts; HNS does not control them and claims no endorsement by them.

2.1 ISO/IEC JTC 1/SC 42 — Artificial Intelligence

Reference	Title / subject
ISO/IEC 42001:2023	AI management system (AIMS) — requirements
ISO/IEC 42005:2025	AI system impact assessment
ISO/IEC 42006	Requirements for bodies auditing and certifying AIMS
ISO/IEC 22989:2022	AI concepts and terminology
ISO/IEC 23894:2023	AI risk management
ISO/IEC 5338	AI system life cycle processes
ISO/IEC TR 5469	Functional safety and AI systems
ISO/IEC 24029-2	Robustness of neural networks — formal methods
ISO/IEC 38507:2022	Governance implications of the use of AI by organizations

2.2 European Union regulatory framework

Reference	Subject
Regulation (EU) 2024/1689	Artificial Intelligence Act (entered into force Aug 2024)
Digital Omnibus on AI (2026)	Provisional agreement of 7 May 2026 amending AI Act deadlines
AI Act Articles 9–15, 17	High-risk requirements; record-keeping; transparency; human oversight; QMS
AI Act Annex III	High-risk use-case categories
Directive (EU) 2022/2555 (NIS2)	Network and information security
CoE CETS No. 225	Council of Europe Framework Convention on AI

2.3 W3C, IEEE, NIST, ITU-T, and CEN-CENELEC

Reference	Subject
W3C PROV-O; RDF; JSON-LD; SPARQL	Provenance ontology and the linked-data stack for audit records
NIST AI RMF 1.0 (2023)	AI Risk Management Framework
NIST-AI-600-1 (2024)	Generative AI Profile
IEEE 7001; IEEE P2863	Transparency; organizational governance of AI

Reference	Subject
IEC 61508; ISO/IEC 27001	Functional safety; information security management
ISO/IEC/IEEE 29119	Software testing (conformance test basis)
CEN-CENELEC JTC 21 (prEN 18228, 18284, 18283)	Draft harmonised standards for the EU AI Act
ITU-T (AI for telecommunications)	External verification requirements in telecom AI

2.4 How these references are used

The referenced instruments play three distinct roles, and keeping the roles separate matters. The ISO/IEC SC 42 family and the governance standards supply the management and life-cycle context into which the HNS technical layer fits. The EU AI Act, NIS2, and the Council of Europe Convention supply the binding obligations the layer is designed to evidence. The linked-data, functional-safety, security, and testing standards supply the engineering substrate — the formats, integrity, and test basis — on which the layer is built. A reference’s role determines how this document treats it: context is aligned with, obligation is evidenced, substrate is conformed to.

2A. Legal Basis for Referenced Standards

2A.1 Legal hierarchy

The instruments above occupy different rungs of bindingness, and this specification keeps them distinct. Binding law (the AI Act; NIS2) imposes obligations directly. Harmonised standards (the CEN-CENELEC drafts), once cited in the Official Journal, confer a presumption of conformity with the law but are not themselves law. International standards (the ISO/IEC family) and framework documents (the NIST RMF) are voluntary unless incorporated by reference into law or contract. HNS, as proposed here, is below all of these: a candidate technical specification that would acquire force only through adoption by one of them.

2A.2 The conformity-of-presumption chain

The mechanism this volume targets is the European one, in which a harmonised standard cited in the Official Journal gives a manufacturer a presumption of conformity with the corresponding legal requirement. The proposal is that an HNS-based technical specification, adopted as or referenced by such a standard, would convert the AI Act’s high-risk requirements (Articles 9–15) into checkable, inference-level criteria. Whether any standard confers presumption of conformity is a decision of the European Commission, not of its drafters; this is stated to avoid overclaiming.

2A.3 HNS component legal-basis matrix

HNS component	Primary legal / standards hook	Nature of the hook
HNS-36/144/864 (Vols 2–3)	AI Act Art. 15 (accuracy, robustness)	proposed technical means
SMS-6 (Vol 4)	AI Act Art. 9 (risk); CoE CETS 225	proposed technical means
EVA (Vol 5)	AI Act Art. 12 (logging), Art. 13 (transparency)	proposed technical means
ECS (Clause 9)	AI Act Art. 14 (human oversight)	proposed technical means
Integrated stack (Clause 10)	ISO/IEC 42001 §5; ISO/IEC 38507	proposed technical means

2A.4 The limit of the chain

The presumption-of-conformity chain has a limit this specification states rather than glosses. A technical specification supports a legal requirement only once a recognized body adopts it and, in the European case, only once its reference is published in the Official Journal; until then the mapping is aspirational. Moreover, presumption of conformity is rebuttable: it shifts the burden of proof, it does not settle the question. HNS is therefore best understood as a candidate input to that chain, whose value is realized only through the standards process set out in Clause 15.

2B. JSON-LD, RDF, and the HNS Ontology

2B.1 The linked-data stack

Conformance evidence under this specification is machine-readable by construction. Records are expressed in JSON-LD, interpreted as RDF, conformant to the W3C PROV-O provenance ontology, and queryable in SPARQL. This choice means that an audit record is not a proprietary log but a graph any third party can parse, relate, and query with standard tooling — the property that makes independent verification practical.

2B.2 The HNS namespace

HNS terms are published under a stable namespace prefix (here written `hns:`) so that each coordinate, operator, and verdict is a dereferenceable term rather than an opaque string. A cell, an operator, a grounding layer, and an audit field each have a defined IRI.

2B.3 Cell definition in JSON-LD

A coordinate cell is defined as a typed entity carrying its layer, category, quadrant, and operator:

```
{
  "@context": "https://naturalstructureworks.com/hns/context",
  "@id": "hns:cell/L6-C5-Q3-02",
  "@type": "hns:Coordinate",
  "hns:layer": "L6", "hns:category": "C5",
  "hns:quadrant": "Q3", "hns:operator": "02"
}
```

2B.4 EVA audit record in JSON-LD / PROV-O

Each verification step is recorded as a PROV-O entity bound to the coordinate and the verdict (the schema is normative in Annex B):

```
{
  "@context": "https://www.w3.org/ns/prov",
  "@type": "prov:Entity",
  "eva:step": 42,
  "eva:configuration": "L6 x C5 x Q3 . 02",
  "eva:assertedOperator": "02",
  "eva:supportedOperator": "01",
  "eva:groundingScore": 0.46,
  "eva:attenuation": 0.46,
  "eva:flag": "UnsupportedCausality/confounding",
  "eva:ecsAction": "GATE_HOLD",
}
```

```
"prov:wasGeneratedBy": "eva:sidecar",  
"eva:signature": "tpm:sha256:9f2c..."  
}
```

2B.5 SPARQL query examples

Because records are RDF, conformance questions become queries.
Two examples:

```
# Query 1 - all records that failed verification  
SELECT ?step ?config ?flag WHERE {  
  ?r eva:flag ?flag ; eva:step ?step ;  
    eva:configuration ?config .  
  FILTER(?flag != "none")  
}  
  
# Query 2 - session risk profile by natural layer  
SELECT ?layer (AVG(?s) AS ?meanGrounding) WHERE {  
  ?r eva:configuration ?config ; eva:groundingScore ?s .  
  BIND(SUBSTR(?config,1,2) AS ?layer)  
} GROUP BY ?layer
```

3. Terms and Definitions

coordinate — the position of a claim in the HNS-36/144/864 system (Volumes 2–3).

grounding score (s) — the composite verdict of the Social Meta Layers, $s \in [0,1]$ (Volume 4).

attenuation — the pre-decode scaling $g(s)$ applied to a candidate output (Volume 3, 06).

EVA — External Verification Architecture: the external, tamper-evident recording layer (Volume 5).

ECS — External Control System: the external control gate and emergency override defined in Clause 9.

audit record — a JSON-LD / PROV-O entity evidencing the verification of one inference step.

conformance — satisfaction of the requirements of Clause 11 of this document.

presumption of conformity — the legal effect, under EU law, of a harmonised standard cited in the Official Journal.

4. HNS Overview (Normative Recap)

This clause recapitulates the HNS components only to the depth this specification needs. The normative source for each is the volume cited; in case of conflict, that volume and the Unified Structural Coordinate Table govern.

4.1 The Multi-Axis Verification process

Reliability is treated as the conjunction of three axes: Axis 1 (generative) proposes; Axis 2 (structural / causal) positions and audits; Axis 3 (grounded / executive) grounds. EVA records the result and ECS gates the output. The axes are approximately independent, which is why their conjunction adds assurance none provides alone.

4.2 The six Natural Layers (canonical)

ID	Name	Lineage
L1	Physical / Substrate	Physics / BFO
L2	Cellular / Informational	Computational neuroscience
L3	Cognitive / System 1	Kahneman Type 1
L4	Operational / System 2	Kahneman Type 2
L5	Micro-Social / Interaction	Habermas
L6	Macro-Social / Institutional	Institutional theory / BFO

4.3 The six Abstract Categories (canonical)

ID	Name	Lineage
C1	Continuant / Entity	Aristotle / BFO
C2	Occurrent / Process	Aristotle / BFO
C3	Relational / Boundary	Aristotle / Kant
C4	Quality / State	Kant
C5	Transformation / Operation	Pearl (do-operator)
C6	Meta-Level / Schema	Kant

NOTE *Naming reconciliation. Earlier drafts of this specification used different layer and category names (e.g. “Substrate / Process...” for the categories). The canonical names above, fixed in the Unified Structural Coordinate Table and Volume 2, are authoritative and supersede those drafts.*

5. HNS-36, HNS-144, and HNS-864 (Normative Recap)

5.1 HNS-36 and HNS-144

HNS-36 is the product of the six Natural Layers and six Abstract Categories (36 base cells); HNS-144 extends it by four Onto-Modal Quadrants (Subjective/Objective $\times$ Necessary/Contingent). The coordinate positions a claim and observes its modal stance, suppressing Layer Jump, Category Ambiguity, and Metaphor Contamination. The full specification is Volume 2; Annex A reproduces the normative 36-cell reference.

5.2 HNS-864 and the structural risk score

HNS-864 extends HNS-144 by six Inference & Control Operators (O1–O3, Pearl’s hierarchy; O4–O5, perceptual control; O6, attenuation). The analytical engine computes, for each candidate, the operator asserted against the operator the evidence supports; a mismatch drives a structural risk score, and the grounding score s drives the attenuation $O6 = g(s)$. The full specification is Volume 3.

$$output = g(s) \cdot raw_output, \quad g(0) = 0, \quad g \text{ monotone increasing}, \quad g(s) \in [0,1]$$

5.3 SMS-6 grounding

The six Social Meta Layers (Volume 4) ground the output from the somatic base to the universal scale; their conjunction is the grounding score s consumed above. SMS-5 (Governance) and SMS-6 (Universal) are the layers most directly engaged by this volume’s technical mapping and conformance criteria..

8. EVA — External Verification Architecture

8.1 Role in this specification

EVA (Volume 5) is the recording and verification layer. It satisfies three structural conditions — Verifiability, Transparency, Physical Immutability — by a sidecar that records each step in JSON-LD/PROV-O and a hardware root of trust that signs the record. In this specification EVA is the source of the audit record against which conformance is judged. EVA records and proves; it does not, by itself, stop output.

8.2 Mandatory audit-record fields

An implementation claiming conformance SHALL emit, for each verified step, at least the following fields (schema normative in Annex B): the coordinate configuration; the asserted and supported operators; the grounding score; the attenuation applied; any failure flag and sub-form; the per-layer SMS verdicts; the generating activity; and a hardware signature.

8.3 Mapping to EU AI Act Articles 12 and 13

AI Act provision	EVA realization
Art. 12 — record-keeping / logging	The JSON-LD audit record gives the required log a schema and a granularity (per inference step)
Art. 13 — transparency / information to deployers	The third-party-readable, tamper-evident record makes the system’s behaviour inspectable
Art. 15 — accuracy, robustness	Operator-mismatch and grounding scores provide an inference-level robustness signal

8.4 The boundary of EVA

EVA’s power is epistemic, and its boundary should be stated so that Clause 9 is not misread as part of it. EVA records faithfully, signs immutably, and exposes to third parties; it does not stop output, does not correct the model, and does not guarantee that a faithfully recorded verdict is correct — the baseline-correctness limit of Volume 5 carries into this specification. The control that acts on EVA’s verdict is the separate tier of Clause 9; conflating the recorder with the controller would overstate what either does.

9. ECS — External Control System

Where EVA records and proves, the External Control System acts. ECS is the control tier of the stack: an external gate on output delivery, with an emergency override, mapped to the human-oversight requirement of EU AI Act Article 14. It is introduced here, in Volume 6, as a component distinct from — and downstream of — the verification of Volume 5..

9.1 The three enforcement tiers

The series distinguishes three mechanisms, complementary not redundant. First, attenuation (Volume 3, O6): a pre-decode, graded, “soft” suppression that shapes what is generated. Second, EVA (Volume 5): faithful, immutable recording that proves what happened but does not stop it. Third, ECS (this clause): an external, “hard” gate that can hold or halt delivery and place a human in the loop. A complete enforcement story uses all three; ECS is the backstop the other two deliberately leave open.

9.2 The gate requirement

Under this specification, no AI output in a high-risk deployment SHOULD be delivered to a user without passing through the control gate, and in the strictest conformance level (Clause 11) SHALL NOT. The gate consumes the EVA verdict for the step: a record below the deployment’s grounding or risk threshold is held, routed for human review, or blocked, and the action taken (for example GATE_HOLD) is itself written to the record.

9.3 Mapping to EU AI Act Article 14

Art. 14 requirement	ECS realization (proposed)
14(1) — human oversight designed in	The control gate is part of the output pipeline by architecture, not an add-on
14(4)(a) — understand capacities and limits	The held record carries the coordinate and the reason for the hold
14(4)(d) — ability to intervene / interrupt	The emergency override halts delivery and requires authenticated reactivation
14(4)(e) — ability to stop the system	The override operates independently of the model’s software state

9.4 Emergency override

The emergency override SHALL operate independently of the AI system’s software state; SHOULD take effect within the latency budget set in the deployment specification (a target of ≤ 500 ms is recommended); SHALL generate a mandatory incident record within 60 seconds of activation; and SHALL require operator authentication before reactivation. A failure-mode analysis SHOULD demonstrate, to the functional-safety expectations of IEC 61508, that no single software failure can disable the override.

NOTE *Status and caution. ECS as specified — a mandatory gate and an independent override — is the author’s proposal, not an adopted requirement, and the latency and single-failure properties are engineering targets, not demonstrated results. An automated external stop is itself a safety-critical component (IEC 61508); “physical enforcement” should be read as a design goal subject to validation, not as a proven guarantee. Volume 5 deliberately placed ECS out of EVA’s scope; this clause supplies it as the separate control tier.*

9.5 Fail-safe design

An external control gate must decide what to do when it is itself uncertain or degraded, and the choice is a safety-design decision with no universally correct answer. A fail-closed gate blocks output when verification is unavailable, favouring safety over availability — appropriate where a wrong output is worse than none. A fail-open gate releases output when the gate is degraded, favouring availability — appropriate where withholding is itself harmful, as in some safety-of-life settings. The deployment specification SHALL state which mode applies per output class, and the chosen mode SHALL be recorded; the human-oversight provisions of Clause 9.3 exist precisely so that this decision is not left implicitly to the machine.

10. The Integrated HNS Stack

The integrated HNS stack is the meta-architecture that composes the coordinate (HNS-36/144/864), the grounding stack (SMS-6), the verification layer (EVA), and the control system (ECS) into one runtime. It engages the governance expectations of ISO/IEC 38507, ISO/IEC 42001 §5, the life-cycle processes of ISO/IEC 5338, and the human-oversight provision of CoE CETS 225.

NOTE *Naming. Earlier drafts called this meta-architecture “HumanOS.” This specification uses “the integrated HNS stack” for the same construct, to avoid introducing a top-level component not present in the Unified Structural Coordinate Table; “HumanOS” is retained only as a legacy name.*

10.1 Runtime data flow

Interface	From	To	Data
I-1	AI model	Axis 2 / Axis 3	candidate output stream
I-2	Axis 2 / Axis 3	EVA	coordinate, operators, grounding score
I-3	EVA	ECS	signed verdict for the step
I-4	ECS	user / deployer	released, held, or blocked output

The flow is a loop: the model proposes; Axes 2 and 3 position and ground; EVA records and signs; ECS gates; and the action returns to the record. Each interface is external to the model in the sense Volume 5 makes precise, and each is auditable after the fact.

11. Conformance Requirements

An implementation claiming conformance to this specification SHALL satisfy the requirements of this clause at the level it declares.

11.1 HNS-36 baseline

The implementation SHALL assign every verified output a coordinate in the canonical HNS-36 system (Annex A), and SHALL record the assignment. Use of HNS-144 and HNS-864 resolution is RECOMMENDED for high-risk deployments.

11.2 EVA logging

The implementation SHALL emit, for each verified step, an audit record conformant to the Annex B schema, in JSON-LD/PROV-O, and SHALL root the integrity of the record in a hardware signature. Records SHALL be retained and made available to accredited third parties.

11.3 ECS safety boundaries

Where the declared level requires it, the implementation SHALL route each output through the control gate and SHALL provide an emergency override meeting Clause 9.4. The boundaries of automated action (hold, review, block) SHALL be configured in the deployment specification and recorded.

11.4 Conformance levels

Level	Requirements	Indicative use
L1 — Recorded	HNS-36 baseline + EVA logging (Verifiability)	internal / development
L2 — Independent	+ Transparency + hardware-signed records	most high-risk deployments
L3 — Controlled	+ ECS gate and emergency override	safety-critical / public-sector

11.5 Conformance test suite

Conformance SHOULD be demonstrated through a documented test suite structured per ISO/IEC/IEEE 29119, exercising representative inputs across the coordinate and verifying that records, gating, and

override behave as specified. The suite and its results form part of the conformance evidence.

11.6 Conformance statement

An implementation claiming conformance SHALL publish a conformance statement recording: the declared level (L1–L3); which of the three conditions are met and by what means; the substrate of the hardware root of trust; the fail-safe mode per output class; the retention and third-party-access policy for records; and the version of this specification and of the Unified Structural Coordinate Table against which conformance is claimed. The statement is itself part of the audit evidence and SHALL be made available to accredited third parties.

12. Implementation Guidance

12.1 A staged adoption path

Adoption is expected to be incremental. A practical path: begin at Level 1 with coordinate assignment and JSON-LD logging in a sidecar; add hardware-signed records and process separation to reach Level 2; and add the control gate and override to reach Level 3 for the highest-stakes deployments. Each stage produces evidence usable for the next.

12.2 Proof-of-concept procedure

A minimal proof of concept assigns the HNS-36 coordinate to a stream of representative outputs, records the verdicts, and demonstrates that a third party can, from the records alone, identify the coordinate of any flagged output and confirm whether a subsequent intervention removed the flag. This is the smallest demonstration that exercises the verification-and-correction cycle the framework rests on.

12.3 A worked conformance walkthrough (informative)

To see the layers act together, follow one output. A high-risk decision-support system proposes the continuation “the applicant was rejected because of the neighbourhood.” Axis 2 positions the claim at ($L6 \times C5 \times Q3$) and reads it as asserting intervention (O2); the evidence supports only association (O1). Axis 3 returns a low governance and economic grounding, since a protected-proxy attribution is in play, and the grounding score falls to $s = 0.4$. The attenuation O6 damps the causal continuation. EVA writes the record — configuration, asserted O2, supported O1, $s = 0.4$, flag `UnsupportedCausality` — with a hardware signature. ECS, at Level 3, reads the sub-threshold record and issues `GATE_HOLD`, routing the decision to a human reviewer and writing the action to the record. A third party later queries the session (Clause 2B.5) and confirms both that the deviation was caught and that the held decision was reviewed. A single output exercises every requirement of Clause 11.

13. Security and Safety Considerations

13.1 Structural failure taxonomy

The framework names a small set of structural failures it is built to surface: Layer Jump (illegitimate transfer across levels), Category Ambiguity (conflated categories or normative kinds), Metaphor Contamination (a figure read as a mechanism), Scope Drift (unannounced change of scope), and Unsupported Causality (a causal claim above its supported operator). Each maps to a coordinate signature and to a record flag, which is what makes them externally auditable rather than matters of judgement.

13.2 Relationship to ISO/IEC 27001, IEC 61508, and NIS2

The verification and control layer is itself an asset to be secured. The integrity of records and the availability of the control gate fall within an information-security management system (ISO/IEC 27001); the emergency override is a functional-safety function (IEC 61508); and, for operators of essential services, both fall within the security obligations of NIS2. This specification assumes, and does not replace, those regimes.

13.3 Threat model for the verification layer

The verification layer is itself a target, and a specification that ignored this would be naive. Three threats are primary. Record forgery — an attacker rewrites or fabricates audit records — is countered by the hardware signature and process separation (Volume 5's Physical Immutability and Transparency). Gate bypass — output routed around the control gate — is countered by making the gate part of the delivery pipeline by architecture (Clause 9.2) rather than an optional filter. Baseline substitution — the reference baseline silently replaced with a permissive one — is countered by binding the conformance statement to a named, versioned baseline (Clause 11.6) a third party can check. None of these countermeasures is absolute; each raises the cost of an attack and, more importantly, makes a successful attack leave a detectable trace.

14. Ethical and Societal Considerations

Three considerations bear stating. First, externality is a safeguard against capture: a verification controlled by the verified party is no safeguard, which is why both the recorder and the reference baseline must be independent and public. Second, recording is not neutral: an inference-level record is a powerful artifact, and its retention, access, and privacy must be governed so that a safety mechanism does not become a surveillance one. Third, the control gate carries its own risk: an automated stop can fail open or fail closed, and the human-oversight provisions exist precisely so that the final judgement remains with a person. The framework is offered in the belief that provable, externally checkable safety serves the public better than unverifiable assurance — a belief, not a proof.

15. Standardization Status and Pathway

15.1 Current status

HNS is not, at the time of writing, part of any adopted standard or law. It is a published, version-controlled framework with a draft New Work Item prepared for submission to ISO/IEC JTC 1/SC 42. The mappings in this volume are proposals for how HNS would support the cited instruments if adopted, not descriptions of existing requirements.

15.2 The regulatory window

The timing is favourable to the proposal and is stated plainly, not triumphantly. The EU AI Act's high-risk obligations have been deferred — under the May 2026 Digital Omnibus agreement, Annex III obligations move to December 2027 — with the deferral linked to the availability of harmonised standards and support tools. The harmonised standards themselves have slipped from their 2025 deadline toward 2026, and the regulator has acknowledged asking for standardization in areas where no settled technical state of the art exists. That is the opening this specification addresses: a defined need for an inference-level technical layer that does not yet have an agreed form.

15.3 Pathway

The pathway is the one set out in Volume 5: publish as a citable preprint; submit a New Work Item to SC 42 through the national body; align with CEN-CENELEC for the EU AI Act; build an implementation track record through proofs of concept; and transfer governance to a neutral multi-stakeholder body so that the standard's independence outlasts its author.

15.4 Limitations

The limitations are real and are not hidden. HNS has not been reviewed by any standards body, and adoption is uncertain and not within the author's gift. The baseline-correctness problem of Volume 5 stands: the stack records and enforces faithfully, but the correctness of its verdicts depends on the correctness of the baseline, which is

itself revisable. The feasibility of per-step verification and gating within a latency budget is an engineering claim to be demonstrated. And the enforcement the framework offers is conditional on deployment: a provider who controls the whole stack can decline to adopt it, which is exactly why the standards and conformance verification route, rather than the purely technical one, is the subject of this volume..

Annex A — HNS-36 Cell Reference (Normative)

The 36 base cells are the product of the six Natural Layers (rows) and six Abstract Categories (columns). Cell identifiers are of the form L×C. The full per-cell specification is Volume 2; this annex fixes the canonical grid.

	C1	C2	C3	C4	C5	C6
L1	L1C1	L1C2	L1C3	L1C4	L1C5	L1C6
L2	L2C1	L2C2	L2C3	L2C4	L2C5	L2C6
L3	L3C1	L3C2	L3C3	L3C4	L3C5	L3C6
L4	L4C1	L4C2	L4C3	L4C4	L4C5	L4C6
L5	L5C1	L5C2	L5C3	L5C4	L5C5	L5C6
L6	L6C1	L6C2	L6C3	L6C4	L6C5	L6C6

Columns: C1 Continuant, C2 Occurrent, C3 Relational, C4 Quality, C5 Transformation, C6 Meta-Level. Rows: L1 Physical, L2 Cellular, L3 Cognitive, L4 Operational, L5 Micro-Social, L6 Macro-Social.

Annex B — EVA Audit Record Schema (Normative)

Each verified step SHALL be recorded as a JSON-LD/PROV-O entity with at least the following fields:

- `eva:configuration` — the HNS coordinate ($L \times C \times Q \cdot O$); string.
- `eva:assertedOperator` / `eva:supportedOperator` — operators 01–06; string.
- `eva:groundingScore` — composite SMS grounding $s \in [0,1]$; number.
- `eva:attenuation` — applied factor $g(s) \in [0,1]$; number.
- `eva:flag` — failure type and sub-form, or “none”; string.
- `eva:smsVerdicts` — per-layer grounding verdicts; object.
- `eva:ecsAction` — control action taken (RELEASE / GATE_HOLD / BLOCK); string.
- `prov:wasGeneratedBy` / `eva:signature` — generating activity; hardware signature; string.

Annex C — Audit Log Examples (Informative)

A failing record (causal over-claim held by the gate):

```
{ "eva:step": 42, "eva:configuration": "L6 x C5 x Q3 . 02",  
  "eva:assertedOperator": "02", "eva:supportedOperator": "01",  
  "eva:groundingScore": 0.46, "eva:attenuation": 0.46,  
  "eva:flag": "UnsupportedCausality/confounding",  
  "eva:ecsAction": "GATE_HOLD" }
```

A passing record (released):

```
{ "eva:step": 43, "eva:configuration": "L4 x C5 x Q3 . 02",  
  "eva:assertedOperator": "02", "eva:supportedOperator": "02",  
  "eva:groundingScore": 0.95, "eva:attenuation": 0.95,  
  "eva:flag": "none", "eva:ecsAction": "RELEASE" }
```

Annex D — EU AI Act Mapping (Normative)

Proposed mapping of HNS components to the high-risk requirements of the AI Act. This is a proposal for adoption, not a statement of existing conformity.

AI Act provision	HNS component	Means
Art. 9 — risk management	SMS-6; structural failure taxonomy	inference-level risk signal
Art. 10 — data governance	coordinate assignment; provenance record	structural classification
Art. 12 — record-keeping	EVA audit record	per-step JSON-LD log
Art. 13 — transparency	EVA third-party-readable record	inspectable behaviour
Art. 14 — human oversight	ECS gate and override	hold / review / stop
Art. 15 — accuracy, robustness	operator-mismatch; grounding score	inference-level metric

NOTE *Timeline. Under the May 2026 Digital Omnibus agreement (subject to formal adoption), Annex III high-risk obligations apply from 2 December 2027 and Annex I product-embedded obligations from 2 August 2028, with the high-risk timeline linked to the availability of harmonised standards.*

D.1 Annex III high-risk categories

For reference, the AI Act’s Annex III enumerates the high-risk use-case areas to which the obligations mapped above principally apply:

- biometrics, where permitted; safety components of critical infrastructure;
- education and vocational training; employment, worker management, and access to self-employment;
- access to essential private and public services and benefits, including creditworthiness and insurance;
- law enforcement; migration, asylum, and border control; administration of justice and democratic processes.

The HNS components of this annex are proposed as a common technical means of evidencing the Article 9–15 obligations across these areas, irrespective of domain.

Annex E — Runtime Architecture (Informative)

The integrated stack composes the components of Clause 10 into a single pipeline: model → Axis 2 / Axis 3 (coordinate, operators, grounding) → attenuation → EVA (record, sign) → ECS (gate, override) → release. The sidecar and the control gate run in process spaces separate from the model; the hardware root of trust signs records and anchors the override. The architecture is the engineering expression of the three enforcement tiers of Clause 9.1.

Annex F — Glossary (Informative)

MAV — Multi-Axis Verification: the conjunction of the generative, structural, and grounded axes.

HNS-36 / 144 / 864 — the nested structural coordinate (Volumes 2–3).

SMS-6 — the six Social Meta Layers; the Axis-3 grounding stack (Volume 4).

EVA — External Verification Architecture; the recording / verification layer (Volume 5).

ECS — External Control System; the control gate and emergency override (Clause 9).

integrated HNS stack — the meta-architecture composing all of the above (legacy name: HumanOS).

presumption of conformity — the EU-law effect of a harmonised standard cited in the Official Journal.

conformance level — L1 Recorded, L2 Independent, or L3 Controlled (Clause 11.4).

Annex G — Standards × HNS Component Alignment (Normative)

A representative alignment of international instruments to HNS components. The mapping states intended support, proposed for adoption; it does not assert that any body has adopted HNS. The set is illustrative of the coverage, not exhaustive.

Instrument	HNS component	Alignment (proposed)
ISO/IEC 42001:2023 §5–10	Integrated stack; EVA	AIMS controls evidenced at inference level
ISO/IEC 42005:2025	SMS-6; coordinate	impact assessment fed by structural signals
ISO/IEC 42006	conformance (Cl. 11)	third-party certification of AIMS
ISO/IEC 22989:2022	terms (Cl. 3)	shared concepts and terminology
ISO/IEC 23894:2023	SMS-6; risk taxonomy	risk management process
ISO/IEC 5338	integrated stack	AI system life-cycle processes
ISO/IEC TR 5469	ECS	functional safety of AI systems
ISO/IEC 24029-2	HNS-864; operators	robustness assessment
ISO/IEC 38507:2022	integrated stack	governance implications
ISO/IEC 27001	EVA; ECS	security of records and control assets
IEC 61508	ECS override	functional-safety integrity
ISO/IEC/IEEE 29119	conformance (Cl. 11.5)	test-suite basis
W3C PROV-O / JSON-LD / SPARQL	EVA record (Cl. 2B)	machine-readable provenance
NIST AI RMF 1.0	MAV; SMS-6	MEASURE / MANAGE functions
NIST-AI-600-1	failure taxonomy	generative-AI risk coverage
IEEE 7001	EVA	transparency of autonomous systems
EU AI Act Art. 9–15	see Annex D	high-risk requirements
EU AI Act Art. 12 / 13	EVA	logging; transparency
EU AI Act Art. 14	ECS	human oversight
CEN-CENELEC prEN 18228 / 18284	SMS-6; coordinate	draft harmonised: risk; data quality
NIS2 (EU 2022/2555)	EVA; ECS	security obligations for essential services
CoE CETS 225	integrated stack; ECS	human-oversight convention

References

- European Parliament and Council. (2024). Regulation (EU) 2024/1689 — Artificial Intelligence Act.
- Council of the EU & European Parliament. (2026). Provisional agreement on the Digital Omnibus on AI (7 May 2026).
- ISO/IEC. (2023). ISO/IEC 42001:2023 — AI management system.
- ISO/IEC. (2025). ISO/IEC 42005:2025 — AI system impact assessment.
- ISO/IEC. (2023). ISO/IEC 23894:2023 — AI risk management.
- ISO/IEC. (2022). ISO/IEC 22989:2022 — AI concepts and terminology.
- ISO/IEC. (2022). ISO/IEC 38507:2022 — Governance implications of the use of AI.
- NIST. (2023). AI Risk Management Framework (AI RMF 1.0).
- NIST. (2024). NIST-AI-600-1 — Generative AI Profile.
- W3C. (2013). PROV-O: The PROV Ontology.
- IEC. (2010). IEC 61508 — Functional safety of E/E/PE safety-related systems.
- ISO/IEC. (2022). ISO/IEC 27001 — Information security management systems.
- CEN-CENELEC JTC 21. (2025–2026). Draft harmonised standards in support of the EU AI Act.
- Council of Europe. (2024). Framework Convention on Artificial Intelligence (CETS No. 225).
- Hara, S. (2026). Human Natural Structure Series (Volumes 1–5):
- Volume 1: Origin & Theoretical Foundation
 - Volume 2: Foundational & Observational Specification
 - Volume 3: Analytical Specification
 - Volume 4: Social Meta-Structure Specification
 - Volume 5: External Verification Architecture. NSW.